

# SCIENTIFIC JOURNAL PUBLICATION TREND TRACKING SYSTEM

Group 6

# I. INTRODUCTION

## 1 Overview

### 1.1 Project Information

- Project Name: Scientific Journal Publication Trend Tracking System
- Group Name: Group 6
- Software Type: Web Application
- GitHub Repository:  
<https://github.com/NguyenNgocBaoTram10/Scientific-Journal-Publication-Trend-Tracking-System>

### 1.2 Project team

| Full Name              | Role   | Student ID   | Email                  |
|------------------------|--------|--------------|------------------------|
| Lê Thanh Tú            | Leader | 049206004778 | tult4778@ut.edu.vn     |
| Nguyễn Ngọc Bảo Trâm   | Member | 082306008015 | tramnnb8015@ut.edu.vn  |
| Lê Thị Huyền Trân      | Member | 077306003635 | tranlth3635@ut.edu.vn  |
| Trần Trung Hiếu        | Member | 089206008473 | hieutt8473@ut.edu.vn   |
| Trần Văn Học           | Member | 052206005618 | hoctv5618@ut.edu.vn    |
| Nguyễn Đoàn Nhật Quang | Member | 079206010585 | quangndn0585@ut.edu.vn |

## 2 Project Scope and Limitations

### 2.1 Major Features

#### 2.1.1 Major Features for User

- Register and log in to the system.
- Search scientific publications using keywords, authors, journals, or research fields.
- View detailed publication information.
- Save publications to bookmarks.
- Follow research topics of interest.

- Receive notifications about new publications and research trends.
- View publication trend analysis and statistical dashboards.
- Generate and export analytical reports (PDF/CSV).

### **2.1.2 Major Features for Administrator**

- Log in and log out.
- Manage user accounts.
- Configure academic API sources (Semantic Scholar, OpenAlex, and Crossref).
- Monitor the data synchronization process.
- Manage system notifications.
- Generate system reports.

## **2.2 Project Limitations**

### **2.2.1 Limited Use of Full-Text Data**

- The system only uses publication metadata such as title, abstract, keywords, authors, and publication year.
- It does not process full-text content, which limits the depth of analysis and understanding of research papers.

### **2.2.2 Dependence on External APIs**

- The system relies on external APIs such as OpenAlex, and Crossref.
- Any API downtime, structural changes, or request limitations may affect system stability and data collection performance.

### **2.2.3 Simplified Trend Analysis and Update Frequency**

- Trend analysis is mainly based on publication frequency and growth rate.
- It does not apply advanced Machine Learning or AI models for prediction.
- Data is synchronized periodically (daily or weekly), so real-time trends may not be immediately reflected.

## II. PROJECT MANAGEMENT PLAN

### 1 Overview

#### 1.1 Scope and estimation

| #                                                  | WBS Item                                                             | Complexity | Duration (Days) |
|----------------------------------------------------|----------------------------------------------------------------------|------------|-----------------|
| <b>1. Project Initiation</b>                       |                                                                      |            |                 |
| 1.1                                                | Define project scope                                                 | Medium     | 1               |
| 1.2                                                | Collect and analyze requirements                                     | Complex    | 2               |
| <b>2. Project Management and Environment Setup</b> |                                                                      |            |                 |
| 2.1                                                | Set up development environment (GitHub, Database, Backend, Frontend) | Simple     | 1               |
| 2.2                                                | Design database (ERD, Database Schema)                               | Complex    | 2               |
| 2.3                                                | Configure security (JWT, RBAC, Password Hashing)                     | Complex    | 2               |
| <b>3. System Design</b>                            |                                                                      |            |                 |
| 3.1                                                | Design Use Case Diagram and Activity Diagram                         | Medium     | 1               |
| 3.2                                                | Design Sequence Diagram                                              | Medium     | 1               |
| 3.3                                                | Design DFD and ERD                                                   | Medium     | 1               |
| <b>4. System Development</b>                       |                                                                      |            |                 |
| 4.1                                                | User Registration and Login                                          | Medium     | 2               |
| 4.2                                                | Search Scientific Publications                                       | Complex    | 2               |
| 4.3                                                | View Publication Details                                             | Medium     | 1               |
| 4.4                                                | Bookmark Management                                                  | Medium     | 1               |
| 4.5                                                | Research Topic Following                                             | Medium     | 1               |
| 4.6                                                | Notification Management                                              | Medium     | 1               |
| 4.7                                                | Publication Trend Analysis                                           | Complex    | 3               |
| 4.8                                                | Develop Statistical Dashboard                                        | Complex    | 2               |
| 4.9                                                | Display Trending Topics                                              | Medium     | 1               |
| 4.10                                               | Generate Statistical Reports (PDF/CSV)                               | Complex    | 2               |

| #                                  | WBS Item                                                       | Complexity | Duration (Days)        |
|------------------------------------|----------------------------------------------------------------|------------|------------------------|
| <b>5. Administration Functions</b> |                                                                |            |                        |
| 5.1                                | User Management                                                | Complex    | 2                      |
| 5.2                                | Configure Academic API Sources                                 | Complex    | 2                      |
| 5.3                                | Synchronize Data from Semantic Scholar, OpenAlex, and Crossref | Complex    | 3                      |
| <b>6. Testing</b>                  |                                                                |            |                        |
| 6.1                                | Prepare Test Plan                                              | Medium     | 1                      |
| 6.2                                | Design Test Cases                                              | Medium     | 2                      |
| 6.3                                | Execute Testing                                                | Complex    | 2                      |
| 6.4                                | Fix Defects After Testing                                      | Complex    | 2                      |
| <b>7. Project Finalization</b>     |                                                                |            |                        |
| 7.1                                | Complete Project Documentation                                 | Medium     | 2                      |
| 7.2                                | Deploy the System                                              | Medium     | 1                      |
| <b>Total Duration</b>              |                                                                |            | <b>45 Working Days</b> |

## 1.2 Project Objective

- **Timeline:** The project must be completed before **July 3, 2026**.
- **Allocated Effort:** 45 man-days.

## 1.3 Project Risks

| # | Risk Description                                                                                            | Impact                                                                                                        | Possibility | Response Plan                                                                                                           |
|---|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------|
| 1 | External academic APIs (Semantic Scholar, OpenAlex, Crossref) become unavailable or change their endpoints. | The system cannot retrieve publication data, causing incomplete search results and inaccurate trend analysis. | High        | Implement API fallback mechanisms, monitor API status regularly, and cache previously retrieved data whenever possible. |

| # | Risk Description                                                                            | Impact                                                                                  | Possibility | Response Plan                                                                                                              |
|---|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------|
| 2 | Large volumes of publication data lead to slow system performance.                          | Users experience long response times when searching publications or viewing dashboards. | Medium      | Optimize database indexing, implement pagination, caching, and asynchronous background synchronization.                    |
| 3 | Security vulnerabilities expose user accounts or stored data.                               | Unauthorized access, data leakage, and loss of user trust.                              | Medium      | Use JWT authentication, RBAC authorization, password hashing, HTTPS, and regular security testing.                         |
| 4 | Trend analysis results become inaccurate because of outdated or incomplete datasets.        | Generated reports and statistics may mislead users' research decisions.                 | Medium      | Schedule automatic data synchronization and validate collected data before analysis.                                       |
| 5 | System integration between frontend, backend, and database encounters compatibility issues. | Development delays and unexpected system failures during deployment.                    | Medium      | Perform integration testing continuously and maintain version compatibility among all components.                          |
| 6 | Insufficient testing before deployment leaves critical defects unresolved.                  | System crashes, incorrect functionality, and poor user experience.                      | High        | Prepare comprehensive Test Plans, Test Cases, conduct functional, integration, and user acceptance testing before release. |
| 7 | Project schedule delays due to underestimated implementation effort.                        | Failure to complete the project within the planned timeline.                            | Medium      | Monitor project progress weekly, prioritize critical features, and adjust the implementation schedule when necessary.      |

## 2 Project Deliverables

| Sprint          | Week   | Duration           | Deliverables                                                                                                                                            | Note |
|-----------------|--------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Sprint 1</b> | Week 1 | 14/05 – 20/05/2026 | <ul style="list-style-type: none"> <li>• Define project scope</li> <li>• Analysis requirement</li> </ul>                                                |      |
| <b>Sprint 2</b> | Week 2 | 21/05 – 27/05/2026 | <ul style="list-style-type: none"> <li>• Finish Product Overview</li> <li>• Finish User Requirement</li> <li>• Finish Functional Requirement</li> </ul> |      |
| <b>Sprint 3</b> | Week 3 | 28/05 – 03/06/2026 | <ul style="list-style-type: none"> <li>• Finish Database Design</li> <li>• Finish Software design description</li> </ul>                                |      |
| <b>Sprint 4</b> | Week 4 | 04/06 – 10/06/2026 | <ul style="list-style-type: none"> <li>• Finish Software requirement specification</li> </ul>                                                           |      |
| <b>Sprint 5</b> | Week 5 | 11/06 – 17/06/2026 | <ul style="list-style-type: none"> <li>• Finish Software design description (continued)</li> </ul>                                                      |      |
| <b>Sprint 6</b> | Week 6 | 18/06 – 24/06/2026 | <ul style="list-style-type: none"> <li>• Finish Software testing document</li> <li>• Finish Release package and user guide</li> </ul>                   |      |
| <b>Sprint 7</b> | Week 7 | 25/06 – 01/07/2026 | <ul style="list-style-type: none"> <li>• Testing</li> <li>• Finish document</li> </ul>                                                                  |      |

### 3. Responsibility Assignment

*Do ~D; Review ~R; Support ~S*

| Responsibility                        | Thanh Tú | Bảo Trâm | Huyền Trân | Trung Hiếu | Văn Học | Nhật Quang |
|---------------------------------------|----------|----------|------------|------------|---------|------------|
| Project Planning & Tracking           | S        | D        | S          | S          | S       | S          |
| Prepare Project Introduction Document | R        | D        | R          | R          | R       | R          |
| Prepare Software Requirement Document | S        | D        | S          | S          | S       | S          |
| Prepare Software Design Document      | D        | D        | D          | D          | D       | D          |
| Prepare Test Document                 | D        | D        | D          | D          | D       | D          |
| Prepare Software User Guides          | D        | D        | D          | D          | D       | D          |
| Prepare Final Report                  | D        | D        | D          | D          | D       | D          |

#### 4. Project Communications

| Communication Item  | Who    | Purpose                                                                                                                                    | When           | Type                   |
|---------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------|
| Team Weekly Meeting | Group  | <ul style="list-style-type: none"> <li>Report work project</li> <li>Assign work for next week</li> <li>Control project deadline</li> </ul> | Every Sunday   | Meeting on Google Meet |
| Weekly Report       | Member | <ul style="list-style-type: none"> <li>Report work process</li> </ul>                                                                      | 1 day / week   | Report via Zalo        |
| Group Meeting       | Group  | <ul style="list-style-type: none"> <li>Report work process</li> </ul>                                                                      | Every Thursday | Offline in class       |

## 5. Configuration Management

### 5.1 Document Management

- Document tool: Confluence
- Task management tool: Jira



## 5.2 Source Code Management

- Source code management tool: GitHub

## 5.3 Tools & Infrastructure

| Category                | Tools / Infrastructure          |
|-------------------------|---------------------------------|
| Programming Languages   | PHP, HTML5, CSS3, JavaScript    |
| Database                | MySQL                           |
| Database Administration | phpMyAdmin                      |
| Academic APIs           | OpenAlex API, Crossref REST API |
| API Testing             | Postman                         |
| Version Control         | GitHub                          |
| Project Management      | Jira, Confluence                |
| Diagramming             | Draw.io, PlantUML               |
| IDE / Code Editor       | Visual Studio Code              |
| Local Server            | WAMP Server                     |

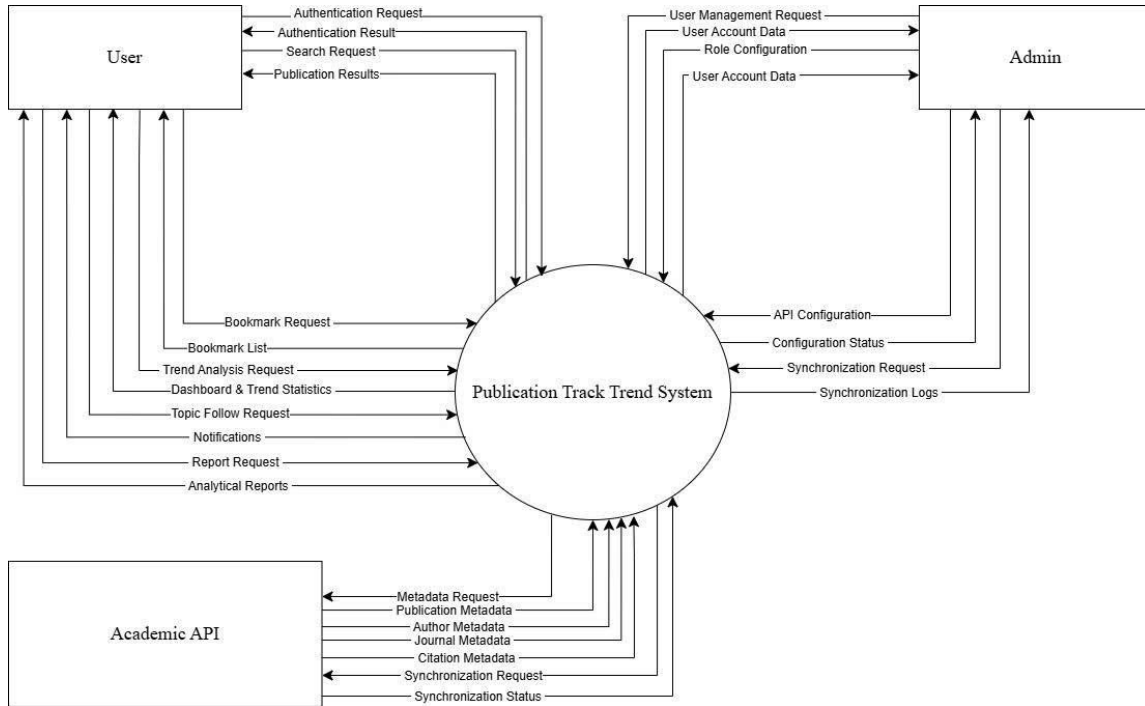
# III. SOFTWARE REQUIREMENT SPECIFICATION

## 1 Product Overview

The Scientific Journal Publication Trend Tracking System (SJPTTS) is a web-based application developed to help researchers, students, lecturers, and academic institutions search, monitor, and analyze scientific publication trends. The system integrates academic data from OpenAlex, and Crossref to provide comprehensive and up-to-date publication information.

The system allows users to search scientific publications using keywords, authors, journals, research fields. Users can view detailed publication information, bookmark publications, follow research topics, receive notifications about newly published papers, analyze publication trends through interactive dashboards, and export reports in PDF or CSV format.

Administrators can manage user accounts, configure academic API sources, synchronize publication data, monitor system operations, and maintain the overall performance of the platform. The system aims to provide an efficient solution for discovering scientific literature and supporting research through publication trend analysis.



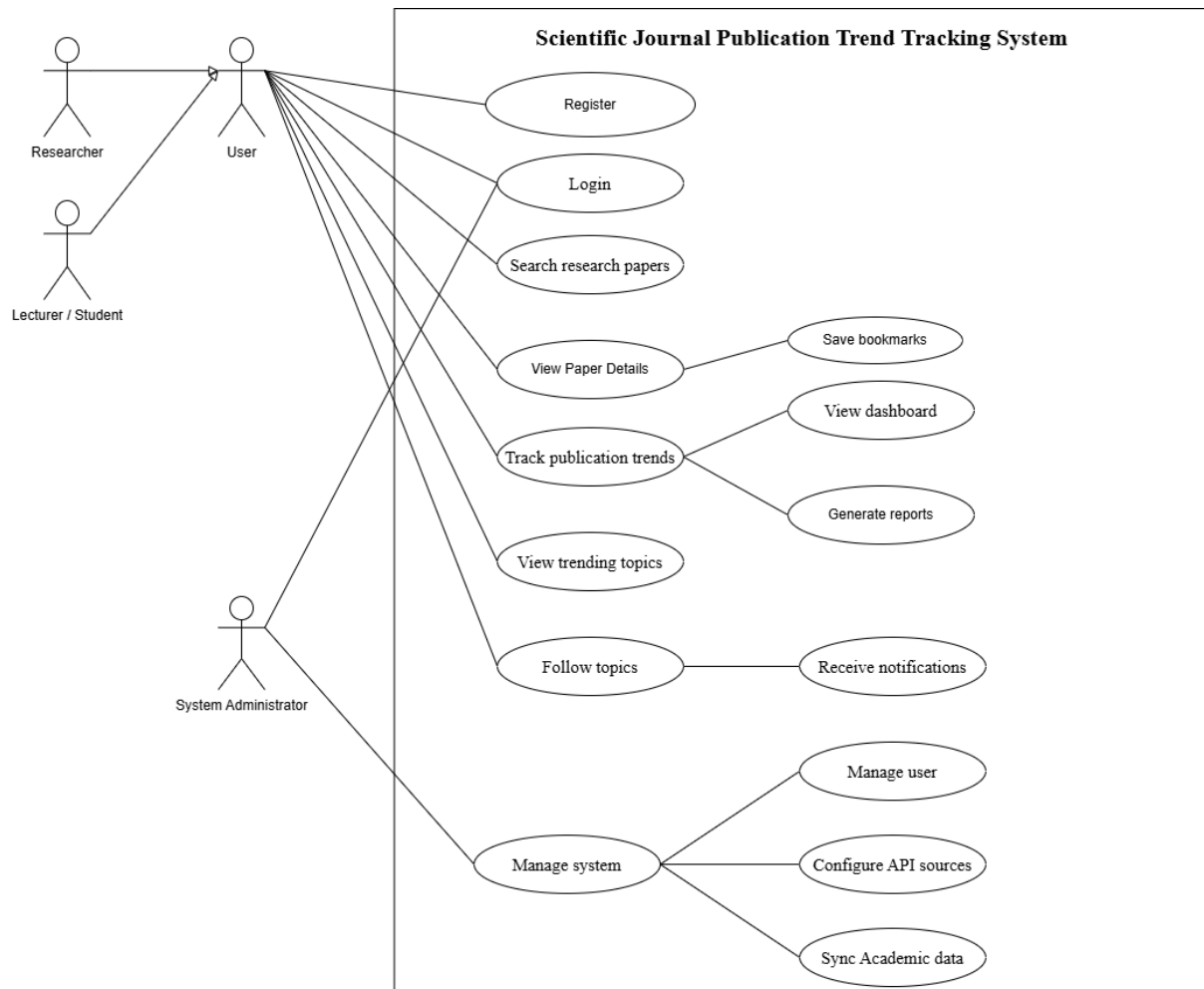
## 2 User Requirements

### 2.1 Actors

| # | Actor                             | Description                                                                                                                                                                                               |
|---|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | User                              | System user who can search scientific publications, view publication details, save bookmarks, follow research topics, receive notifications, and analyze publication trends.                              |
| 2 | Administrator                     | System administrator responsible for managing users, configuring the system, setting up and validating academic data sources (OpenAlex, Crossref), synchronizing data, and monitoring system operations.  |
| 3 | Publication Trend Tracking System | The core system that handles all functionalities such as paper search, trend analysis, dashboard generation, trending topic detection, bookmark management, notification delivery, and report generation. |
| 4 | Academic API                      | External services such as OpenAlex and Crossref that provide scientific publication metadata (authors, journals, citations) for system synchronization and analysis.                                      |

## 2.2 Use Case Diagrams

### 2.2.1 Diagram



### 2.2.2 Use Case Descriptions

| Main UC Group  | Main Agent (Actor) | Sub-UCs         | Brief Description of the Process                                                                             | Output                                                                    |
|----------------|--------------------|-----------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Manage Account | User               | Register, Login | Users create an account and authenticate to access the Scientific Journal Publication Trend Tracking System. | A registered account is created and a secure user session is established. |

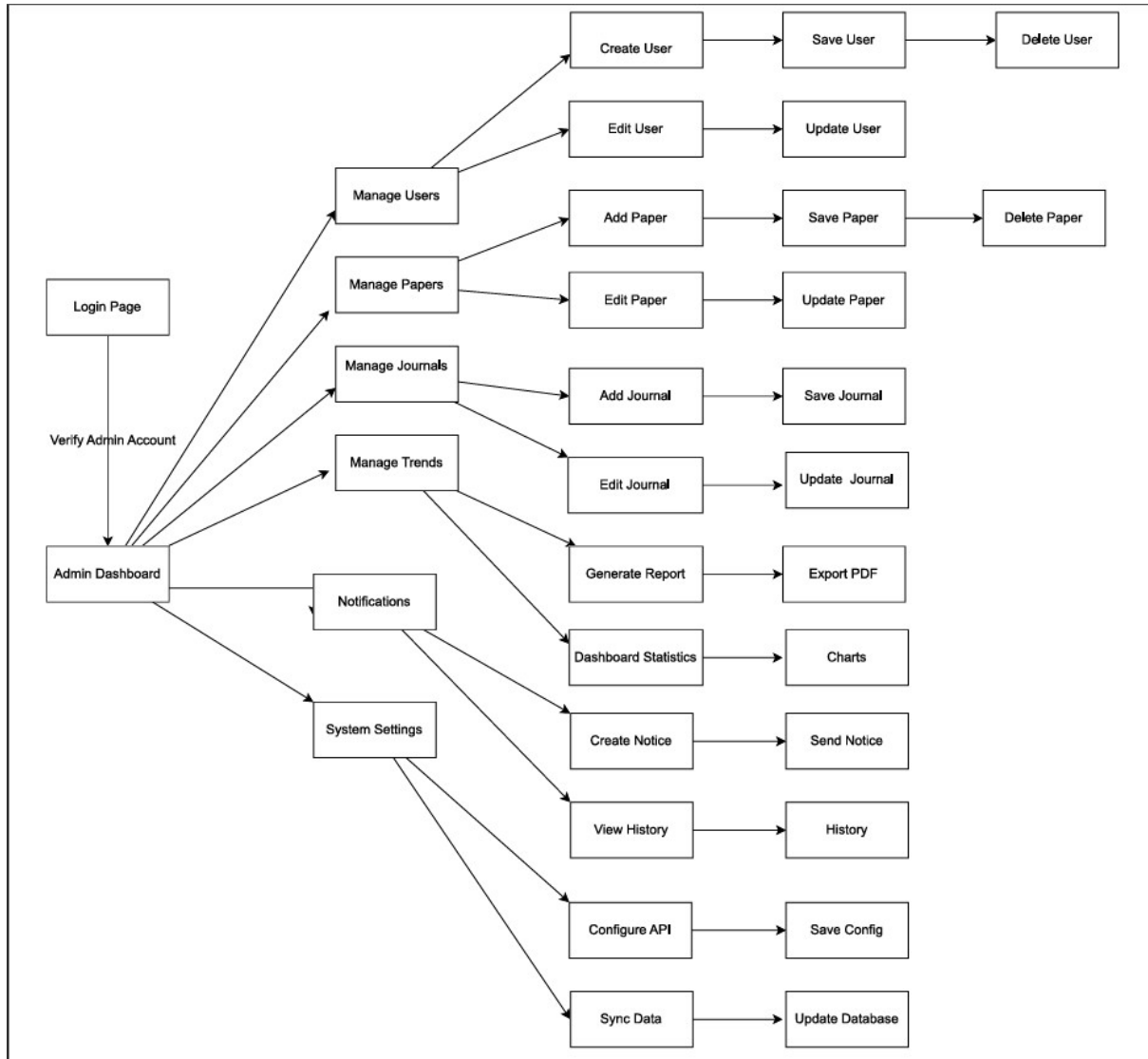
| Main UC Group            | Main Agent (Actor) | Sub-UCs                                                         | Brief Description of the Process                                                                                                                                      | Output                                                                                            |
|--------------------------|--------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Search Publications      | User               | Search Research Papers, View Paper Details                      | Users search scientific publications using keywords, author names, or journal names and view detailed publication meta-data.                                          | Matching publications and detailed publication information are displayed.                         |
| Manage Bookmarks         | User               | Save Bookmarks                                                  | Users save selected publications into their personal bookmark collection for future reference.                                                                        | The selected publication is stored in the user's bookmark list.                                   |
| Track Publication Trends | User               | Track Publication Trends, View Dashboard, View Trending Topics  | Users analyze publication trends by keyword, topic, author, journal, and time range. The system generates statistical charts and identifies trending research topics. | Trend statistics, dashboards and trending topics are displayed.                                   |
| Follow Topics            | User               | Follow Topics, Receive Notifications                            | Users follow research topics to receive notifications about tracked activities                                                                                        | The selected topic is added to the user's followed-topic list and notifications become available. |
| Generate Reports         | User               | Generate Reports                                                | Users generate analytical reports based on publication trend analysis and selected report parameters.                                                                 | An analytical report summarizing publication trends is generated.                                 |
| Manage System            | Administrator      | Manage System, Configure API Sources, Synchronize Academic Data | Administrators manage system settings, configure academic APIs, synchronize publication metadata and maintain system operations.                                      | System settings and publication metadata are successfully updated.                                |
| Manage Users             | Administrator      | Manage Users                                                    | Administrators manage user accounts, roles, permissions and account status.                                                                                           | User information and access permissions are updated.                                              |
| Configure API Sources    | Administrator      | Configure API Sources                                           | Administrators configure, update and test external academic APIs such as OpenAlex, Crossref and Semantic Scholar.                                                     | API configuration settings are successfully validated and stored.                                 |

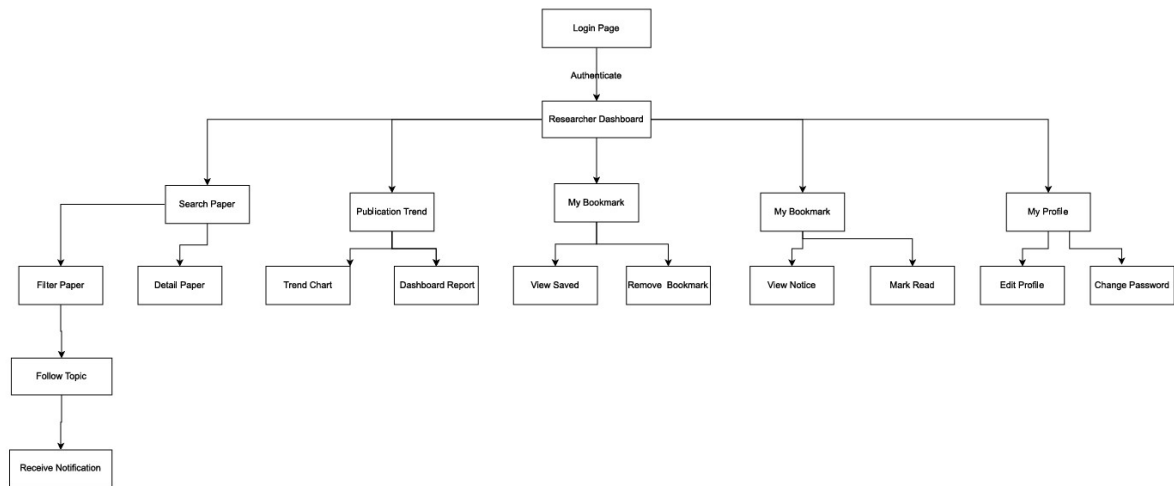
| Main UC Group             | Main Agent (Actor) | Sub-UCs                   | Brief Description of the Process                                                                                                      | Output                                                                             |
|---------------------------|--------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Synchronize Academic Data | Administrator      | Synchronize Academic Data | Administrators synchronize publication metadata from external academic APIs into the local database for searching and trend analysis. | The publication metadata database is updated with the latest synchronized records. |

## 3 Functional Requirement

### 3.1 System Functional Overview

#### 3.1.1 Screens Flow





| Functional Group    | Screen Name            | Detailed Description                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User Authentication | Login                  | Purpose: Allows users to log into the system.<br>Components: Username, Password, Login button, and Register link. Flow: The user enters credentials → The system authenticates the user → The main system interface is displayed. Rule: Username and password are required; invalid credentials will generate an error message.                                                       |
| User Authentication | Register               | Purpose: Allows new users to create an account.<br>Components: Full Name, Email, Username, Password, and Register button. Flow: The user enters the required information → The system validates the email and username → A new account is created → The user is redirected to the Login page. Rules: Email and username must be unique; passwords are securely hashed before storage. |
| Paper Management    | Search Research Papers | Purpose: Allows users to search scientific publications by keyword, author, or journal. Components: Search box, filters, Search button, and result list. Flow: The user enters search criteria → The system searches publication metadata → Matching publications are displayed.                                                                                                      |

| Functional Group           | Screen Name              | Detailed Description                                                                                                                                                                                                                                                                                        |
|----------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Paper Management           | View Paper Details       | Purpose: Displays detailed metadata of a selected research paper. Components: Title, Authors, Abstract, Keywords, Publication Year, Journal Name, Citation Information, and Save Bookmark button. Flow: The user selects a paper → The system retrieves publication metadata → Paper details are displayed. |
| Paper Management           | Bookmarks                | Purpose: Manages the user's bookmarked papers. Components: Bookmark list, Delete Bookmark button. Flow: The user views or removes bookmarked papers. Rule: Duplicate bookmarks are not allowed.                                                                                                             |
| Publication Trend Analysis | Track Publication Trends | Purpose: Analyzes publication trends based on keywords, research topics, authors, or journals. Components: Search filters, time range selector, and Analyze button. Flow: The user enters analysis criteria → The system analyzes publication metadata → Statistical results and charts are displayed.      |
| Publication Trend Analysis | Dashboard                | Purpose: Displays an overview of publication statistics. Components: Publications by year chart, Topic distribution chart, Journal distribution chart, and Growth trend chart. Flow: The system generates dashboard visualizations after trend analysis is completed.                                       |
| Publication Trend Analysis | Trending Topics          | Purpose: Displays the most popular research topics. Flow: The system analyzes publication data → Ranks research topics → Displays trending topics.                                                                                                                                                          |
| Topic Management           | Follow Topics            | Purpose: Allows users to follow research topics of interest. Components: Topic list and Follow button. Flow: The user selects a topic → The system stores it in the followed-topic list.                                                                                                                    |
| Notifications              | Notifications            | Purpose: Displays system notifications or notifications when following a new topic. Flow: The system generates notifications → The user views notification details and marks them as read.                                                                                                                  |



| Functional Group      | Screen Name               | Detailed Description                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reporting             | Generate Reports          | Purpose: Generates reports based on publication trend analysis. Flow: The user clicks "Generate Report" → The system displays options: Preview & Print / Save as PDF, Direct PDF Download, and Download Report (HTML) → The system executes the request based on the selected option and completes the generation, display, or export of the analytical report. |
| System Administration | Manage System             | Purpose: Provides administrative functions for system management. Components: User Management, API Configuration, and Academic Data Synchronization. Flow: The administrator selects a management function → The system performs the requested operation.                                                                                                       |
| System Administration | Manage Users              | Purpose: Manages user accounts. Components: User list, Change User Role, Activate Account, and Deactivate Account. Flow: The administrator selects a user → Updates account information → Saves the changes.                                                                                                                                                    |
| System Administration | Configure API Sources     | Purpose: To define and manage external academic data sources. Flow: The administrator enters source details (Name, URL, Priority, Status, Description). The administrator clicks "Add API Source" to save it to the system. The system displays the list, allowing the administrator to track, edit, or delete entries via the "Action" column.                 |
| System Administration | Synchronize Academic Data | Purpose: Synchronize publication data from external APIs into the system. Components: API List, Synchronize button, Synchronization Log. Workflow: Click Synchronize → The system retrieves data from the APIs → Updates the database → Displays the synchronization result.                                                                                    |

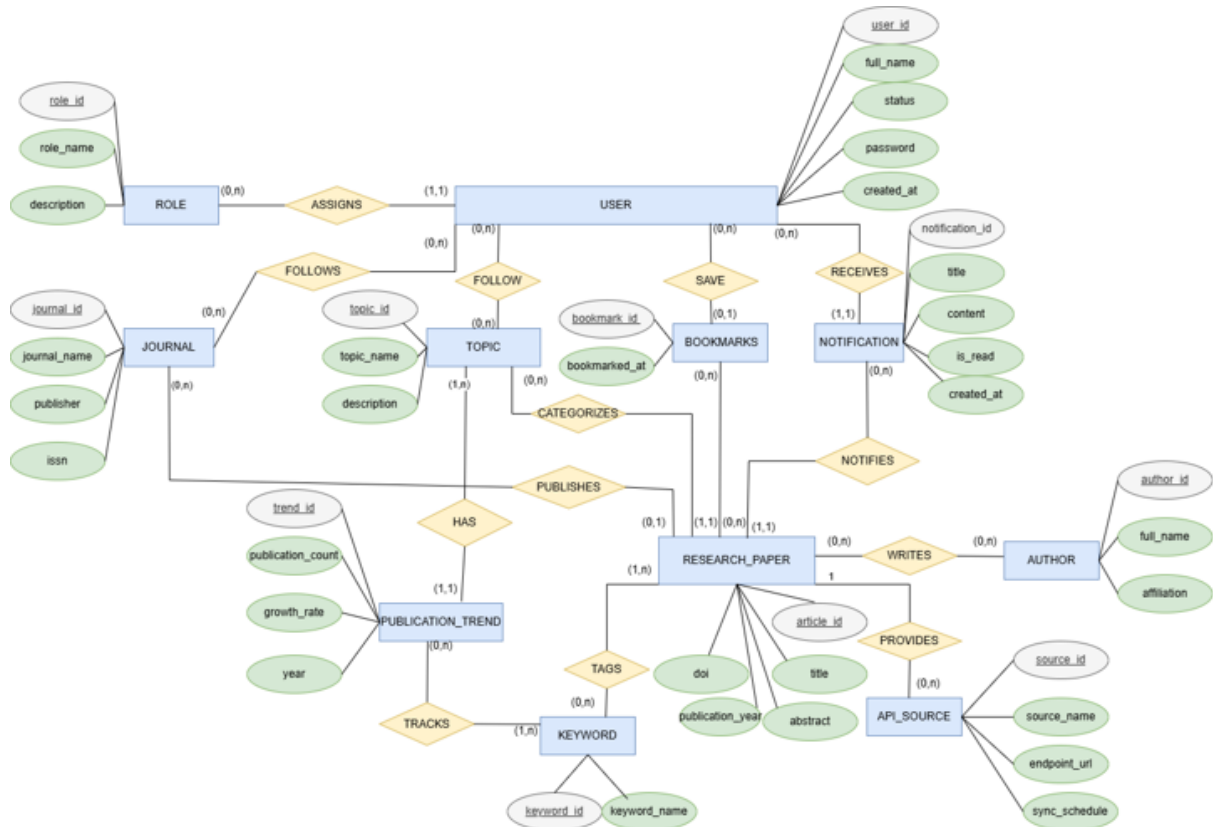
### 3.1.2 Screen Authorization

| Screen                   | Researcher | Lecturer/Student | System Administrator |
|--------------------------|------------|------------------|----------------------|
| Login                    | x          | x                | x                    |
| Logout                   | x          | x                | x                    |
| Register                 | x          | x                |                      |
| Search Research Papers   | x          | x                |                      |
| View Paper Details       | x          | x                |                      |
| Save Bookmarks           | x          | x                |                      |
| Track Publication Trends | x          | x                |                      |
| View Dashboard           | x          | x                |                      |
| Generate Reports         | x          | x                |                      |
| View Trending Topics     | x          | x                |                      |
| Follow Topics            | x          | x                |                      |
| Receive Notifications    | x          | x                |                      |
| View Profile             | x          | x                |                      |
| Edit Profile             | x          | x                |                      |
| View Notifications       | x          | x                |                      |
| Create Report            | x          | x                |                      |
| Manage System            |            |                  | x                    |
| Manage Users             |            |                  | x                    |
| Configure API Sources    |            |                  | x                    |
| Sync Academic Data       |            |                  | x                    |

### 3.1.3 Non-Screen Functions

| # | Feature                                 | System Function | Description                                                                                                                                                                                                                                                                                                         |
|---|-----------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Automated Academic Data Synchronization | Scheduled Job   | Scheduled job that automatically retrieves publication metadata directly from external academic APIs (OpenAlex and Crossref). The retrieved data is used for publication search, statistical analysis, and publication trend tracking.                                                                              |
| 2 | Automated Trend Calculation             | Scheduled Job   | Scheduled job that automatically recalculates publication trend statistics after each data synchronization from external academic APIs. The system analyzes total publications, total citations, Open Access rate, and research topic trends, then updates the data displayed on the dashboard and Trending Topics. |

## 3.2 Entity Relationship Diagram



## 3.3 Authentication

### 3.3.1 Register

**Role:** User

**Function trigger:** The user clicks Đăng kí.

**Function description:**

- Purpose: Create a new user account.
- Data processing: Validate user information and store it in the database.

**Function Details:**

- Whether the username already exists
- Validate password requirements.
- Create a new account.
- Display a success message.
- Sign up and check your email.

## Tạo tài khoản ScholarTrend

Tham gia cộng đồng nghiên cứu và khám phá xu hướng khoa học toàn cầu.

Họ và tên

 Nguyễn Văn A

Email

 example@email.com

Tên đăng nhập

 Username

Mật khẩu

 Password

Đăng ký

Đã có tài khoản? [Đăng nhập](#)

- Email already exists

## Tạo tài khoản ScholarTrend

Tham gia cộng đồng nghiên cứu và khám phá xu hướng khoa học toàn cầu.

Email đã được đăng ký.

Họ và tên

Trần Trung Hiếu

Email

eldenring.674@gamil.com

Tên đăng nhập

Hizeu

Mật khẩu

••••••••

- Username already exists.

## Tạo tài khoản ScholarTrend

Tham gia cộng đồng nghiên cứu và khám phá xu hướng khoa học toàn cầu.

Tên đăng nhập đã tồn tại.

Họ và tên

 Nguyễn Ngọc Bảo Trâm

Email

 Trammmm@gmail.com

Tên đăng nhập

 baotram

Mật khẩu

 .....

Đăng ký

Đã có tài khoản? [Đăng nhập](#)

- Password does not meet the required format.

**ScholarTrend**

## Tạo tài khoản ScholarTrend

Tham gia cộng đồng nghiên cứu và khám phá xu hướng khoa học toàn cầu.

Mật khẩu phải có ít nhất 8 ký tự và bao gồm cả chữ và số.

Họ và tên

 Nguyễn Ngọc Bảo Trâm

Email

 Trammmm@gmail.com

Tên đăng nhập

 baotrammm

Mật khẩu

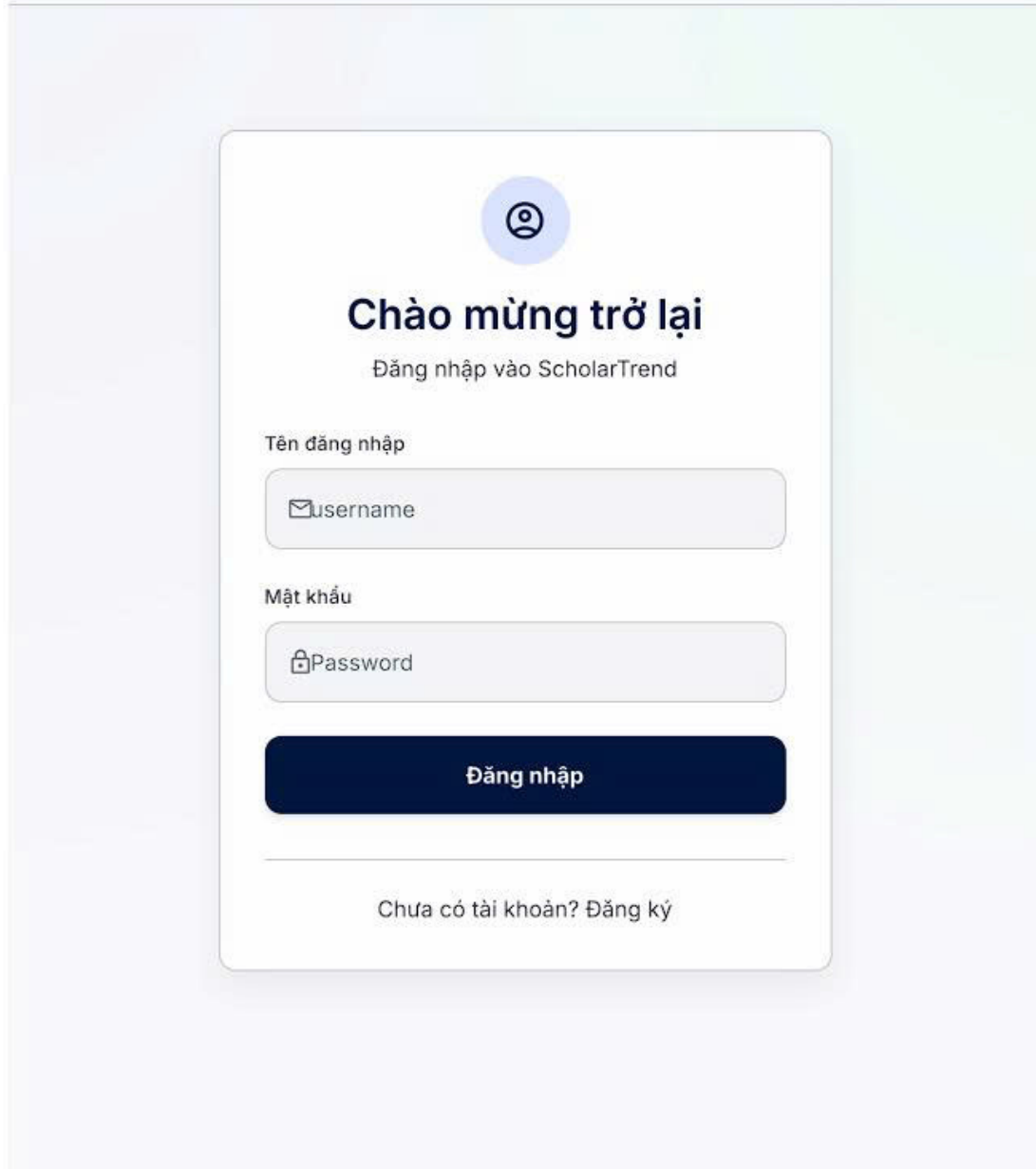
 ....

**Đăng ký**

Đã có tài khoản? [Đăng nhập](#)

- Registration successful.





The image shows a login interface for ScholarTrend. At the top, the header 'ScholarTrend' is displayed. Below it, a central white card contains a blue circular icon with a person silhouette. The card has the heading 'Chào mừng trở lại' (Welcome back) and the subtitle 'Đăng nhập vào ScholarTrend'. There are two input fields: 'Tên đăng nhập' (Username) with an envelope icon and 'Mật khẩu' (Password) with a lock icon. A dark blue button labeled 'Đăng nhập' (Login) is positioned below the fields. At the bottom of the card, a link reads 'Chưa có tài khoản? Đăng ký' (Don't have an account? Register).

### 3.3.2 Login

**Role:** User

**Function trigger:**

- The user enters a username and password to log in.

**Function description:**

- **Purpose:** Allow users to access the system.
- **Data processing:** Authenticate the entered username and password against the system database.

**Function Details:**

- If the login is successful, redirect the user to the Home page.
- If the login fails, display the error message: “Tên đăng nhập hoặc mật khẩu không đúng.”

## ScholarTrend



### Chào mừng trở lại

Đăng nhập vào ScholarTrend

Tên đăng nhập

username

Mật khẩu

Password

Đăng nhập

Chưa có tài khoản? Đăng ký

- Invalid username or password.

ScholarTrend



## Chào mừng trở lại

Đăng nhập vào ScholarTrend

Tên đăng nhập

Mật khẩu

Tên đăng nhập hoặc mật khẩu không đúng.

Đăng nhập

- Login successful.

DỮ LIỆU HỌC THUẬT THỜI GIAN THỰC

# Giải mã xu hướng Khoa học toàn cầu

Theo dõi hàng triệu ấn phẩm, phân tích tốc độ phát triển của các lĩnh vực và đón đầu những đột phá nghiên cứu quan trọng nhất.



Tìm kiếm bài báo nghiên cứu, tác giả hoặc từ khóa...

Khám phá ngay →

## Chủ đề đang nổi bật

Xem tất cả xu hướng ↗



+21% so với cùng kỳ

### Năng lượng Tái tạo

Hiệu suất pin mặt trời thế hệ mới và lưu trữ năng lượng quy mô lớn.

4.758+

Bài báo mới



+17% so với cùng kỳ

### Thu giữ Carbon

Công nghệ thu giữ carbon trực tiếp từ không khí và lưu trữ địa chất.

14.186+

Bài báo mới



+7% so với cùng kỳ

### Chỉnh sửa Gen CRISPR

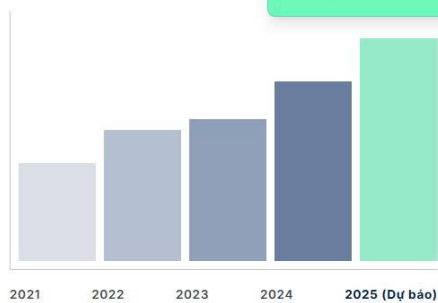
Ứng dụng CRISPR trong điều trị bệnh di truyền và nông nghiệp.

3.977+

Bài báo mới



### Tần suất xuất bản theo năm



TỈ LỆ TĂNG TRƯỞNG  
21,6%  
Trung bình năm

## Phân tích xu hướng chuyên sâu

ScholarTrend hỗ trợ thu thập và phân tích metadata bài báo khoa học từ các nguồn học thuật uy tín. Hệ thống giúp người dùng theo dõi xu hướng nghiên cứu, khám phá các chủ đề nổi bật và hỗ trợ quá trình tìm kiếm, phân tích tài liệu khoa học.

- ✓ Theo dõi số lượng bài báo theo từng năm và xu hướng xuất bản.
- ✓ Phân tích các chủ đề nghiên cứu nổi bật dựa trên dữ liệu học thuật.
- ✓ Hiện thị Dashboard trực quan với biểu đồ thống kê và kết quả phân tích.



## Đừng bỏ lỡ bất kỳ đột phá nào

Theo dõi các chủ đề nghiên cứu để nhận thông báo khi có bài báo mới hoặc khi xu hướng xuất bản của chủ đề thay đổi.

### 3.3.3 Profile Management

#### Update Profile

**Role:** User

**Function trigger:** The user opens the Profile page and clicks **Chỉnh sửa thông tin**.

**Function description:**

- Purpose: Allow users to update their personal information.
- Data processing: Validate and save the updated information.

**Function Details:**

- Update name, Email and other personal information.
- Display a success message after saving.

**ScholarTrend**

**Thông tin tài khoản**  
Quản lý hồ sơ cá nhân và cài đặt bảo mật của bạn.

**HỌ VÀ TÊN**  
Trần Trung Hiếu

**EMAIL**  
eldenring.674@gamil.com

**TÊN ĐĂNG NHẬP**  
Hizeu

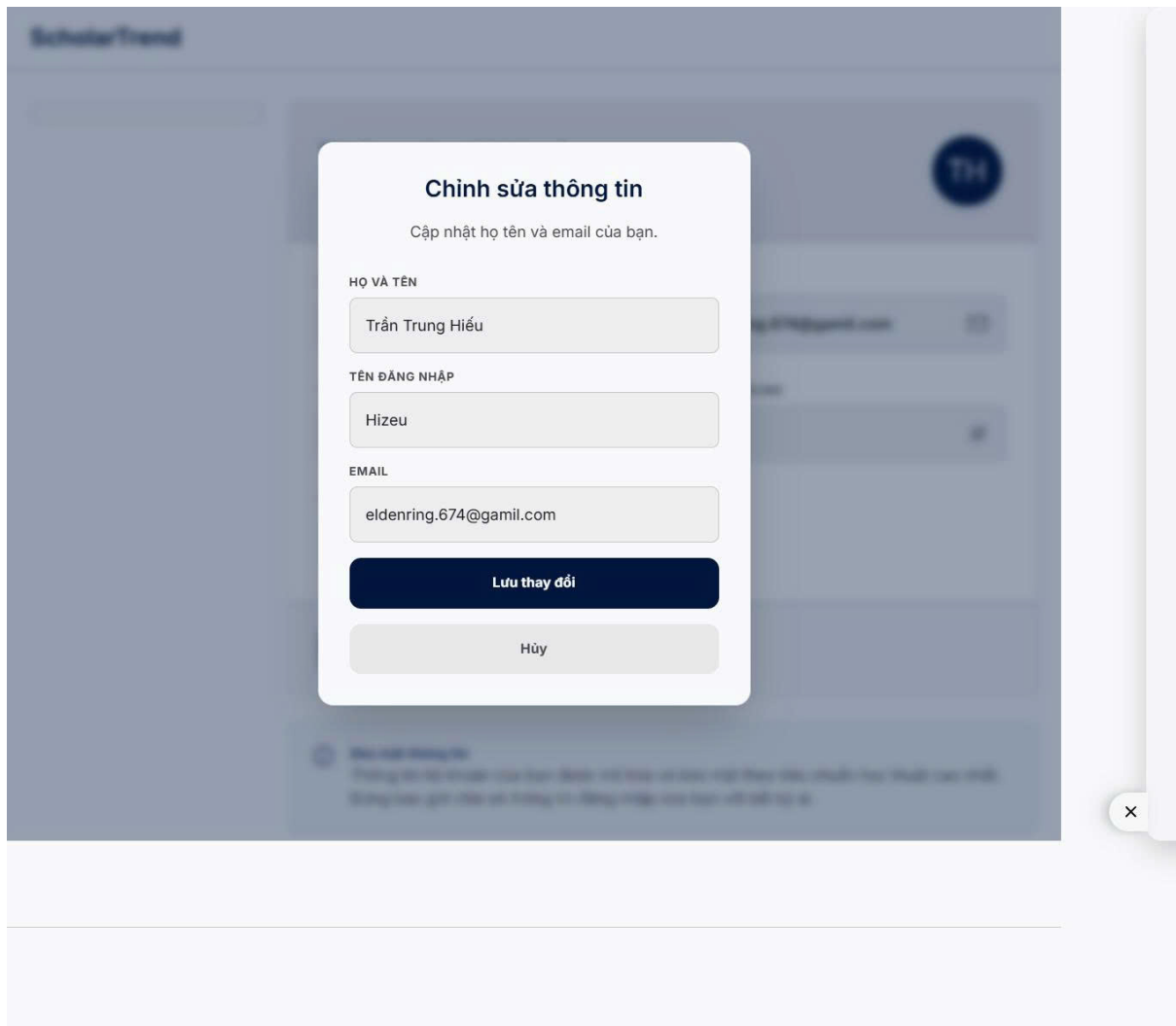
**MÃ NGƯỜI DÙNG**  
31

**TRANG THÁI EMAIL**  
Đã xác minh

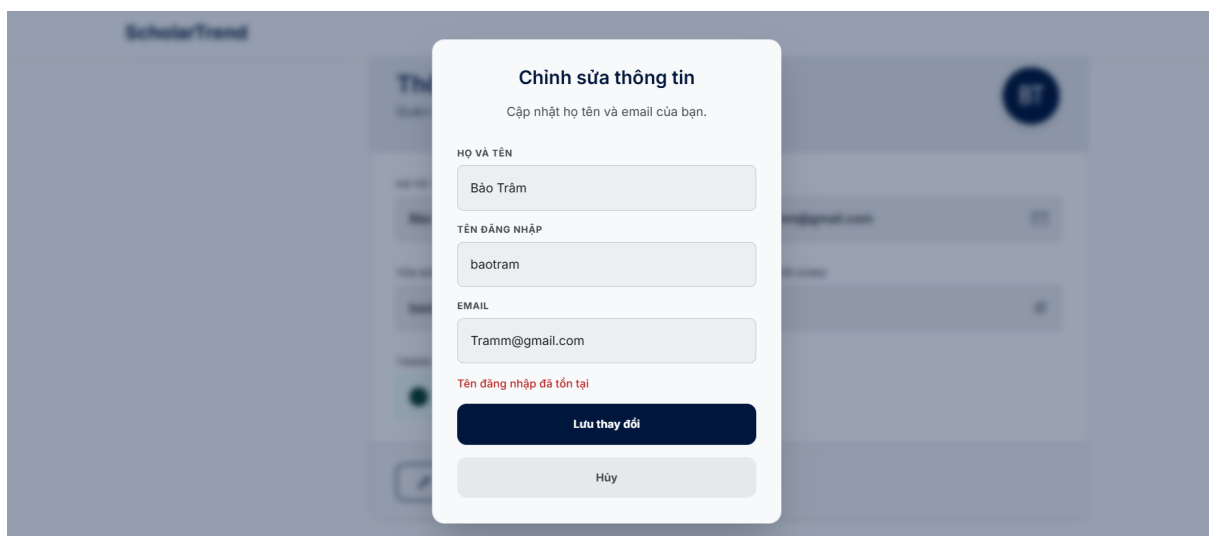
**Chỉnh sửa thông tin** **Đăng xuất**

**Bảo mật thông tin**  
Thông tin tài khoản của bạn được mã hóa và bảo mật theo tiêu chuẩn học thuật cao nhất.  
Đừng bao giờ chia sẻ thông tin đăng nhập của bạn với bất kỳ ai.

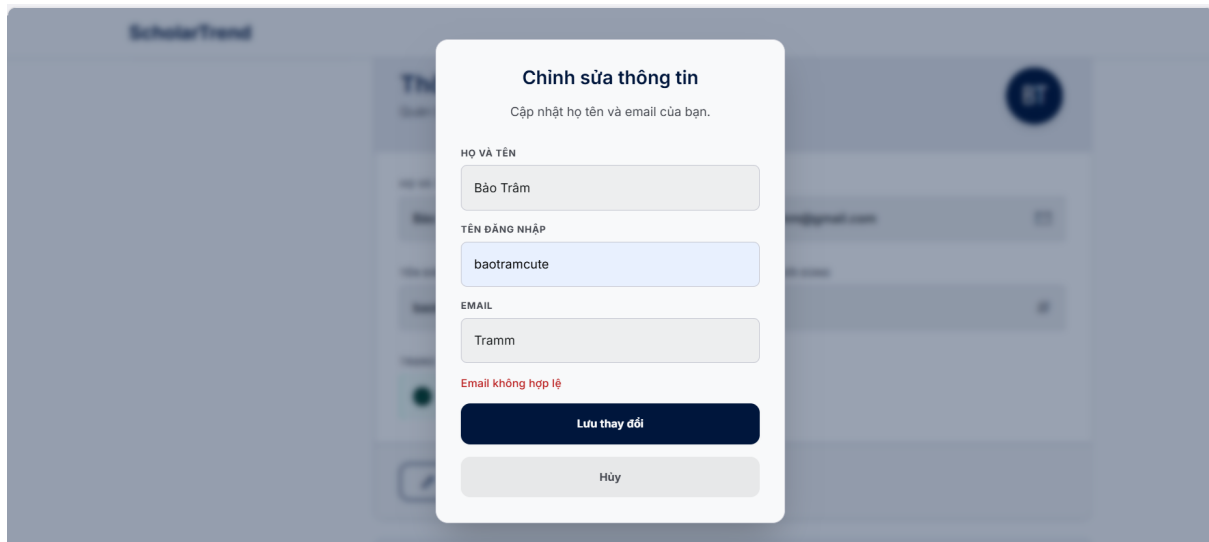
- Edit Profile



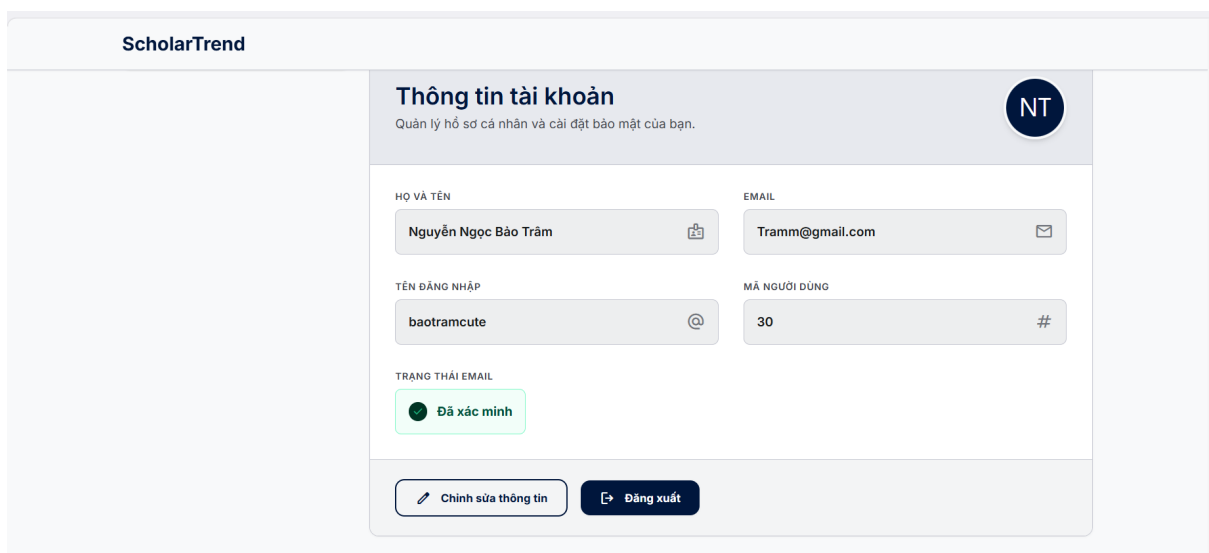
- Username already exists.



- Invalid email address



- Update successful.



### 3.3.4 Logout

**Role:** User

**Function trigger:** The user opens the Profile page and clicks Đăng xuất.

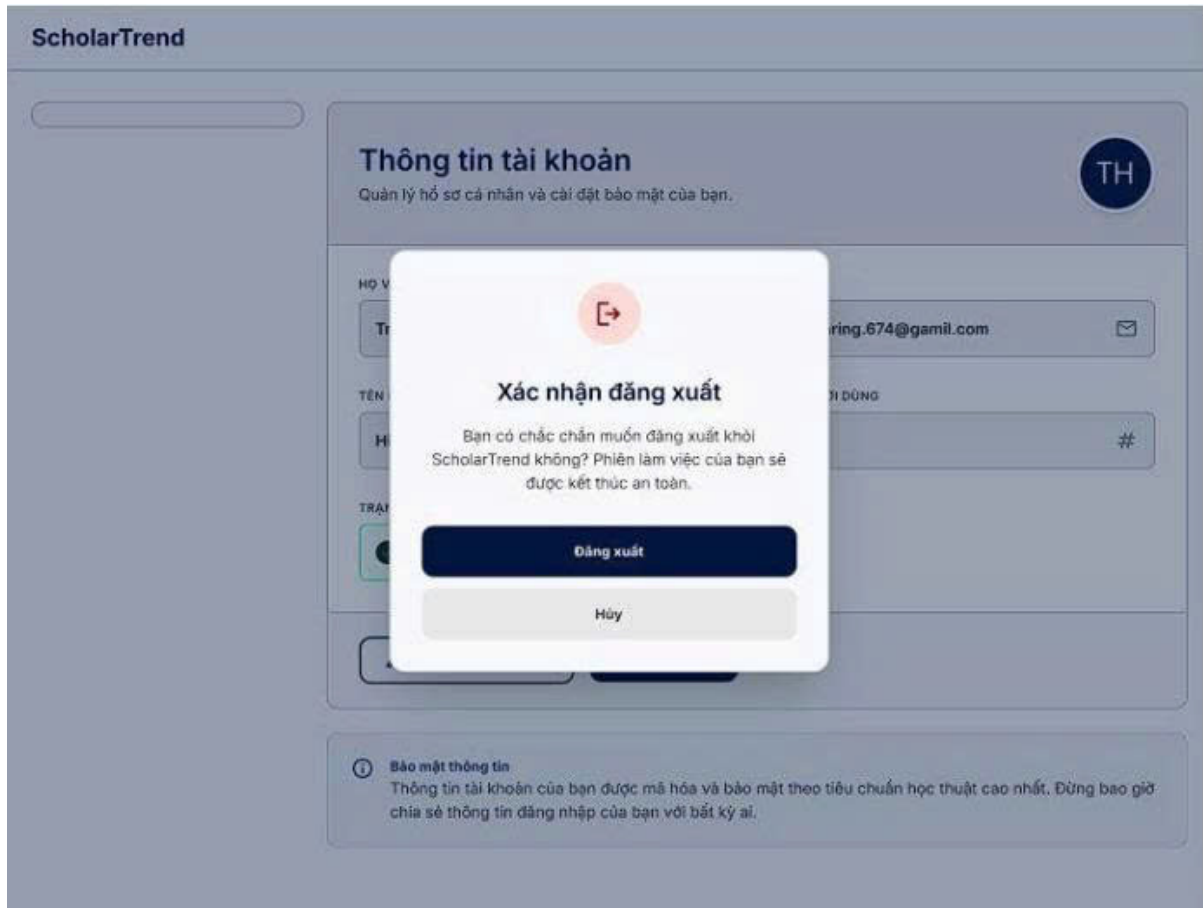
**Function description:**

- Purpose: Allow the user to securely log out of the system.
- Data processing: End the current user session.

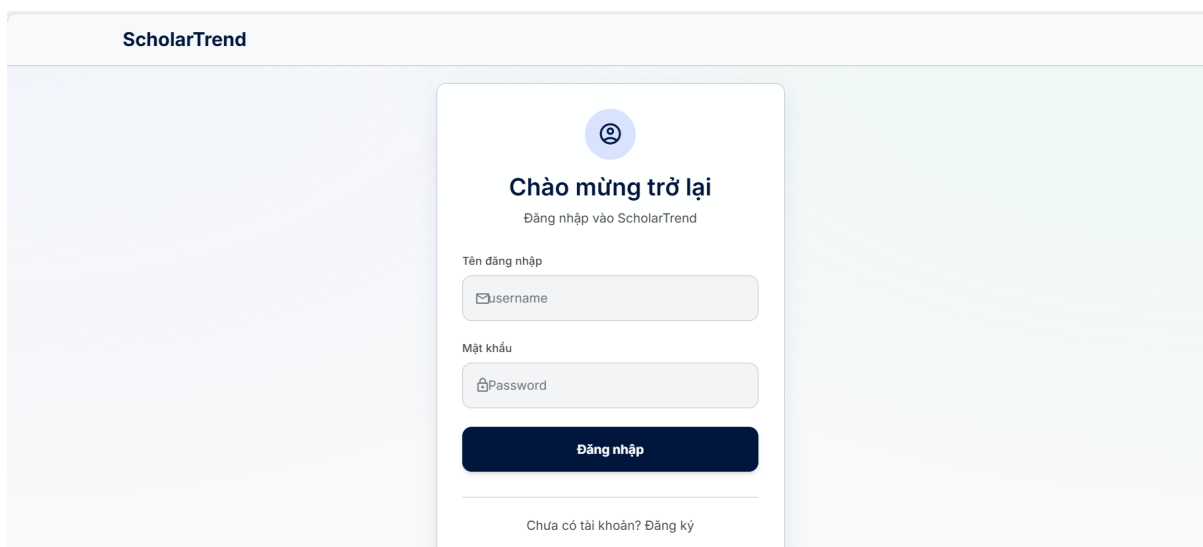
**Function Details:**

- Display a confirmation dialog asking whether the user wants to Đăng xuất.

- If the user selects Đăng xuất, the system logs the user out and redirects them to the Đăng nhập page.
- If the user selects Hủy, the dialog is closed and the user remains on the current page.



- Signed out successfully.





### 3.3.5 Search Papers

**Role:** User

#### **Function Trigger**

User opens “*Tìm kiếm*” → Enters a keyword → Selects filters (Publication Year, Open Access, Paper Type) → Clicks **Search**.

#### **Function Description**

- **Purpose:**

Allow users to search scientific papers from academic databases based on keywords and filtering conditions.

- **Data Processing:**

Validate the search input, send the request to academic databases (e.g., Crossref API), retrieve matching publications, apply selected filters, and display the search results.

#### **Function Details**

- **Validation:**

- Keyword cannot be empty.
- Start year must not be greater than End year.

- **Business Rules:**

- Search supports Title, Keyword and DOI.
- Users may filter by publication year.
- Users may filter by paper type.
- Users may choose to display only Open Access publications.
- Results are displayed with pagination (10 items per page).
- Each result includes Title, Authors, Year, DOI and Citation Count (if available), with the option to save the paper to Favorites.

- **Normal Case:**

- User enters a keyword and selects filters.
- User clicks the **Search** button.
- System queries the academic database.
- Matching publications are returned and displayed.
- User can view paper details, open the DOI link, or save the paper to Favorites.

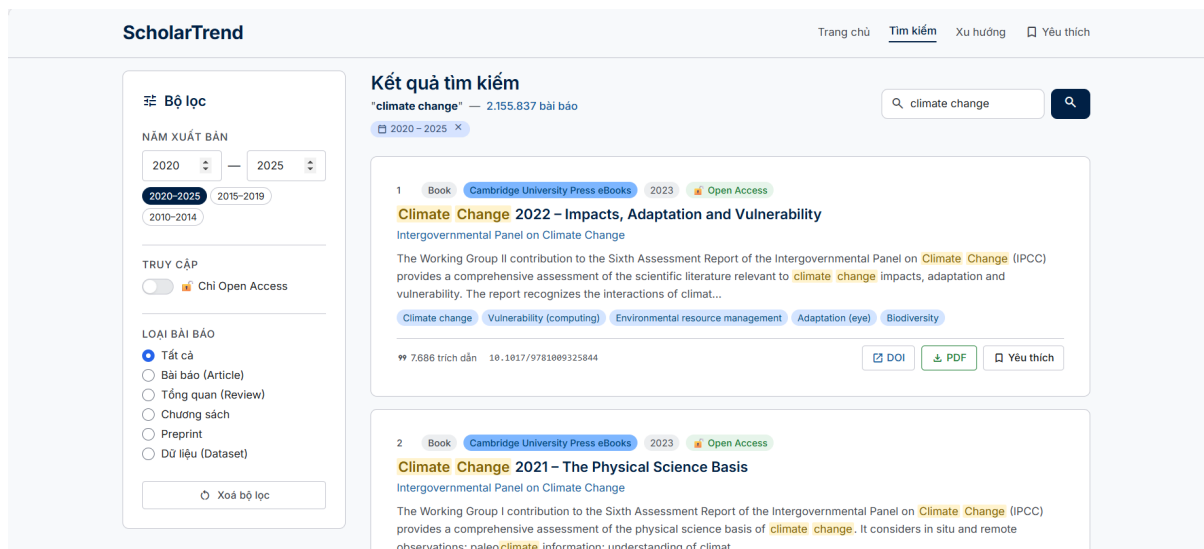
- **Abnormal Case:**

- No papers found → Display “*No matching papers found.*”

- Invalid year range → Request the user to correct the input.
- API unavailable → Display “Unable to retrieve publication data. Please try again later.”

- The start year cannot be greater than the end year.

- Search Results



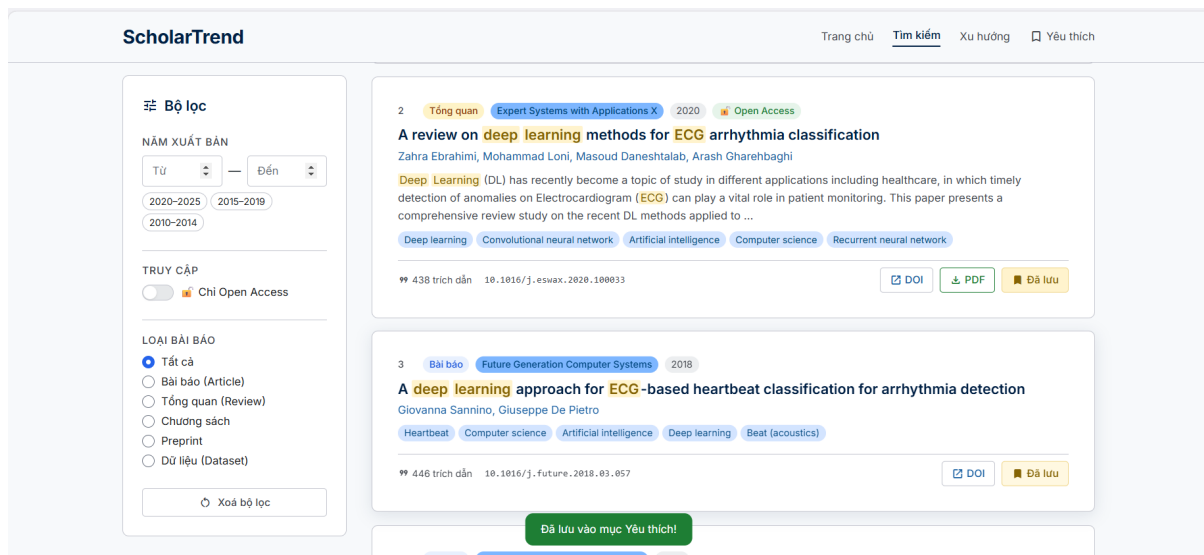
### 3.3.6 Bookmark

The Favorite Papers feature allows users to save publications of interest for future reference and easy access.

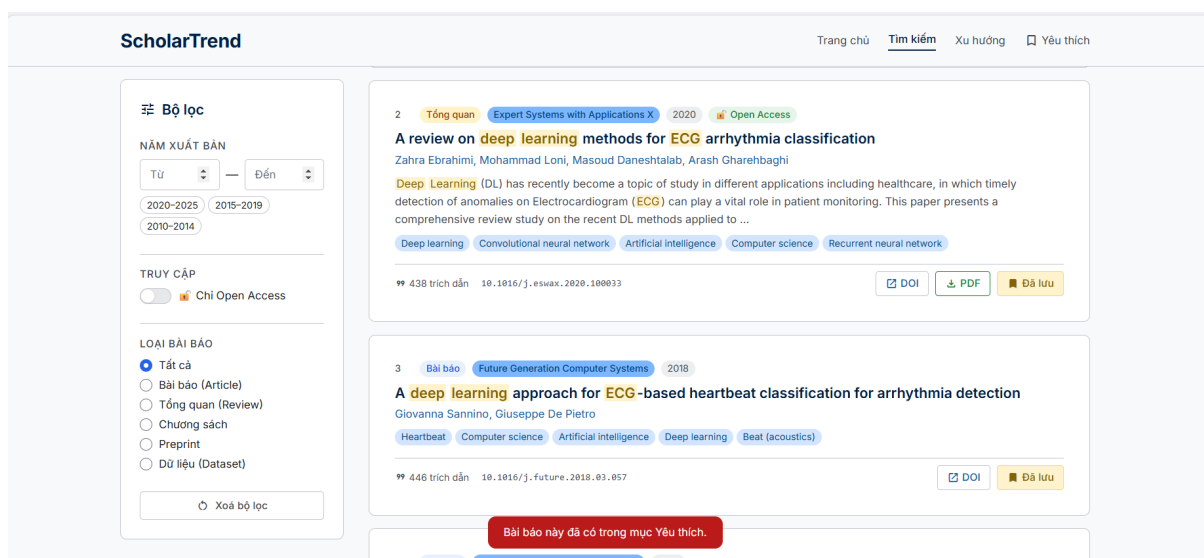
- Users can click the Favorite button on a paper in the Search Results page to add it to their favorite list.
- If the paper is successfully added, the system displays the message "Added to Favorites successfully." and updates the button status to Saved.
- If the selected paper already exists in the favorite list, the system displays the message "This paper is already in your Favorites." and prevents duplicate entries.
- Users can access the Favorites page from the navigation bar to view all saved papers.
- Each saved paper displays the following information:
  - Publication title.
  - Publication year.
  - Data source (e.g., OpenAlex).
  - Saved date and time.
  - DOI button for accessing the original publication.
  - Delete button for removing the paper from the favorite list.
- Users can remove a paper from the favorite list by clicking the Delete button. The system immediately updates the favorite list.
- Users can click the Find More Papers button to return to the search page and continue searching for publications.

## Expected Result

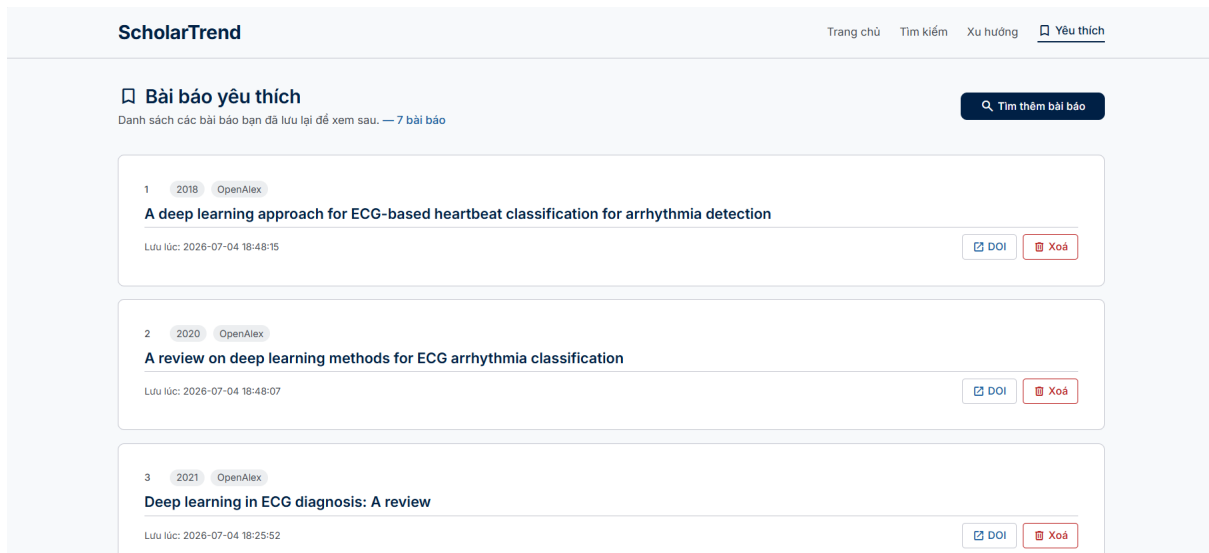
- Each paper can only be saved once in the favorite list.
- Users can easily manage, review, and access their saved publications.
- All changes to the favorite list are synchronized with the database and reflected immediately in the user interface.



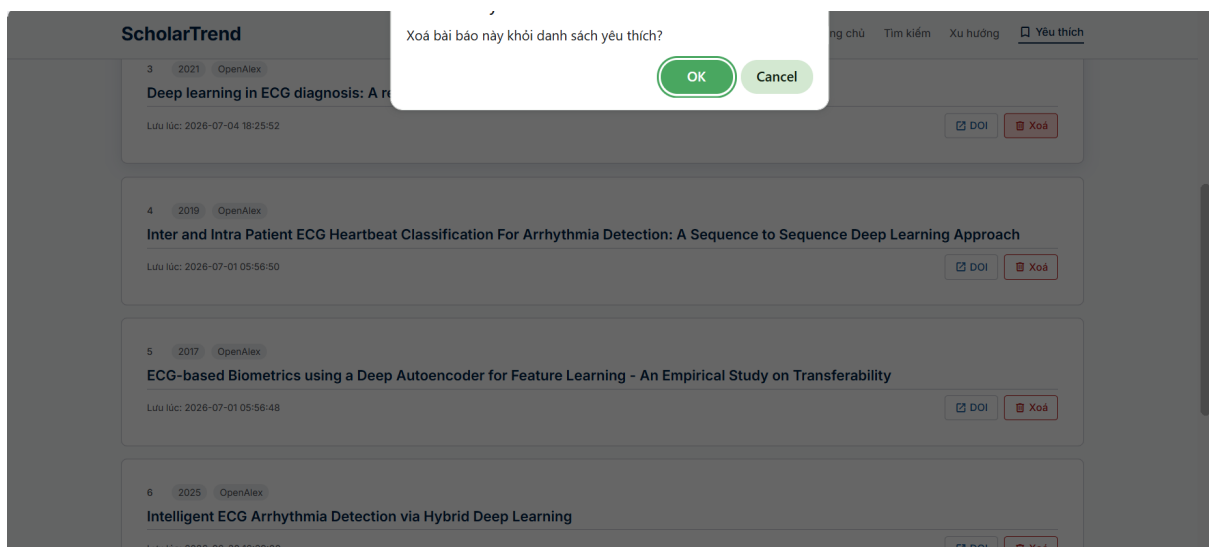
- Marked as Favorite.



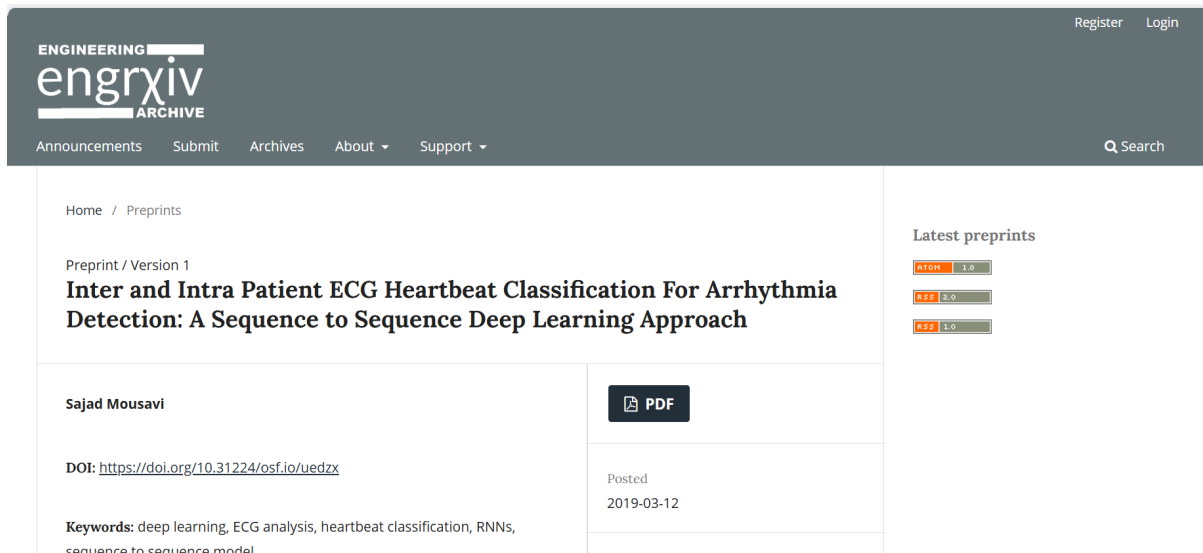
- Bookmarked List



- Remove from Bookmarks.



- Link to the bookmarked paper.



### 3.3.7 Receive Notifications

### 3.3.8 Research Trend Analysis

#### Function Trigger

User opens *Research Trend Analysis* from the main menu.

#### Function Description

**Actors:** Researcher, Lecturer, Student

#### Purpose

Allow users to analyze research trends of a specific research topic within a selected time range.

#### Data Processing

- Validate the input topic and time range.
- Retrieve publication data from the database.
- Aggregate statistical information.
- Calculate research trend metrics.
- Generate analytical charts.
- Display the research trend dashboard.

#### Interface

The **Research Trend Dashboard** includes:

- Search bar for entering a research topic.

- Time range selector.
- **Analyze** button.
- Two tabs:
  - Dashboard
  - Trending Topics
- Dashboard widgets:
  - Total Publications
  - Total Citations
  - Open Access Ratio
  - Trend Growth
- Charts:
  - Publications by Year
  - Top Keywords
  - Top Journals
  - Top Authors
  - Featured Publications

## **Function Details**

### **Validation**

- Topic keyword cannot be empty.
- Start Year must not be greater than End Year.
- The selected time range must be within the supported system range.

### **Business Rules**

- Statistics are calculated based on publications retrieved by the system.
- Trend Growth is calculated by comparing the selected period with the previous period of the same duration.
- Users can follow or unfollow a trending topic.
- Dashboard data is refreshed whenever the topic or time range changes.
- Trending Topics are ranked according to publication growth and research activity.
- Charts and statistics always correspond to the currently selected topic and time range.

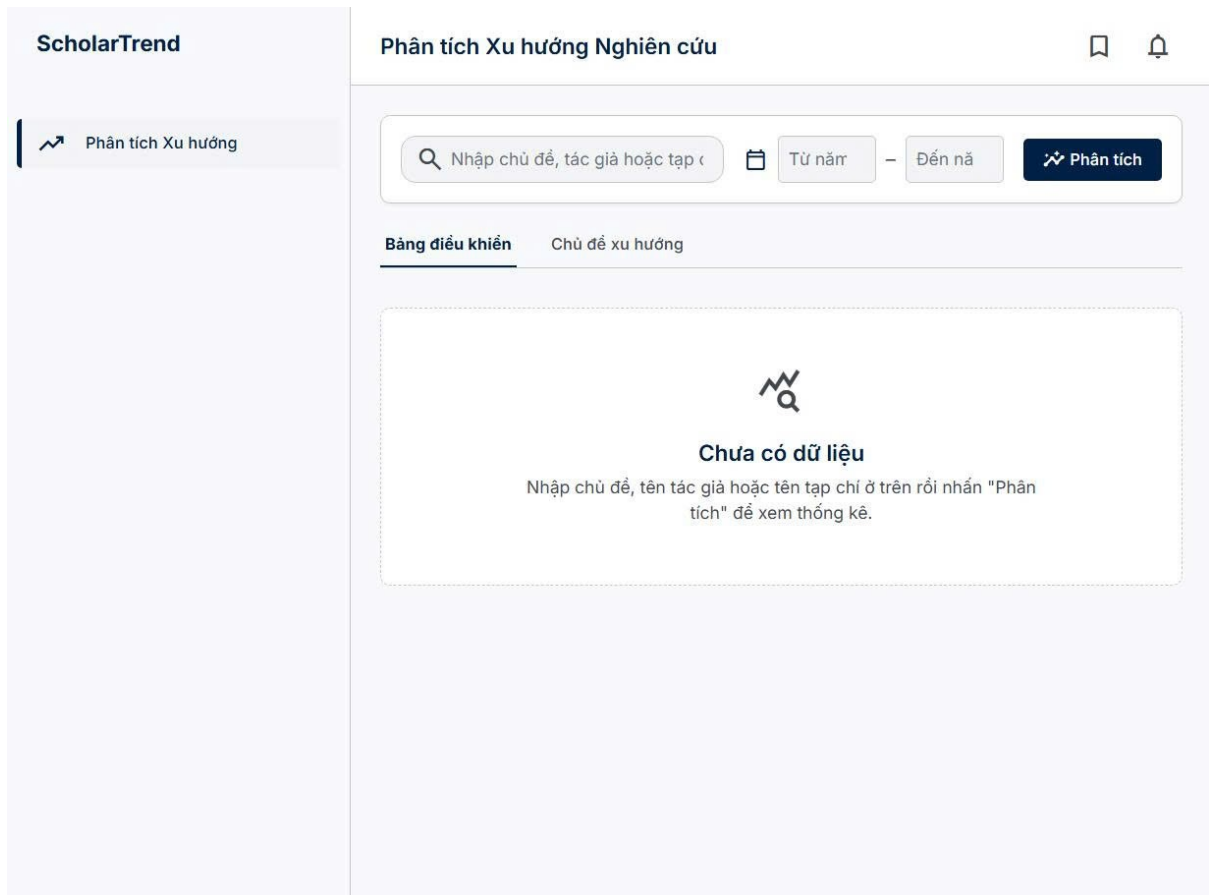
### Normal Flow

1. User enters a research topic.
2. User selects a time range.
3. User clicks the **Analyze** button.
4. The system validates the input.
5. The system retrieves publication data.
6. The system calculates statistics and generates charts.
7. The dashboard displays the analysis results.
8. User switches to the **Trending Topics** tab.
9. User follows or unfollows a trending topic.

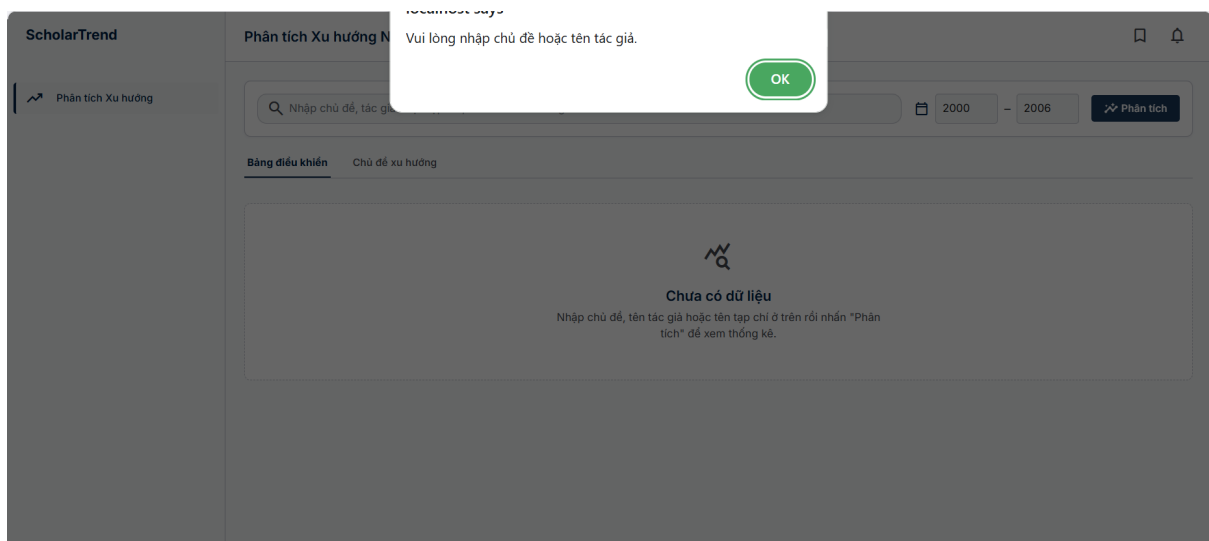
### Alternative / Exception Flow

- No matching data found.
  - Display message: *"No data available."*
- Invalid time range.
  - Request the user to correct the selected time range.
- System or database error.
  - Display message: *"Unable to load data. Please try again later."*

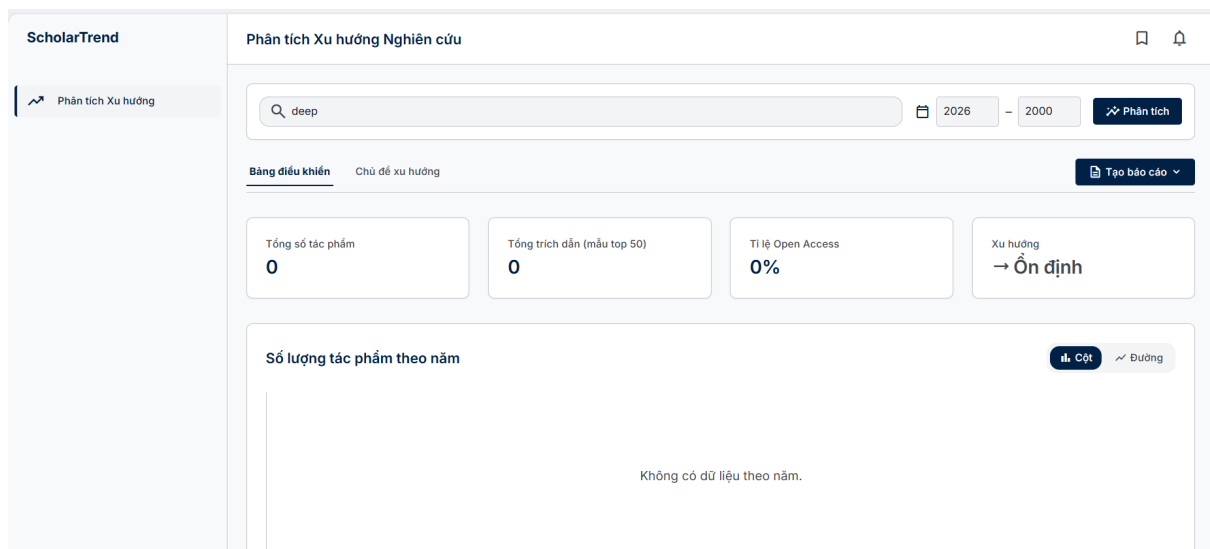




- Please enter a search keyword.



- The start year cannot be greater than the end year.



- Analysis Results.

 khoa học

2020

– 2026

 Phân tíchBảng điều khiển

Chủ đề xu hướng

 Tạo báo cáo ▾

Tổng số tác phẩm

59K

Tổng trích dẫn  
(mẫu top 50)

13K

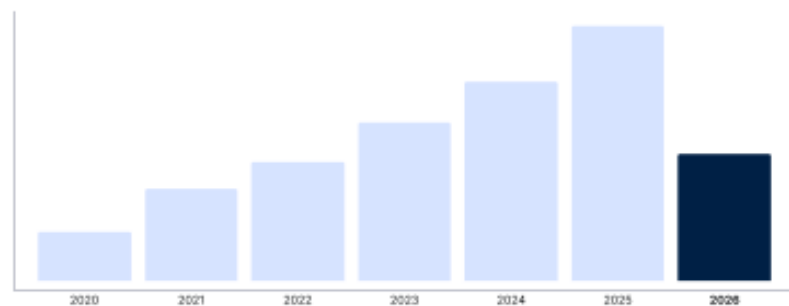
Tỉ lệ Open  
Access

98%

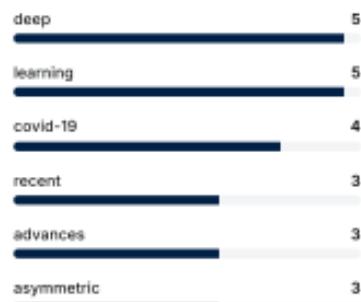
Xu hướng

 Tăng  
(137%)

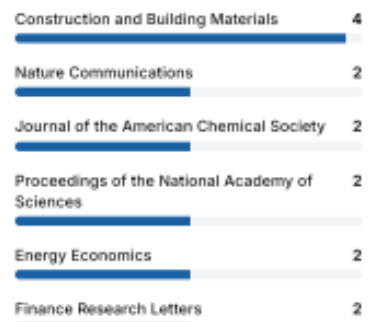
## Số lượng tác phẩm theo năm

 Cột Đường

## Phân bố chủ đề con



## Phân bố tạp chí



## Tác giả nổi bật

| TÁC GIẢ          | SỐ BÀI | TRÍCH DẪN |
|------------------|--------|-----------|
| Sudabeh Alatab   | 1      | 2K        |
| Sadaf G Sepanlou | 1      | 2K        |
| Kevin Ikuta      | 1      | 2K        |
| Homayoon Vahedi  | 1      | 2K        |

ScholarTrend

Phân tích Xu hướng

| TÁC GIẢ             | SỐ BÀI | TRÍCH DẪN |
|---------------------|--------|-----------|
| Catherine Bisignano | 1      | 2K        |
| Saeid Safiri        | 1      | 2K        |
| Anahita Sadeghi     | 1      | 2K        |
| Molly R Nixon       | 1      | 2K        |

Tác phẩm nổi bật

The global, regional, and national burden of inflammatory bowel disease in 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017 **OPEN ACCESS**

2020 · 2K trích dẫn

[Xem DOI →](#)

Deep learning in cancer diagnosis, prognosis and treatment selection **OPEN ACCESS**

2021 · 809 trích dẫn

[Xem DOI →](#)

Spiking Neural Networks and Their Applications: A Review **OPEN ACCESS**

2022 · 570 trích dẫn

[Xem DOI →](#)

### 3.3.9 Export Research Trend Report

Function Trigger User opens *Research Trend Analysis* → Enters a research topic and selects a time range → Clicks Analyze → Clicks Generate Report → Selects an export option (Preview & Print / Save PDF, Download PDF, or Download HTML). Function Description Actors:

Researcher, Lecturer, Student Purpose Allow users to export research trend analysis results into different report formats for printing, sharing, or offline viewing. Data Processing

- Verify that research trend analysis has been completed.
- Validate that report data is available.
- Collect the current analysis results, including:
  - Statistical summaries
  - Charts and visualizations
  - Trending Topics
  - Top Journals

- Top Authors
- Top Keywords
- Featured Publications
- Generate the report in the selected format.
- Prepare the report for preview or download.

#### Function Details Validation

- Research trend analysis must be completed before exporting.
- Report data must be available.

#### Business Rules

- The exported report contains the currently displayed analysis results.
- Users may choose one of the following export options:
  - Preview & Print / Save PDF
  - Download PDF
  - Download HTML
- Charts, statistical summaries, and publication information are included in the exported report.
- The exported report reflects the currently selected research topic and time range.
- The report preserves the dashboard layout and chart formatting.

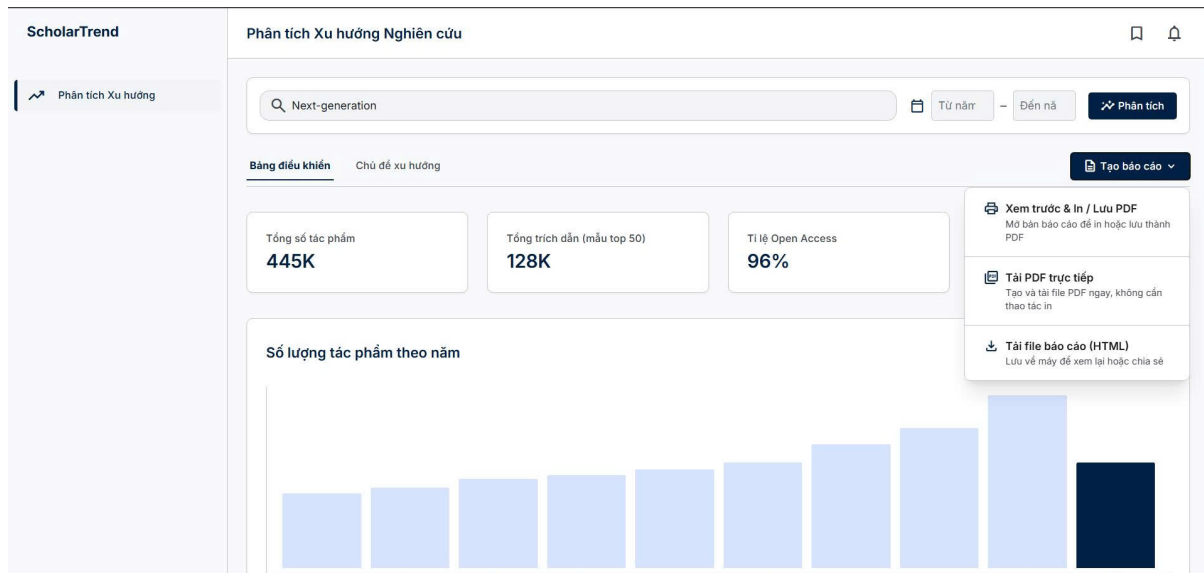
#### Normal Flow

1. User completes the research trend analysis.
2. User clicks the Generate Report button.
3. User selects an export format.
4. The system validates the report data.
5. The system generates the report.
6. The report is previewed or downloaded successfully.

#### Alternative / Exception Flow

- No analysis data available.
  - Display message: *"No report data available."*

- Report generation fails.
  - Display message: *"Unable to generate report. Please try again later."*
- Selected export format is unavailable.
  - Display message: *"Selected export format is unavailable."*



- Export Report.

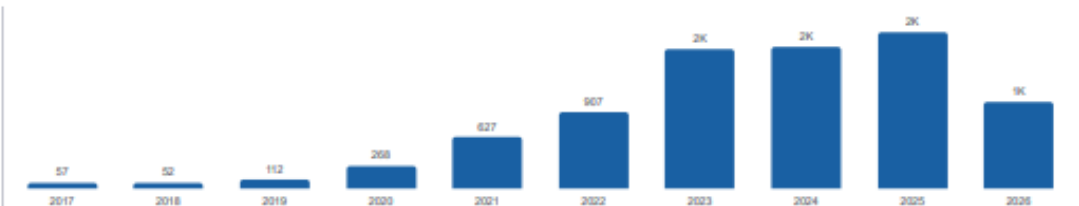
BÁO CÁO PHÂN TÍCH XU HƯỚNG NGHIÊN CỨU

|                           |                   |
|---------------------------|-------------------|
| Chủ đề / tác giả tìm kiếm | y tế              |
| Khoảng thời gian          | Toàn bộ thời gian |
| Ngày tạo báo cáo          | 4/7/2026 19:44    |

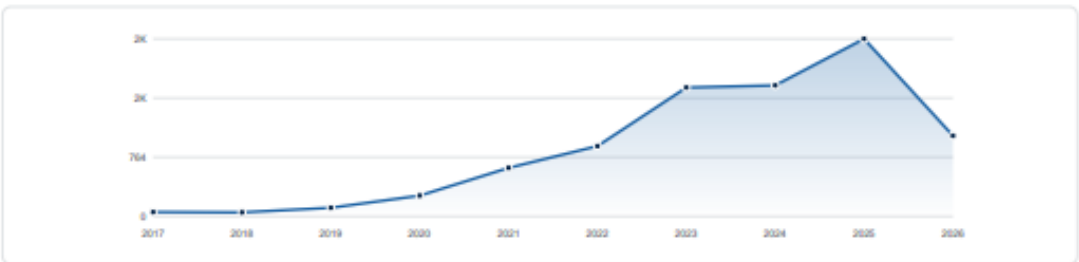
Tổng quan

|                               |                                          |                                 |                                       |
|-------------------------------|------------------------------------------|---------------------------------|---------------------------------------|
| Tổng số tác phẩm<br><b>9K</b> | Tổng trích dẫn (mẫu top 50)<br><b>4K</b> | Tỉ lệ Open Access<br><b>98%</b> | Xu hướng<br><b>Tăng trưởng (580%)</b> |
|-------------------------------|------------------------------------------|---------------------------------|---------------------------------------|

Số lượng tác phẩm theo năm (biểu đồ cột)



Biểu đồ đường xu hướng theo thời gian



Phân bố chủ đề con

|               |   |
|---------------|---|
| connectedness | 9 |
| between       | 9 |
| impact        | 8 |
| spillovers    | 8 |
| green         | 6 |
| asymmetric    | 4 |

Phân bố tạp chí

|                                                   |   |
|---------------------------------------------------|---|
| Finance Research Letters                          | 4 |
| Economic Analysis and Policy                      | 4 |
| Research in International Business and Finance    | 3 |
| Energy Economics                                  | 2 |
| Sustainable Production and Consumption            | 2 |
| The International Journal of Management Education | 2 |

3.3.10 Follow Topic

Function Activation

Users open the **Trend Analysis** page → Enter a research topic and select the time range  
→ Click **Analyze** → Click the **Follow** button to follow a research topic.

### Description

**Actors:** Researchers, Lecturers, Students.

### Purpose

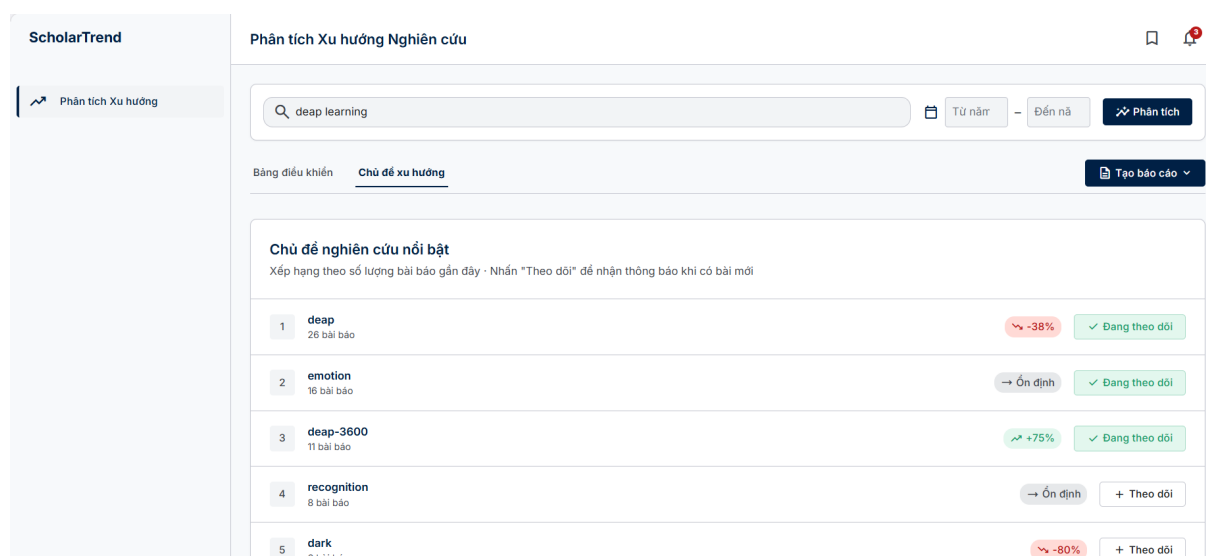
Allow users to follow research topics of interest for easier management and quick access.

### Data Processing

- Verify that the user is logged into the system.
- Check whether the selected topic has already been followed.
- If the topic has not been followed, save it to the user's followed topic list and update its status to **Following**.
- If the topic is already followed, prevent duplicate entries.
- Allow users to view the list of followed topics.
- Allow users to unfollow a topic by clicking the **Unfollow** button.
- Update the followed topic list after each follow or unfollow operation.

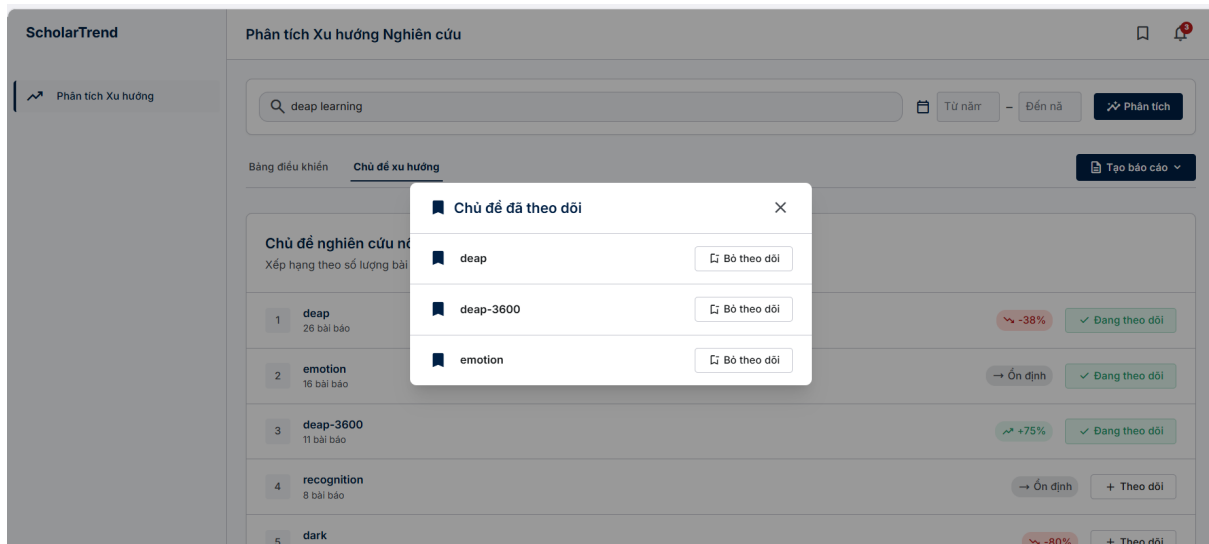
### Result

- The selected topic is added to the followed topic list.
- The followed topic list is updated correctly.
- Users can follow or unfollow topics at any time.

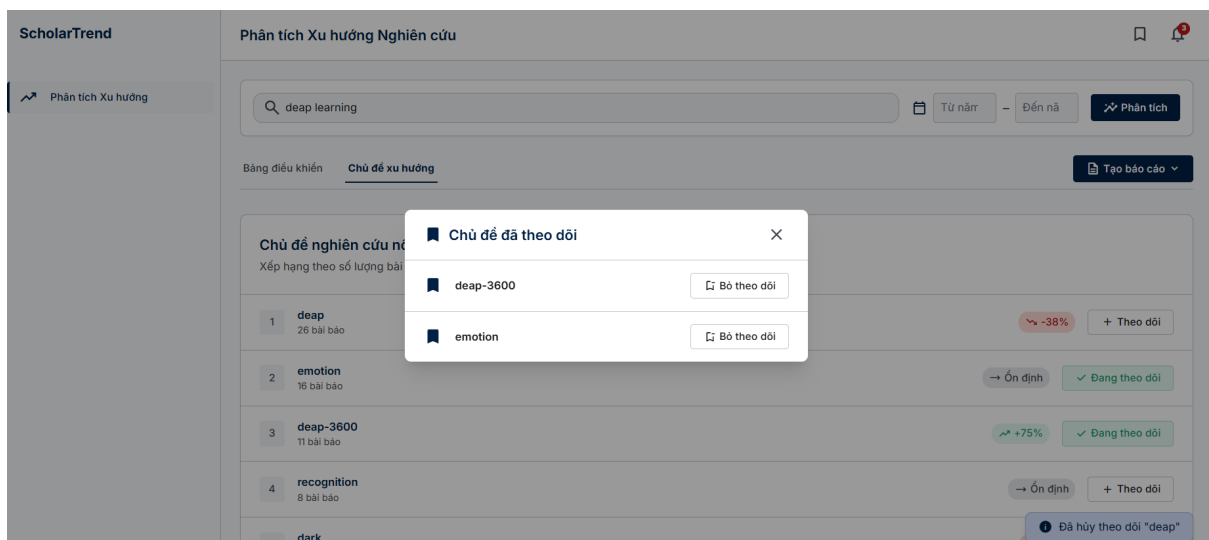


- Follow topics list





- Unfollow topic



### 3.3.11 Receive Notifications

#### Function Activation

Users log in to the system. Notifications are automatically displayed when new notifications are available.

#### Description

**Actors:** Researchers, Lecturers, Students.

#### Purpose

Allow users to receive notifications about followed research topics and system announcements from administrators.

#### Data Processing

- Verify that the user is logged into the system.
- Retrieve unread notifications from the database.
- Display notifications related to followed research topics.
- Display announcements and notifications sent by administrators.
- Allow users to mark notifications as read.
- Update the notification status after it has been read.

## Result

- Users receive notifications about followed topics.
- Users receive announcements from administrators.
- Notification status is updated correctly after being read.

The screenshot displays the ScholarTrend web application. The main content area is titled "Phân tích Xu hướng Nghiên cứu" (Research Trend Analysis). It features a search bar with the text "kinh tế" (economy). Below the search bar, there are tabs for "Bảng điều khiển" (Dashboard) and "Chủ đề xu hướng" (Trending Topics). The "Chủ đề xu hướng" tab is active, showing a list of trending topics under the heading "Chủ đề nghiên cứu nổi bật" (Prominent Research Topics). The list includes five topics: "kinh" (17 bài báo), "động" (11 bài báo), "tăng" (10 bài báo), "phát" (6 bài báo), and "trường" (5 bài báo). Each topic has a corresponding percentage increase and a "Theo dõi" (Follow) button. A notification overlay titled "Thông báo" (Notification) is visible on the right side, showing three notifications about following topics and a link to "Xem tất cả chủ đề đã theo dõi" (View all followed topics).

- Mark as Read.

ScholarTrend

Phân tích Xu hướng

Phân tích Xu hướng Nghiên cứu

🔍 kinh tế

Bảng điều khiển

Chủ đề xu hướng

Chủ đề nghiên cứu nổi bật

Xếp hạng theo số lượng bài báo gần đây · Nhấn "Theo dõi" để nhận thông báo khi có bài mới

1

kinh

17 bài báo

2

động

11 bài báo

3

tăng

10 bài báo

4

phát

6 bài báo

5

trường

+133%

+ Theo dõi

+400%

+ Theo dõi

+300%

+ Theo dõi

Thông báo

Đánh dấu đã đọc

🔔 Đã theo dõi chủ đề mới

Bạn đang theo dõi "động". Bạn sẽ nhận thông báo khi có bài mới hoặc xu hướng thay đổi.

Vừa xong

🔔 Đã theo dõi chủ đề mới

Bạn đang theo dõi "kinh". Bạn sẽ nhận thông báo khi có bài mới hoặc xu hướng thay đổi.

Vừa xong

🔔 Đã theo dõi chủ đề mới

Bạn đang theo dõi "deap-3600". Bạn sẽ nhận thông báo khi có bài mới hoặc xu hướng thay đổi.

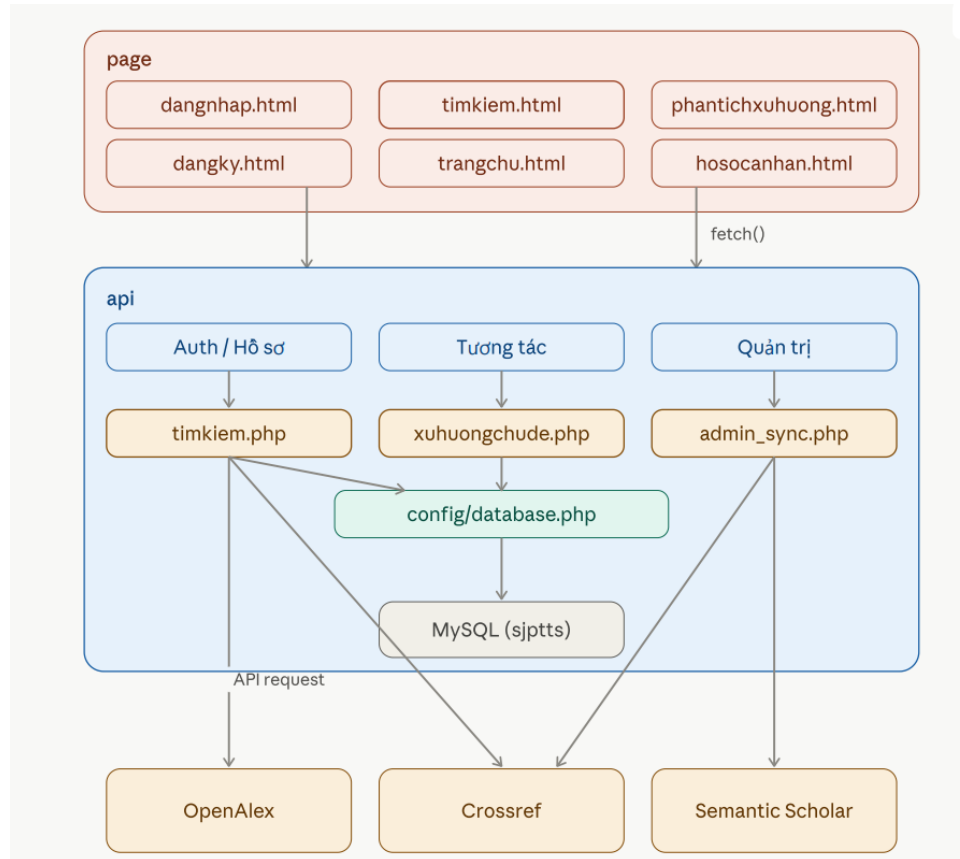
Vừa xong

Xem tất cả chủ đề đã theo dõi

## IV. Software Design Description

### 1 System Design

#### 1.1 System Architecture



##### 1.1.1 Frontend App

The frontend application is developed using HTML, CSS, and JavaScript to provide the user interface for interacting with the system. It is responsible for presenting data, handling user interactions, and communicating with the backend system through HTTP requests.

- **Pages:** Responsible for rendering the main user interfaces, including Login Page, Dashboard Page, Publication Search Page, Trend Analysis Page, Bookmark Page, and Notification Page.
- **Components:** Reusable user interface elements used to build application pages, such as search bars, filter panels, publication cards, charts, tables, and navigation menus.
- **Service:** Responsible for managing communication between the frontend and backend system through HTTP requests and responses, including sending requests and processing returned data.

### 1.1.2 Backend System

The backend system follows a layered architecture consisting of Controller Layer, Service Layer, and Repository Layer. This architecture improves maintainability, scalability, and separation of concerns.

- **Controller Layer:** Handles incoming HTTP requests from the frontend, processes request validation, and forwards requests to the appropriate service layer.
- **Service Layer:** Contains business logic for processing publication data, trend analysis, user personalization, and system administration.
- **Repository Layer:** Responsible for interacting with the database, executing queries, and managing data persistence.

### 1.1.3 External Systems

The system integrates with external services to retrieve academic publication data and store system information.

- **Academic APIs:** External academic data sources such as Semantic Scholar API, OpenAlex API, and Crossref API are used to retrieve publication metadata.
- **MySQL Database:** Stores user information, publication metadata, bookmarks, followed topics, notifications, and system configuration data.

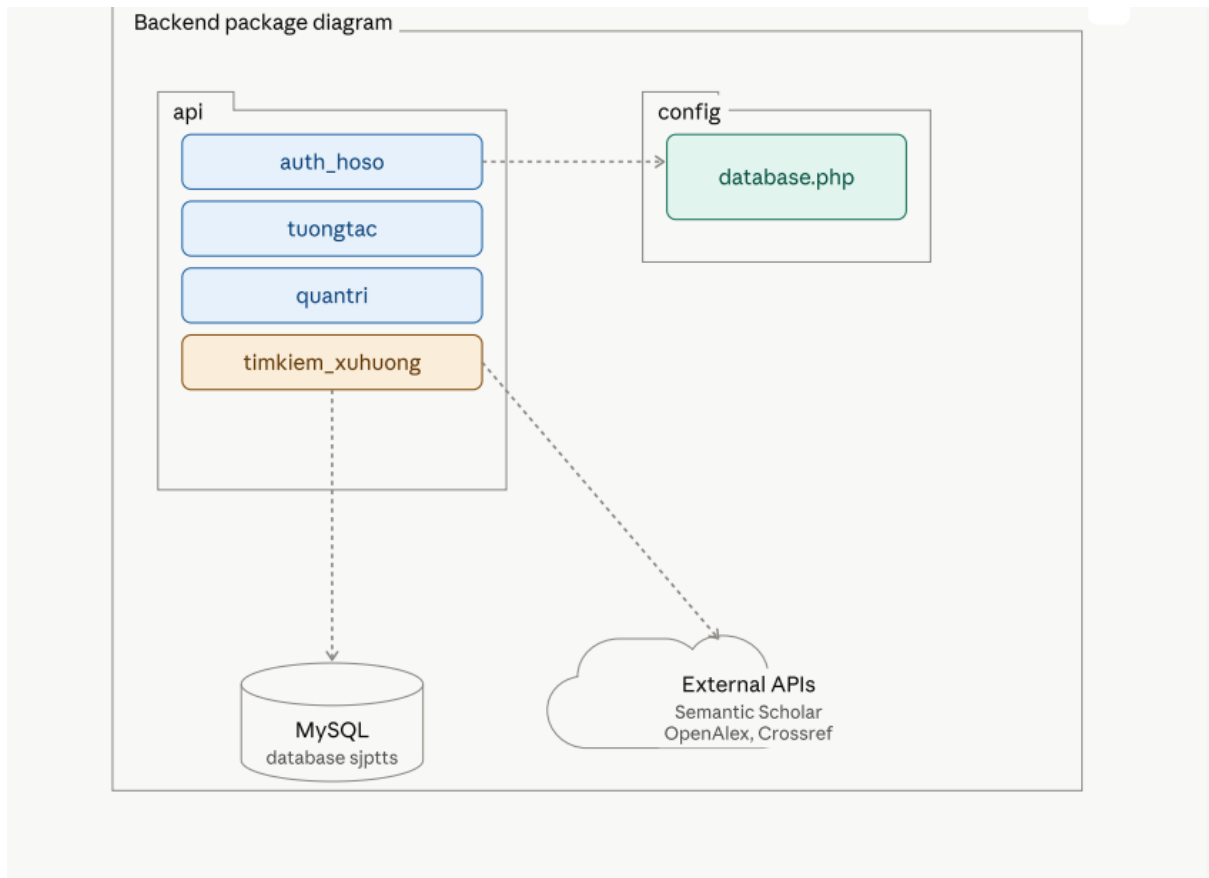
## 1.2 Package Diagram

### 1.2.1 Front End



| No | Page                       | Description                                                                                                                                          |
|----|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | dangnhap.html, dangky.html | Login and registration pages, calling dangnhap.php / dangky.php.                                                                                     |
| 2  | trangchu.html              | Public landing page (before login) — hero banner introducing ScholarTrend, a search box for papers/authors/keywords, and a list of trending topics.  |
| 3  | trangchu1.html             | Home page for logged-in users — full navigation bar (Home, Trends, notifications, avatar), with links to profile, trend analysis, and notifications. |
| 4  | xemchitietbaiba            | Search page and article detail view — calls timkiem.php to search, and yeuthich.php to save favorite papers.                                         |
| 5  | phantichxuhuong            | Trend analysis page by topic — calls phantichxuhuong.php, supports following topics and receiving notifications.                                     |
| 6  | hosocanhon.html            | Account information page — view/edit full name, email, username, email verification status; calls hosocanhon.php, capnhathoso.php, dangxuat.php.     |
| 7  | yeuthich.html              | List of favorited papers, calls yeuthich.php.                                                                                                        |
| 8  | xemthongbao.htm            | Page displaying all notifications, calls thongbao.php.                                                                                               |
| 9  | admin.html                 | System administration page — manages users, topics, API sources, and data synchronization.                                                           |

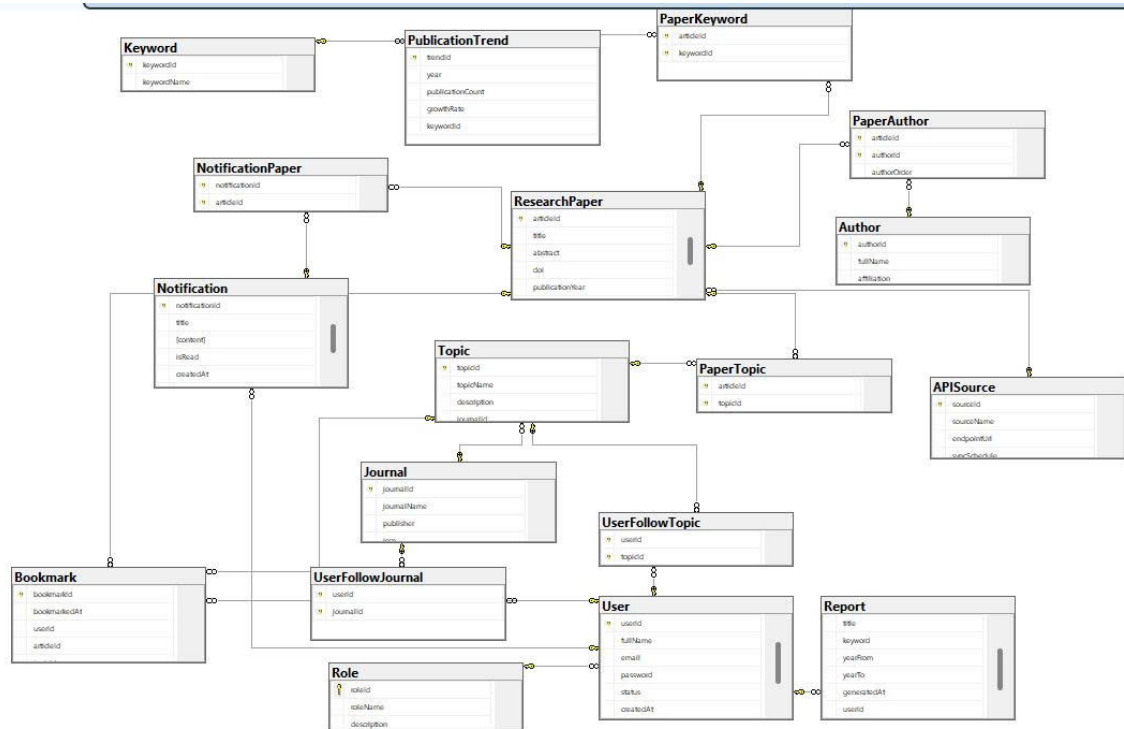
### 1.2.2 Back End



| No | Group / File                                                                                                        | Description                                                                                                                                            |
|----|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | dangnhap.php,<br>dangky.php,<br>dangxuat.php,<br>check_session.ph                                                   | Handles user authentication via PHP sessions, with hashed passwords ( <code>password_verify</code> ).                                                  |
| 2  | hosocanhan.php,<br>capnhathoso.php                                                                                  | Retrieves and updates personal profile information.                                                                                                    |
| 3  | timkiem.php                                                                                                         | Searches for research papers — queries both OpenAlex and Crossref, then merges the results.                                                            |
| 4  | xuhuongchude.php                                                                                                    | Analyzes trending topics — uses Crossref as the primary data source.                                                                                   |
| 5  | phantichxuhuong.                                                                                                    | Processes trend analysis logic for topics already stored in the database.                                                                              |
| 6  | theodoichude.php<br>yeuthich.php,<br>thongbao.php                                                                   | Handles topic following, favorite papers, and user notifications.                                                                                      |
| 7  | admin_users.php,<br>admin_papers.php<br>admin_topics.php<br>admin_api_source<br>admin_notificati<br>admin_check.php | Administrative functions: managing users, papers, topics, API sources, and notifications.                                                              |
| 8  | admin_sync.php                                                                                                      | Synchronizes paper data from OpenAlex and Semantic Scholar into the database.                                                                          |
| 9  | api_proxy.php                                                                                                       | Proxy that calls OpenAlex directly for the dashboard, with file-based caching (10-minute TTL) and rate limiting (5 requests/second).                   |
| 10 | config/database.                                                                                                    | Establishes a PDO connection to the MySQL database ( <code>sjptts</code> ), required by most files in <code>api</code> via <code>require_once</code> . |



## 2 Database Design



## Entities Description

### 2.1 Role Table

| Field       | Type          | Description | PK  | FK |
|-------------|---------------|-------------|-----|----|
| roleId      | INT           | Role ID     | Yes | No |
| roleName    | NVARCHAR(100) | Role name   | No  | No |
| description | NVARCHAR(255) | Description | No  | No |

### 2.2 User Table

| Field     | Type          | Description    | PK  | FK         |
|-----------|---------------|----------------|-----|------------|
| userId    | INT           | User ID        | Yes | No         |
| fullName  | NVARCHAR(150) | Full name      | No  | No         |
| email     | NVARCHAR(150) | Email          | No  | No         |
| password  | NVARCHAR(255) | Password       | No  | No         |
| status    | NVARCHAR(50)  | Status         | No  | No         |
| createdAt | DATETIME2     | Created time   | No  | No         |
| roleId    | INT           | Role reference | No  | Yes → Role |

### 2.3 Journal Table

| Field       | Type          | Description  | PK  | FK |
|-------------|---------------|--------------|-----|----|
| journalId   | INT           | Journal ID   | Yes | No |
| journalName | NVARCHAR(200) | Journal name | No  | No |
| publisher   | NVARCHAR(150) | Publisher    | No  | No |
| issn        | NVARCHAR(20)  | ISSN         | No  | No |

## 2.4 UserFollowJournal (M:N)

| Field     | Type | Description | PK                      | FK               |
|-----------|------|-------------|-------------------------|------------------|
| userId    | INT  | User        | Yes<br>(Com-<br>posite) | Yes →<br>User    |
| journalId | INT  | Journal     | Yes<br>(Com-<br>posite) | Yes →<br>Journal |

## 2.5 Topic Table

| Field       | Type          | Description       | PK  | FK               |
|-------------|---------------|-------------------|-----|------------------|
| topicId     | INT           | Topic ID          | Yes | No               |
| topicName   | NVARCHAR(150) | Topic name        | No  | No               |
| description | NVARCHAR(MAX) | Description       | No  | No               |
| journalId   | INT           | Journal reference | No  | Yes →<br>Journal |

## 2.6 UserFollowTopic (M:N)

| Field   | Type | Description | PK                      | FK             |
|---------|------|-------------|-------------------------|----------------|
| userId  | INT  | User        | Yes<br>(Com-<br>posite) | Yes →<br>User  |
| topicId | INT  | Topic       | Yes<br>(Com-<br>posite) | Yes →<br>Topic |

## 2.7 APISource Table

| Field | Type | Description | PK | FK |
|-------|------|-------------|----|----|
|-------|------|-------------|----|----|

|              |               |               |     |    |
|--------------|---------------|---------------|-----|----|
| sourceId     | INT           | Source ID     | Yes | No |
| sourceName   | NVARCHAR(150) | Source name   | No  | No |
| endpointUrl  | NVARCHAR(500) | API URL       | No  | No |
| syncSchedule | NVARCHAR(100) | Sync schedule | No  | No |

## 2.8 Author Table

| Field       | Type          | Description | PK  | FK |
|-------------|---------------|-------------|-----|----|
| authorId    | INT           | Author ID   | Yes | No |
| fullName    | NVARCHAR(150) | Full name   | No  | No |
| affiliation | NVARCHAR(255) | Affiliation | No  | No |

## 2.9 Keyword Table

| Field       | Type          | Description | PK  | FK |
|-------------|---------------|-------------|-----|----|
| keywordId   | INT           | Keyword ID  | Yes | No |
| keywordName | NVARCHAR(100) | Keyword     | No  | No |

## 2.10 Research Paper Table

| Field           | Type          | Description | PK  | FK              |
|-----------------|---------------|-------------|-----|-----------------|
| articleId       | INT           | Paper ID    | Yes | No              |
| title           | NVARCHAR(500) | Title       | No  | No              |
| abstract        | NVARCHAR(MAX) | Abstract    | No  | No              |
| doi             | NVARCHAR(200) | DOI         | No  | No              |
| publicationYear | INT           | Year        | No  | No              |
| sourceId        | INT           | API Source  | No  | Yes → APISource |

## 2.11 PaperTopic (M:N)

| Field     | Type | Description | PK                      | FK                               |
|-----------|------|-------------|-------------------------|----------------------------------|
| articleId | INT  | Paper       | Yes<br>(Com-<br>posite) | Yes<br>→ Re-<br>search-<br>Paper |
| topicId   | INT  | Topic       | Yes<br>(Com-<br>posite) | Yes →<br>Topic                   |

## 2.12 PaperKeyword (M:N)

| Field     | Type | Description | PK                      | FK                               |
|-----------|------|-------------|-------------------------|----------------------------------|
| articleId | INT  | Paper       | Yes<br>(Com-<br>posite) | Yes<br>→ Re-<br>search-<br>Paper |
| keywordId | INT  | Keyword     | Yes<br>(Com-<br>posite) | Yes →<br>Keyword                 |

## 2.13 PaperAuthor (M:N)

| Field       | Type | Description  | PK                      | FK                               |
|-------------|------|--------------|-------------------------|----------------------------------|
| articleId   | INT  | Paper        | Yes<br>(Com-<br>posite) | Yes<br>→ Re-<br>search-<br>Paper |
| authorId    | INT  | Author       | Yes<br>(Com-<br>posite) | Yes →<br>Author                  |
| authorOrder | INT  | Author order | No                      | No                               |

## 2.14 PublicationTrend Table

| Field            | Type  | Description | PK  | FK            |
|------------------|-------|-------------|-----|---------------|
| trendId          | INT   | Trend ID    | Yes | No            |
| year             | INT   | Year        | No  | No            |
| publicationCount | INT   | Count       | No  | No            |
| growthRate       | FLOAT | Growth rate | No  | No            |
| keywordId        | INT   | Keyword     | No  | Yes → Keyword |

## 2.15 Report Table

| Field       | Type          | Description  | PK  | FK         |
|-------------|---------------|--------------|-----|------------|
| reportId    | INT           | Report ID    | Yes | No         |
| title       | NVARCHAR(300) | Title        | No  | No         |
| keyword     | NVARCHAR(200) | Keyword      | No  | No         |
| yearFrom    | INT           | From year    | No  | No         |
| yearTo      | INT           | To year      | No  | No         |
| generatedAt | DATETIME2     | Created time | No  | No         |
| userId      | INT           | User         | No  | Yes → User |

## 2.16 Notification Table

| Field          | Type          | Description     | PK  | FK |
|----------------|---------------|-----------------|-----|----|
| notificationId | INT           | Notification ID | Yes | No |
| title          | NVARCHAR(300) | Title           | No  | No |
| content        | NVARCHAR(MAX) | Content         | No  | No |
| isRead         | BIT           | Read status     | No  | No |

|           |           |              |    |            |
|-----------|-----------|--------------|----|------------|
| createdAt | DATETIME2 | Created time | No | No         |
| userId    | INT       | User         | No | Yes → User |

## 2.17 NotificationPaper (M:N)

| Field          | Type | Description  | PK                 | FK                   |
|----------------|------|--------------|--------------------|----------------------|
| notificationId | INT  | Notification | Yes<br>(Composite) | Yes → Notification   |
| articleId      | INT  | Paper        | Yes<br>(Composite) | Yes → Research-Paper |

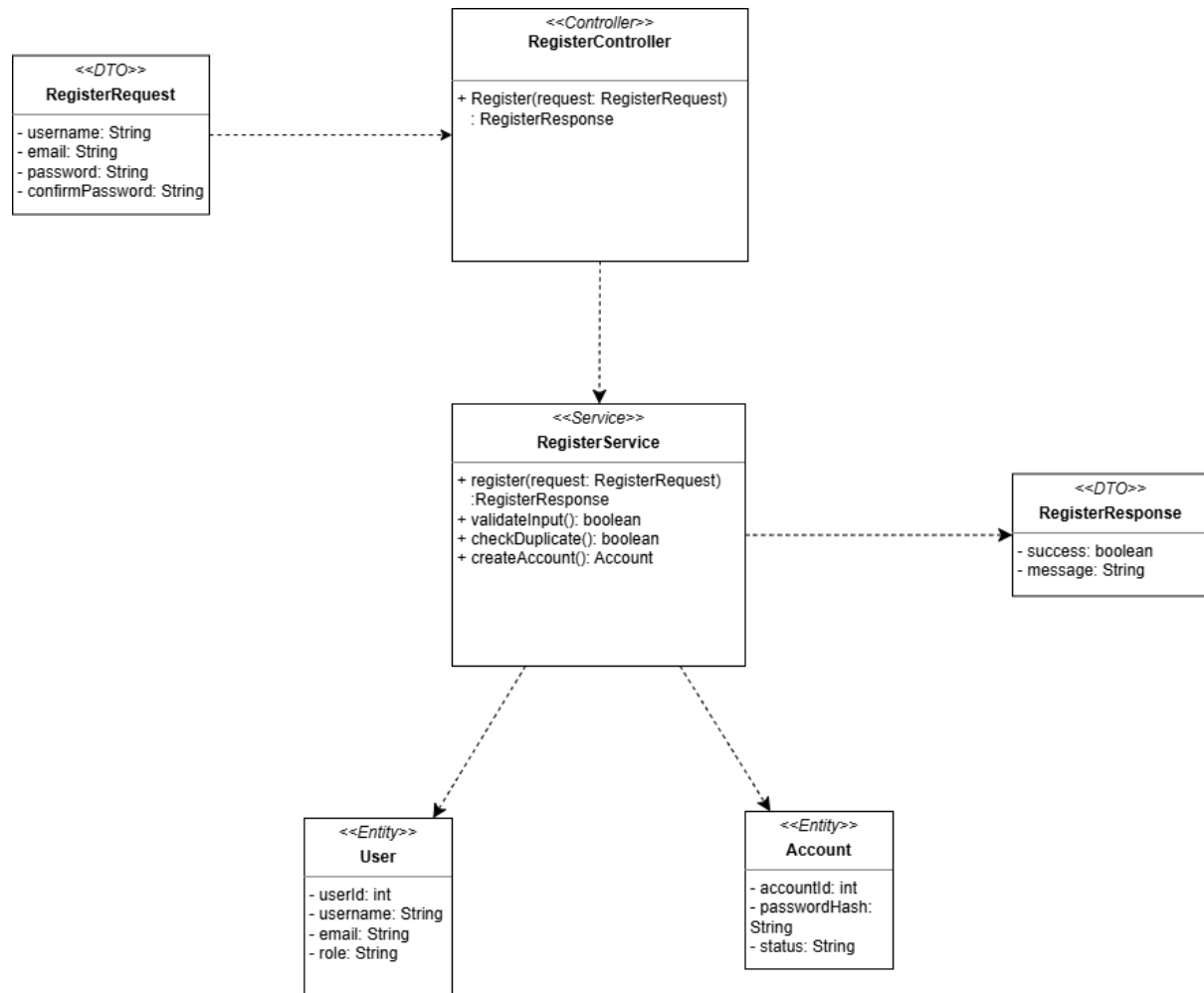
## 2.18 Bookmark Table

| Field        | Type      | Description | PK  | FK                   |
|--------------|-----------|-------------|-----|----------------------|
| bookmarkId   | INT       | Bookmark ID | Yes | No                   |
| bookmarkedAt | DATETIME2 | Time saved  | No  | No                   |
| userId       | INT       | User        | No  | Yes → User           |
| articleId    | INT       | Paper       | No  | Yes → Research-Paper |
| topicId      | INT       | Topic       | No  | Yes → Topic          |

## 3 Detail Design

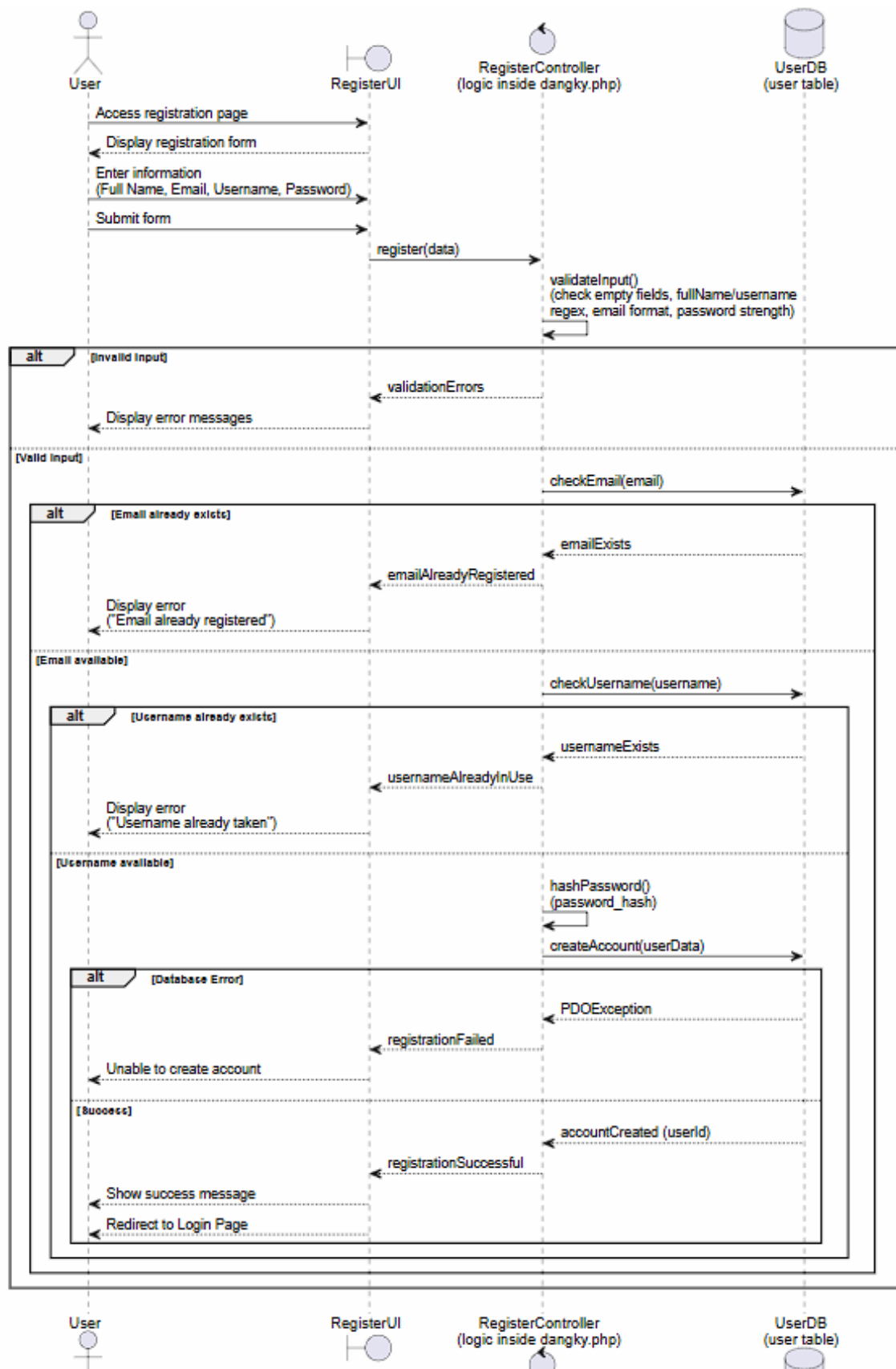
### 3.1 Register

#### 3.1.1 Class Diagram



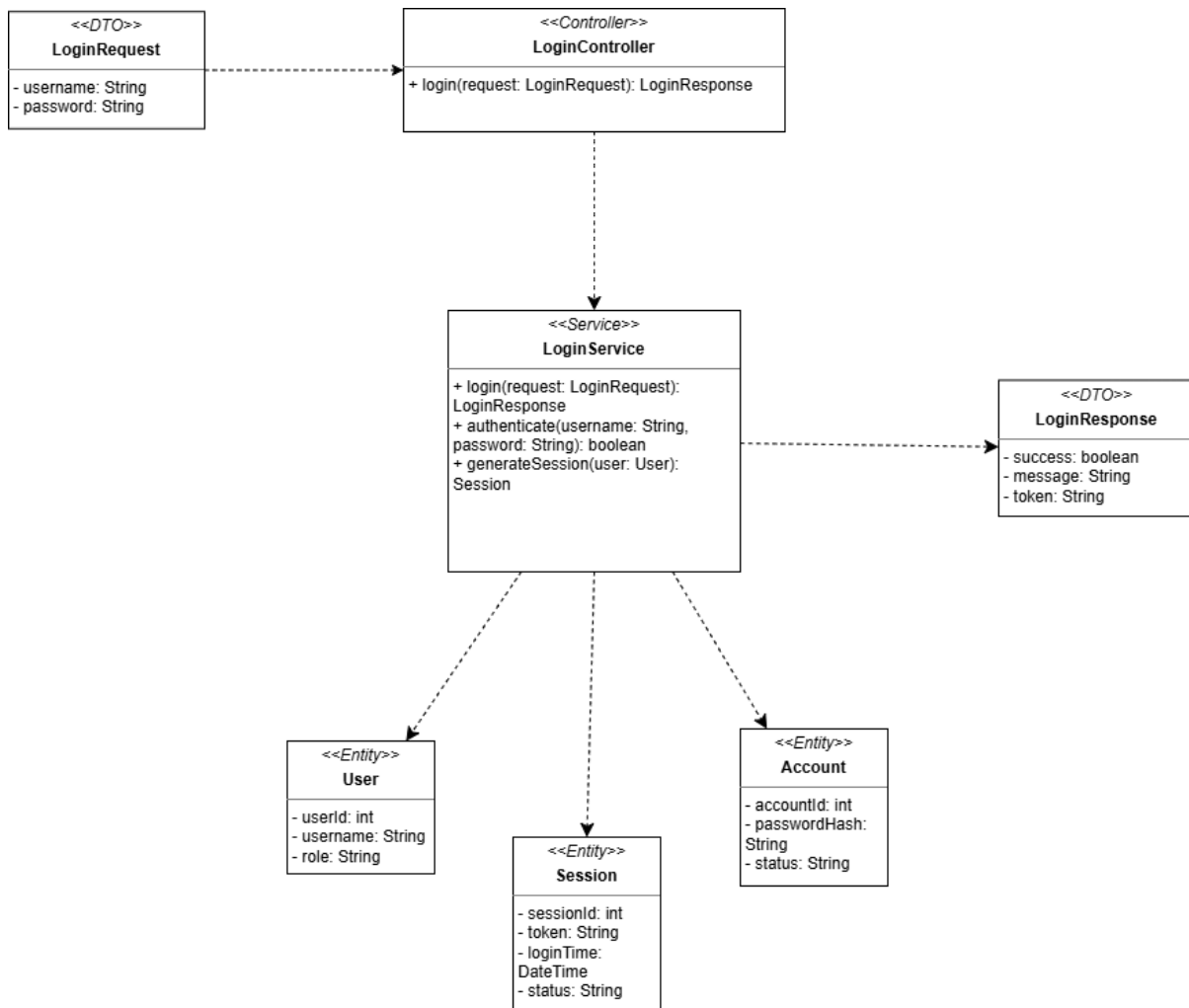


### 3.1.2 Sequence Diagram

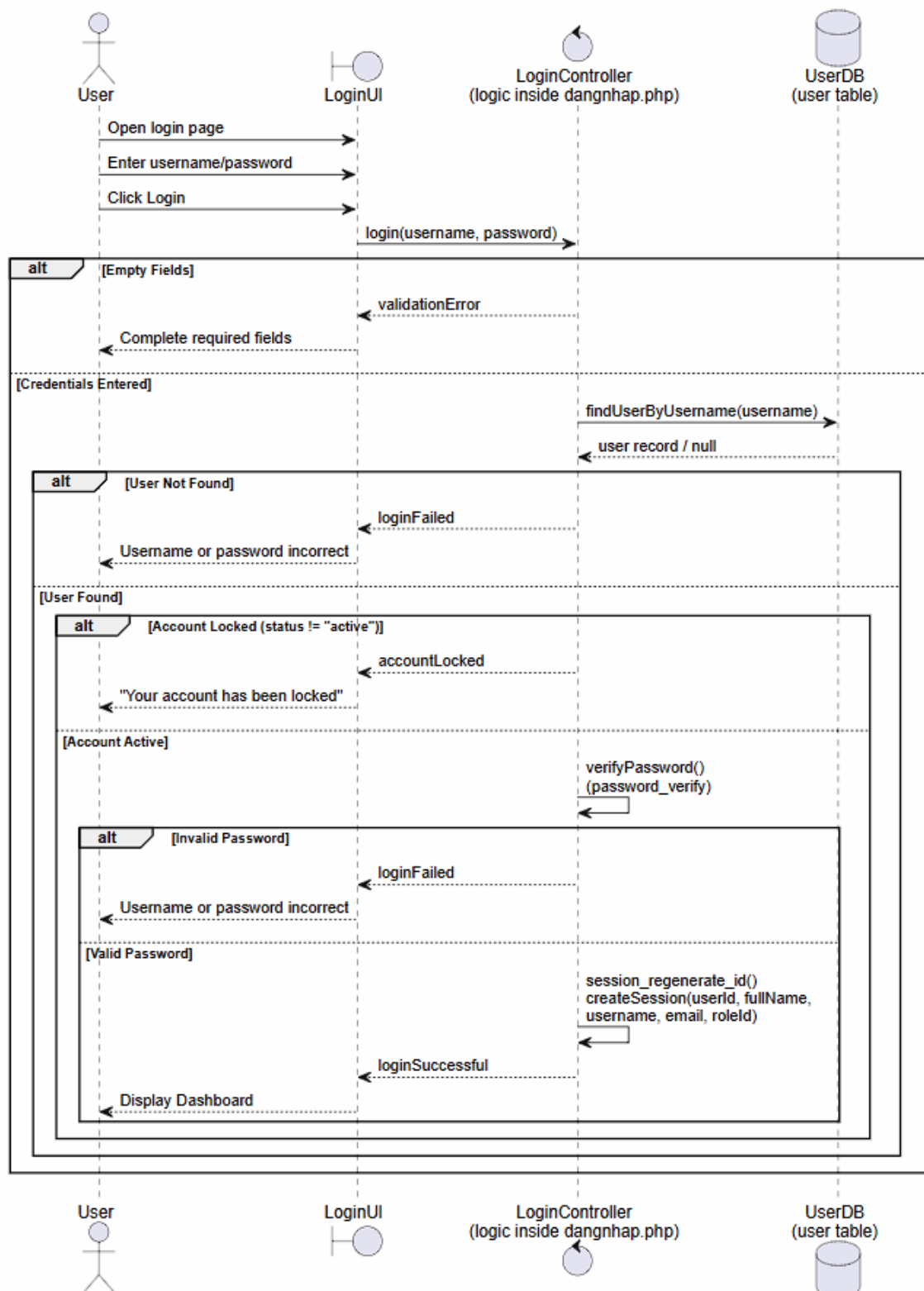


## 3.2 Login

### 3.2.1 Class Diagram

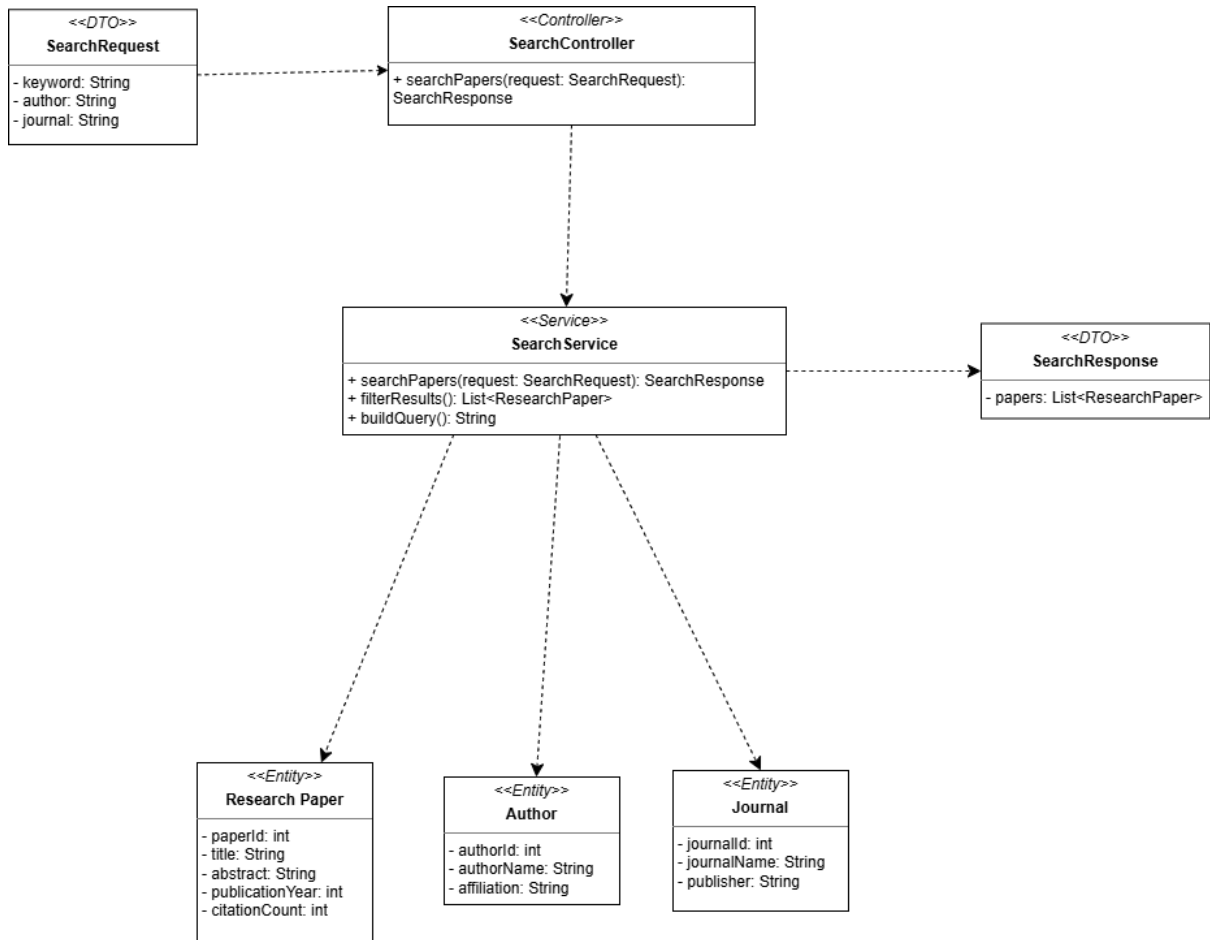


### 3.2.2 Sequence Diagram



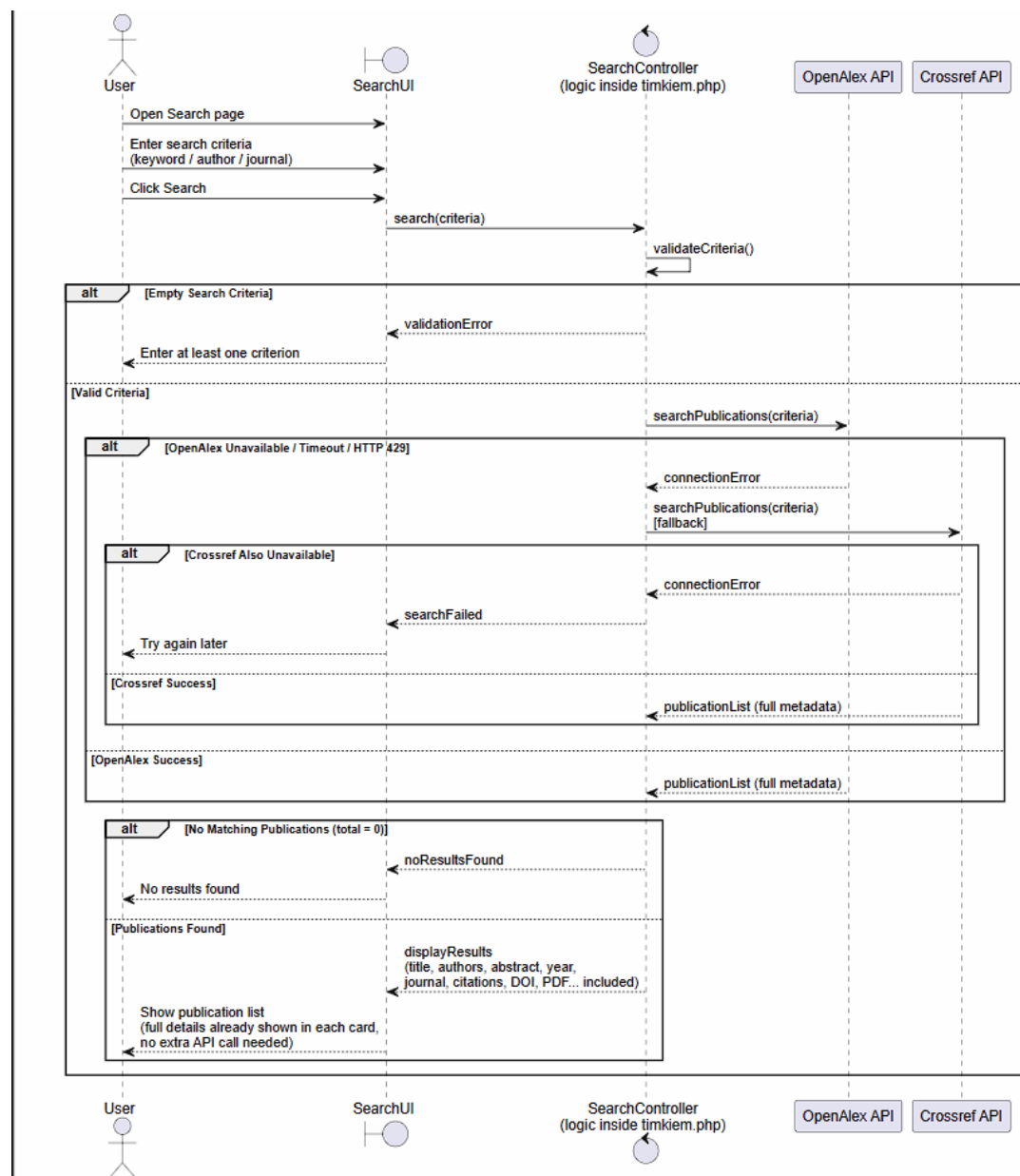
## 3.3 Search Research Papers

### 3.3.1 Class Diagram



Hình 1: Class Diagram: Search Research Papers

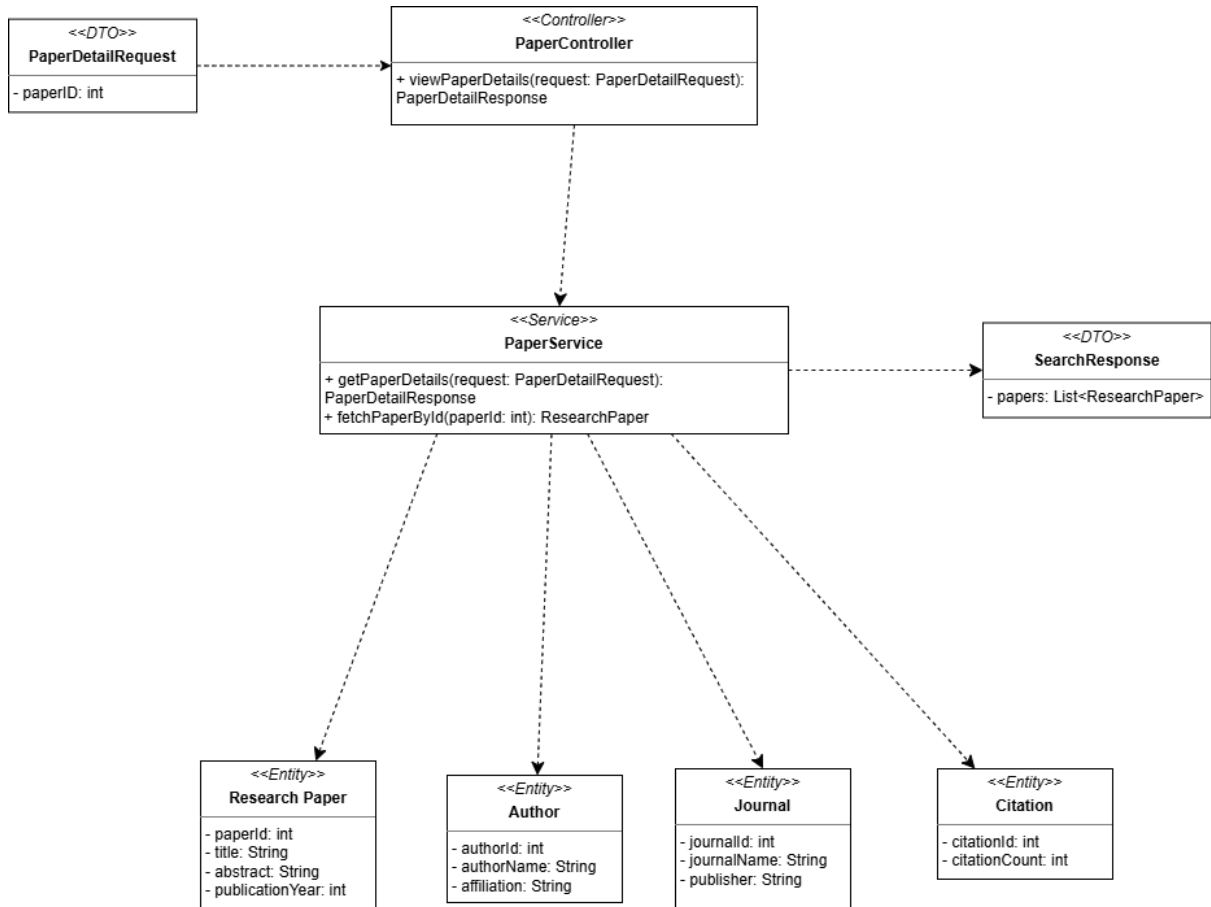
### 3.3.2 Sequence Diagram



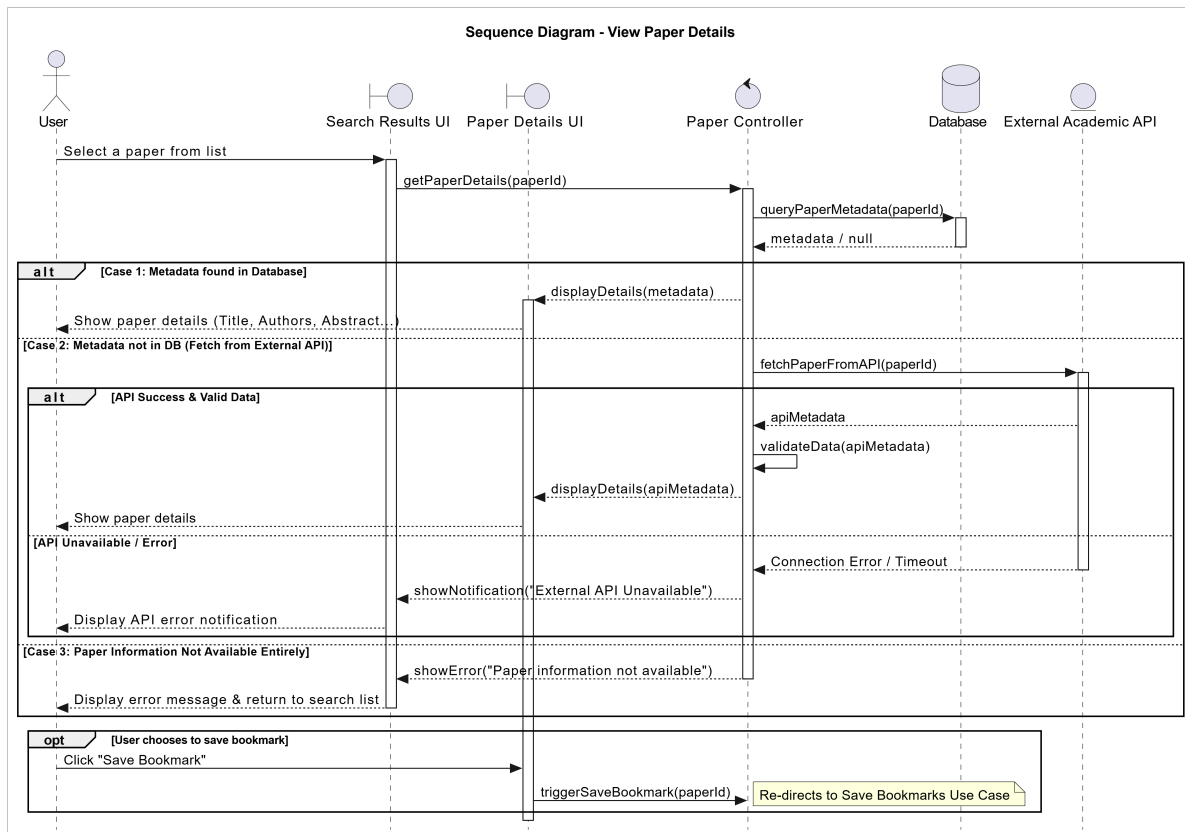
Hình 2: Sequence Diagram

## 3.4 View Paper Details

### 3.4.1 Class Diagram

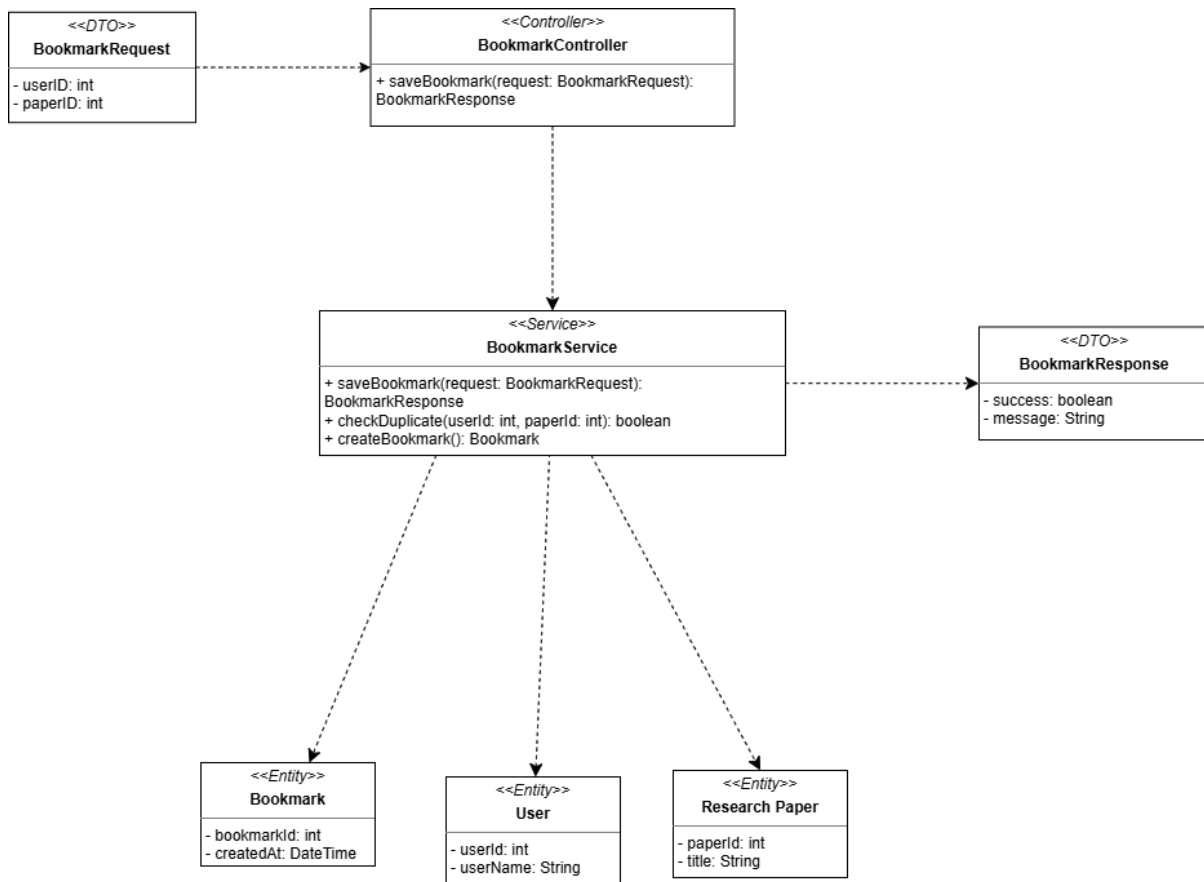


### 3.4.2 Sequence Diagram



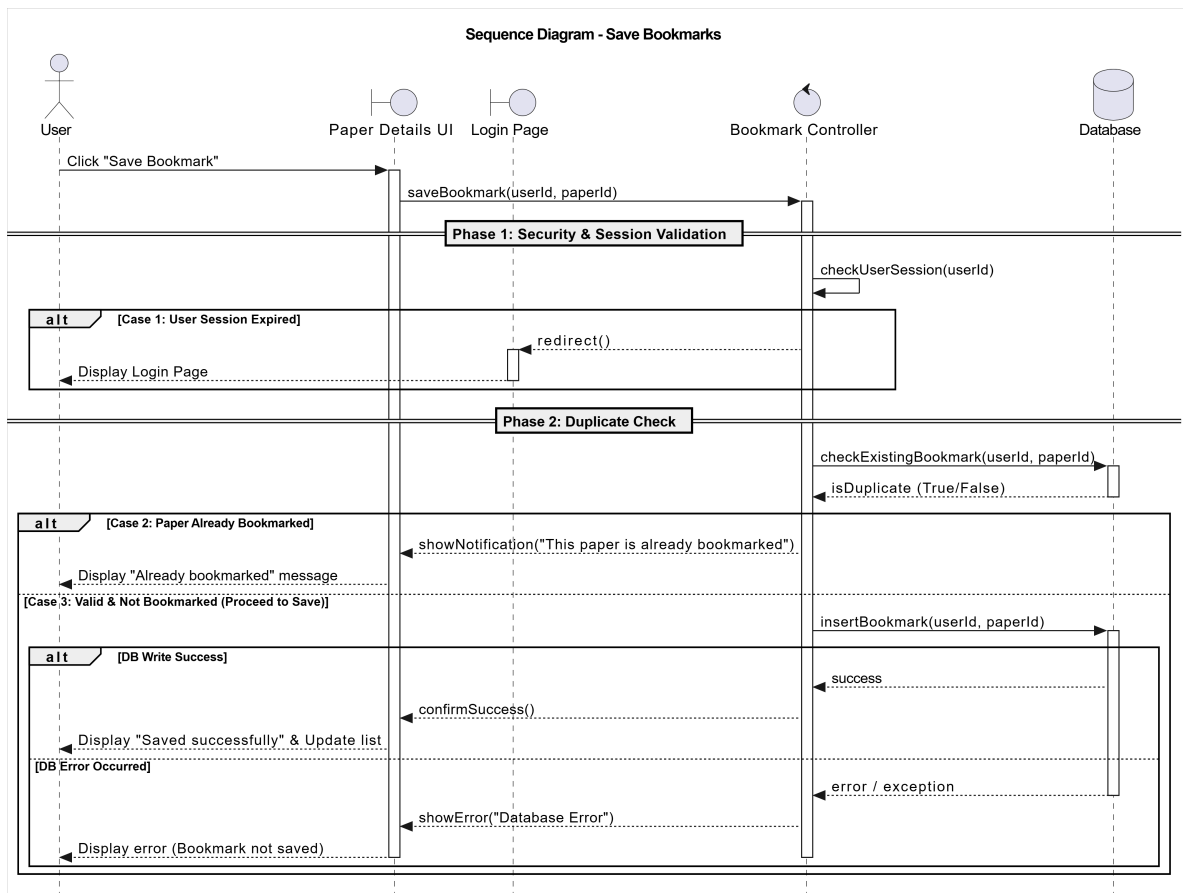
## 3.5 Save Bookmarks

### 3.5.1 Class Diagram





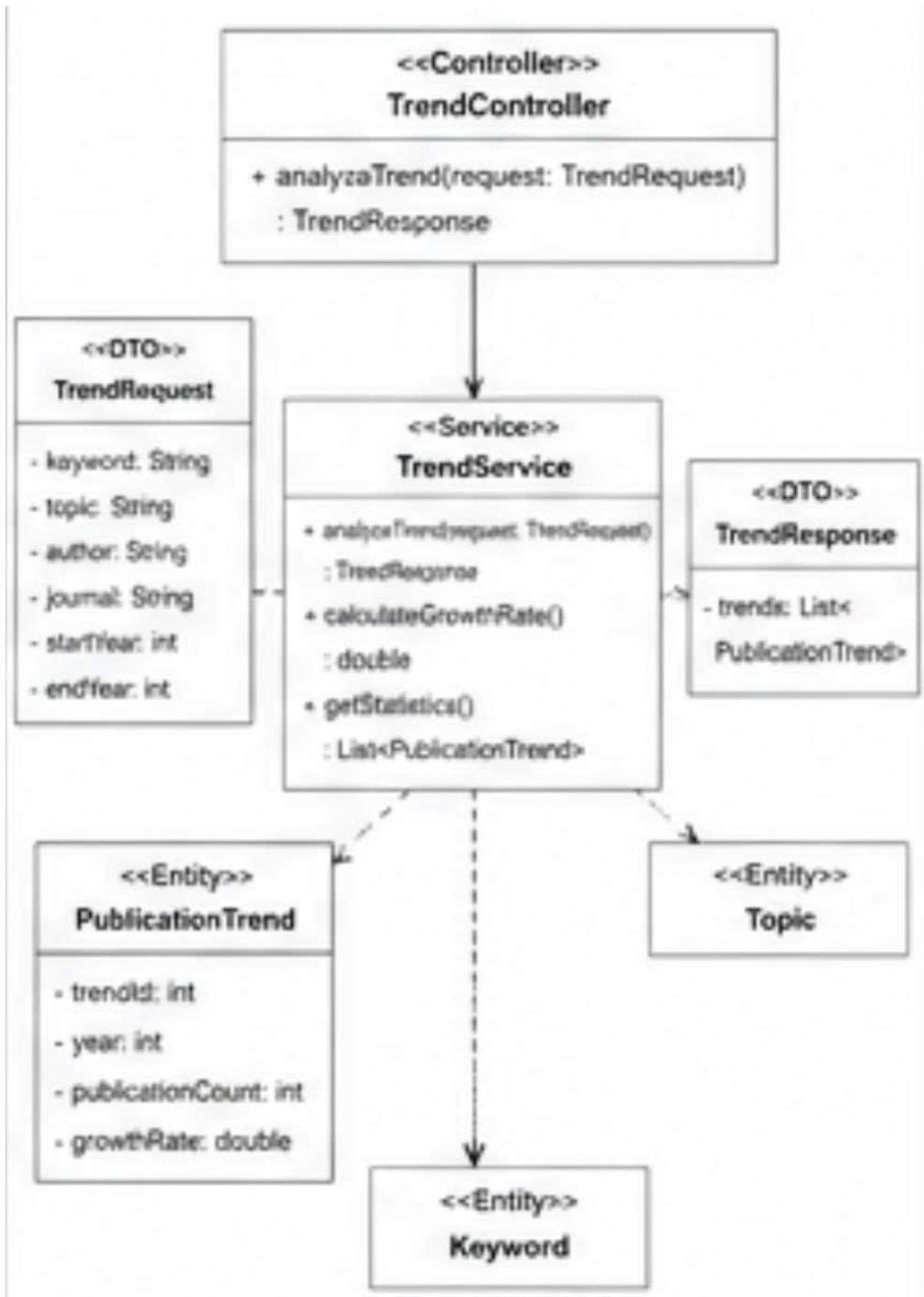
### 3.5.2 Sequence Diagram





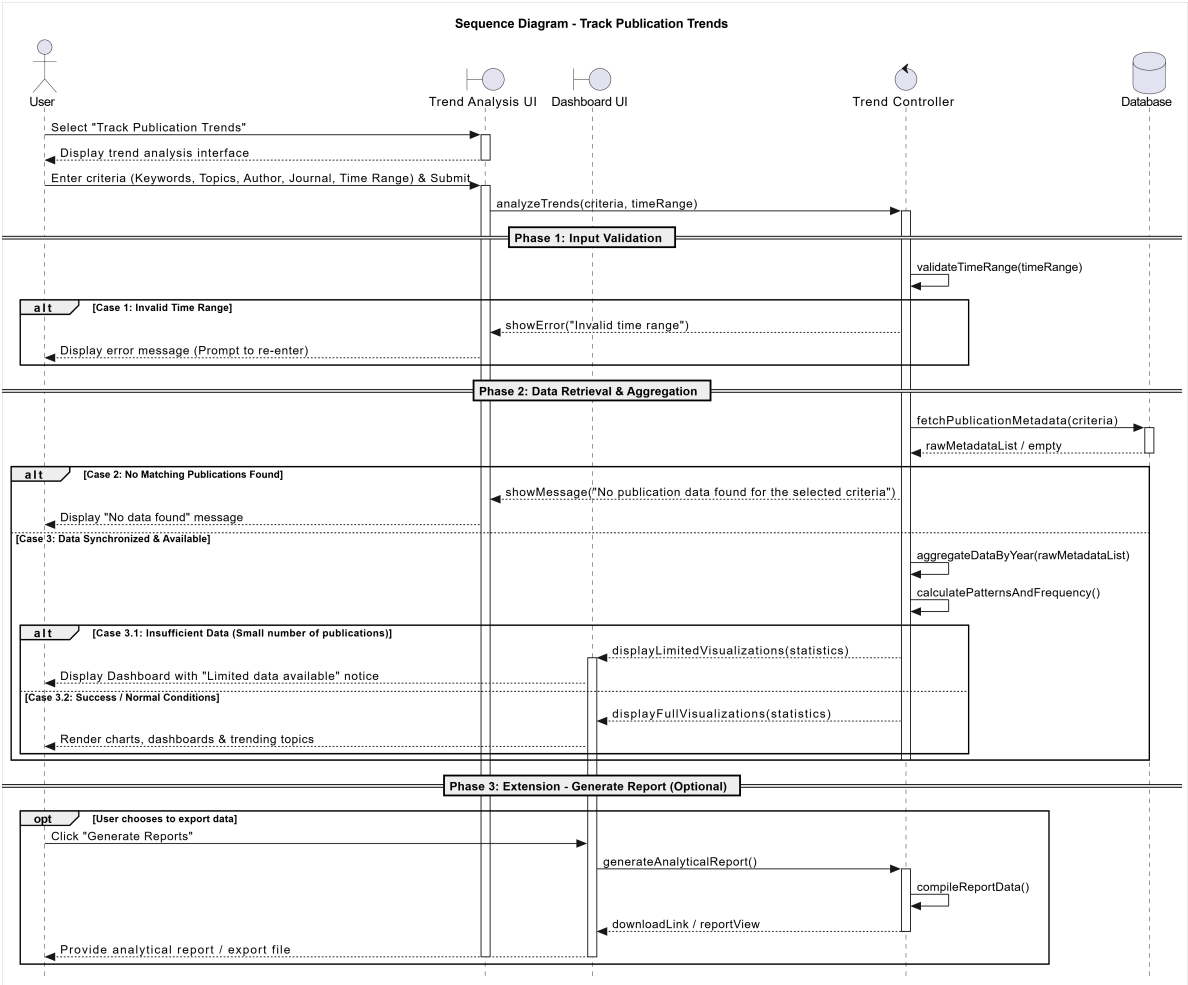
## 3.6 Track Publication Trends

### 3.6.1 Class Diagram



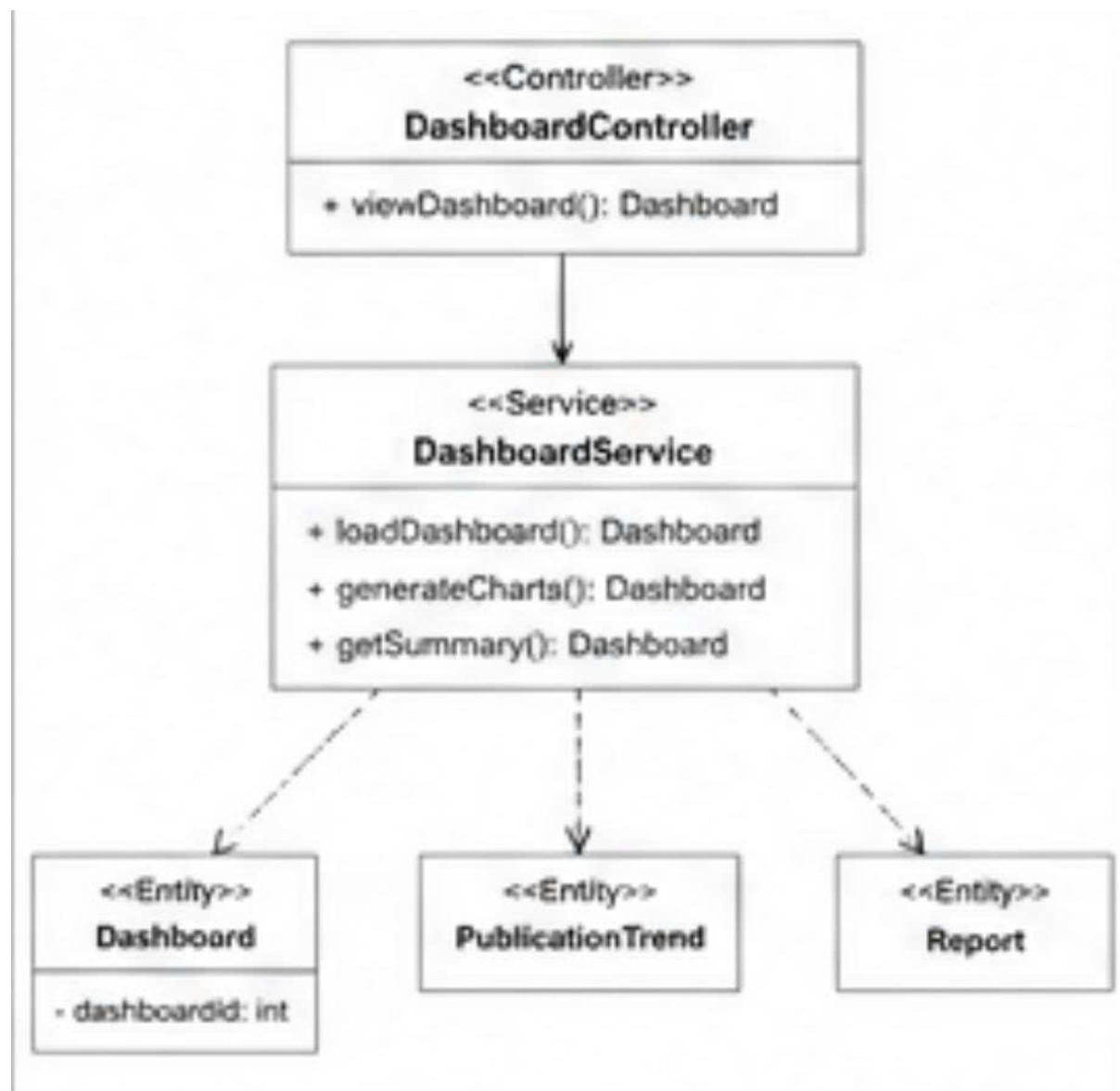
Hình 3: Class Diagram: Track Publication Trends

### 3.6.2 Sequence Diagram



## 3.7 View Dashboard

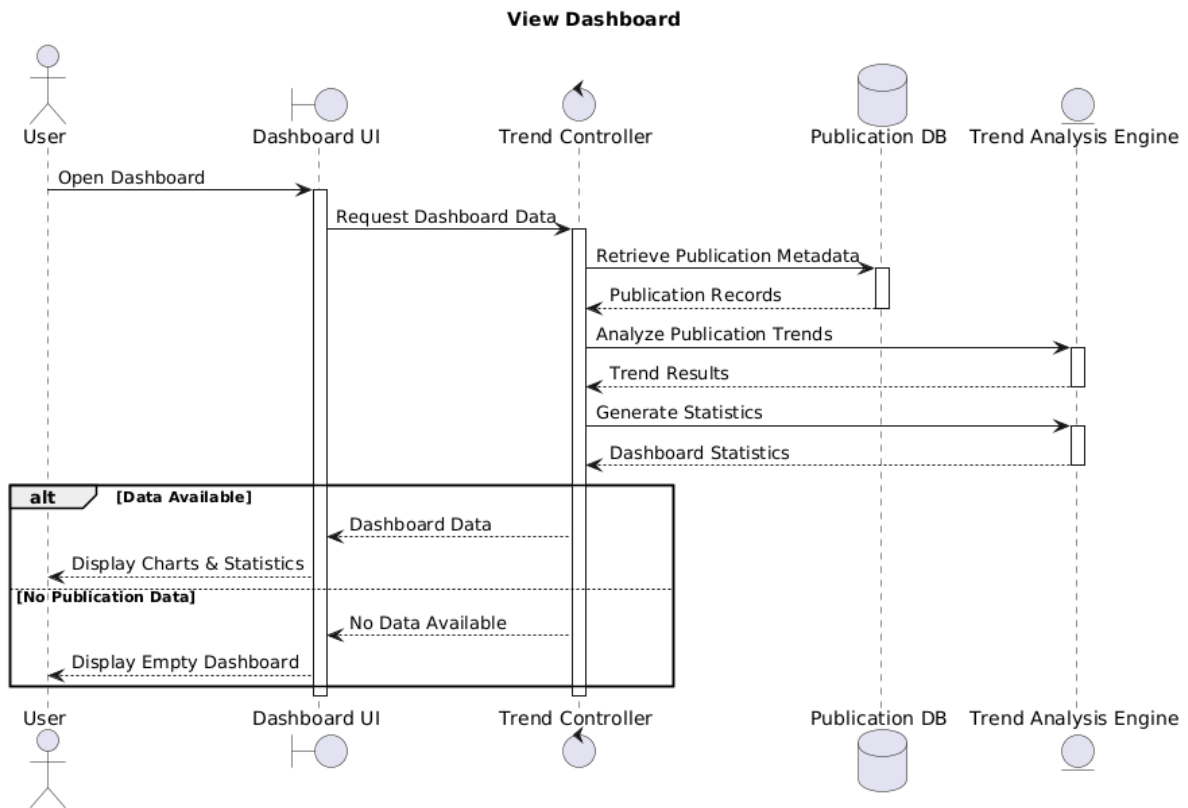
### 3.7.1 Class Diagram



Hình 4: Class Diagram: View Dashboard

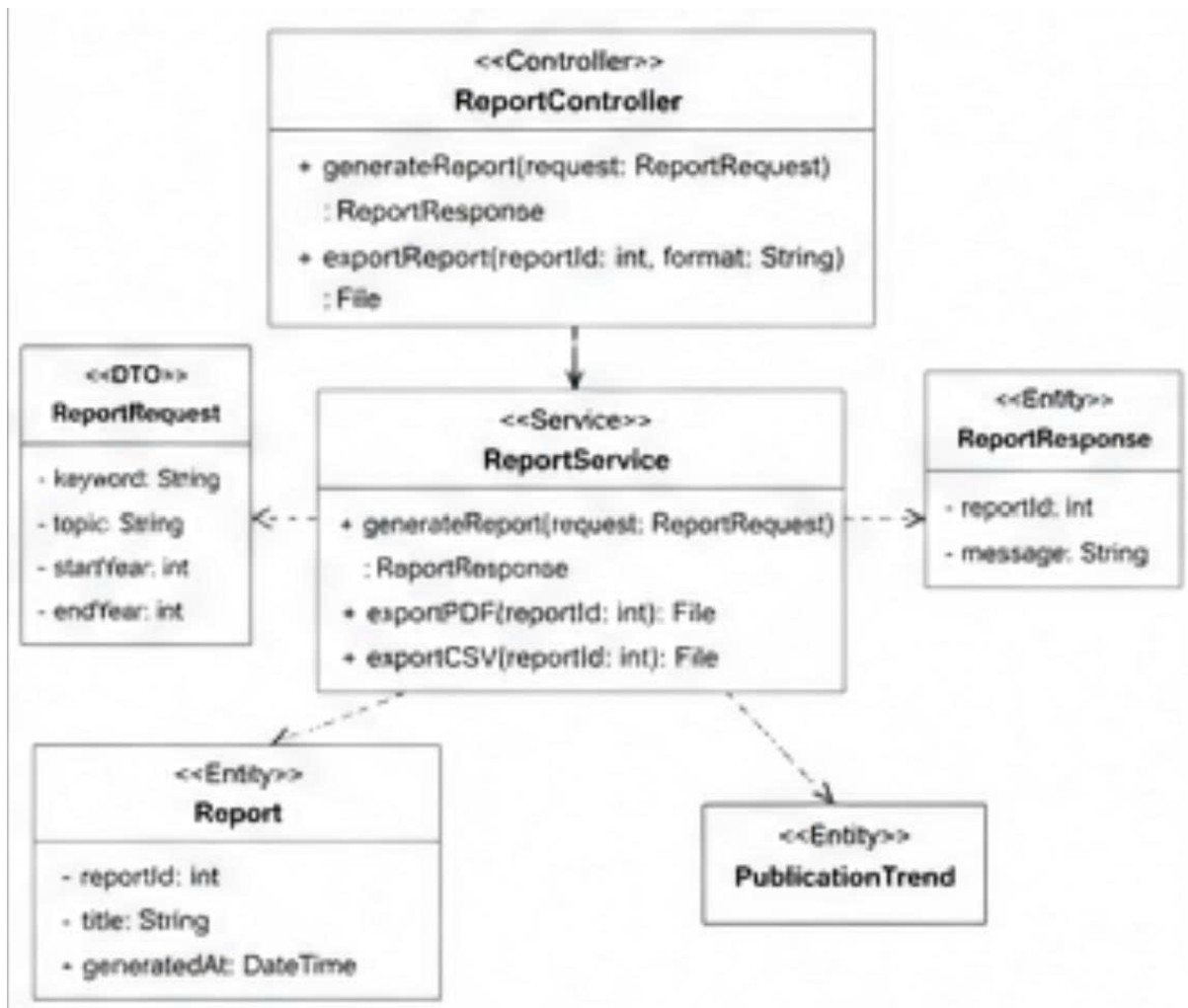
## 3.8 View Dashboard

### 3.8.1 Sequence Diagram



## 3.9 Generate Reports

### 3.9.1 Class Diagram



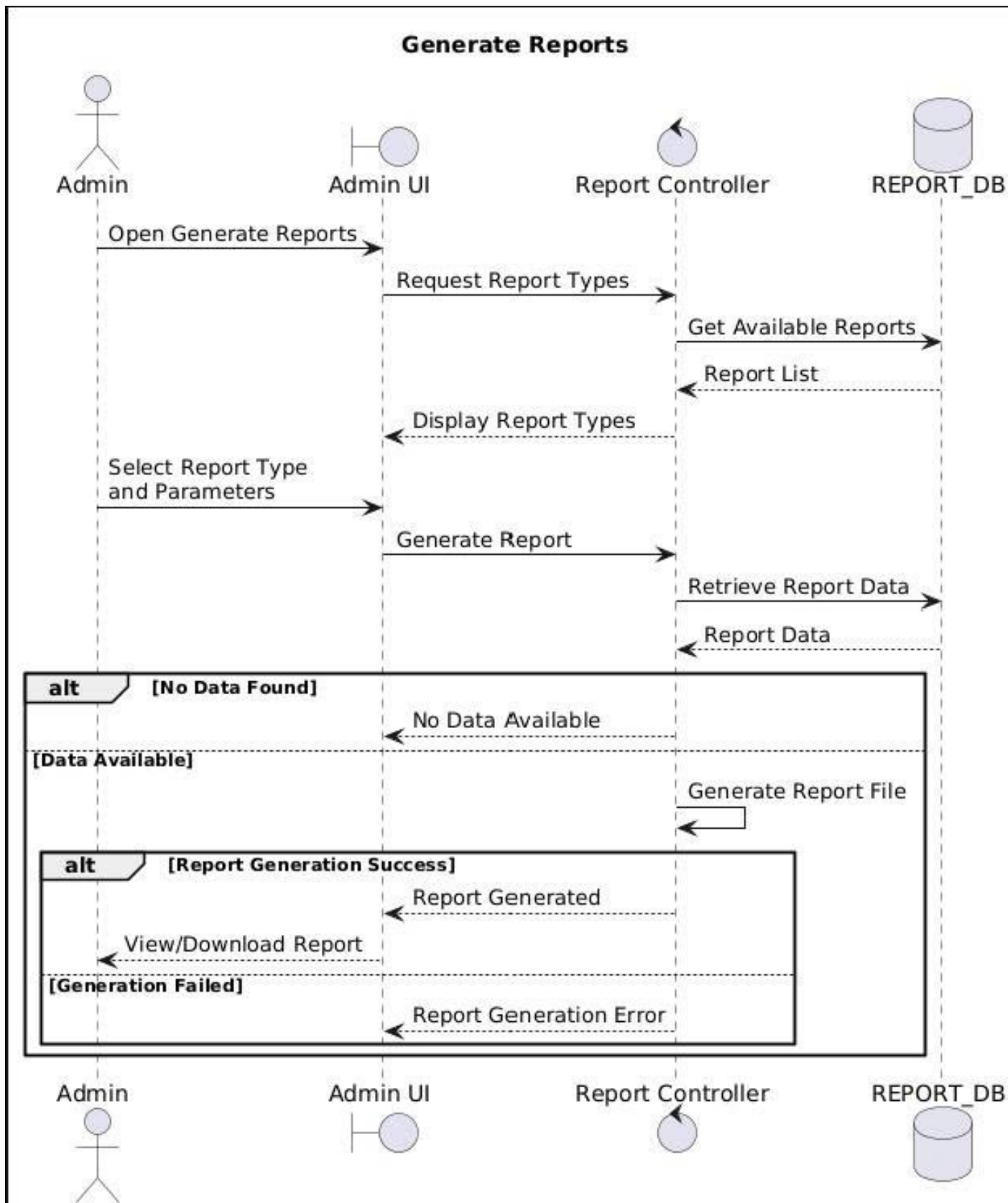
Hình 5: Class Diagram - Generate Reports





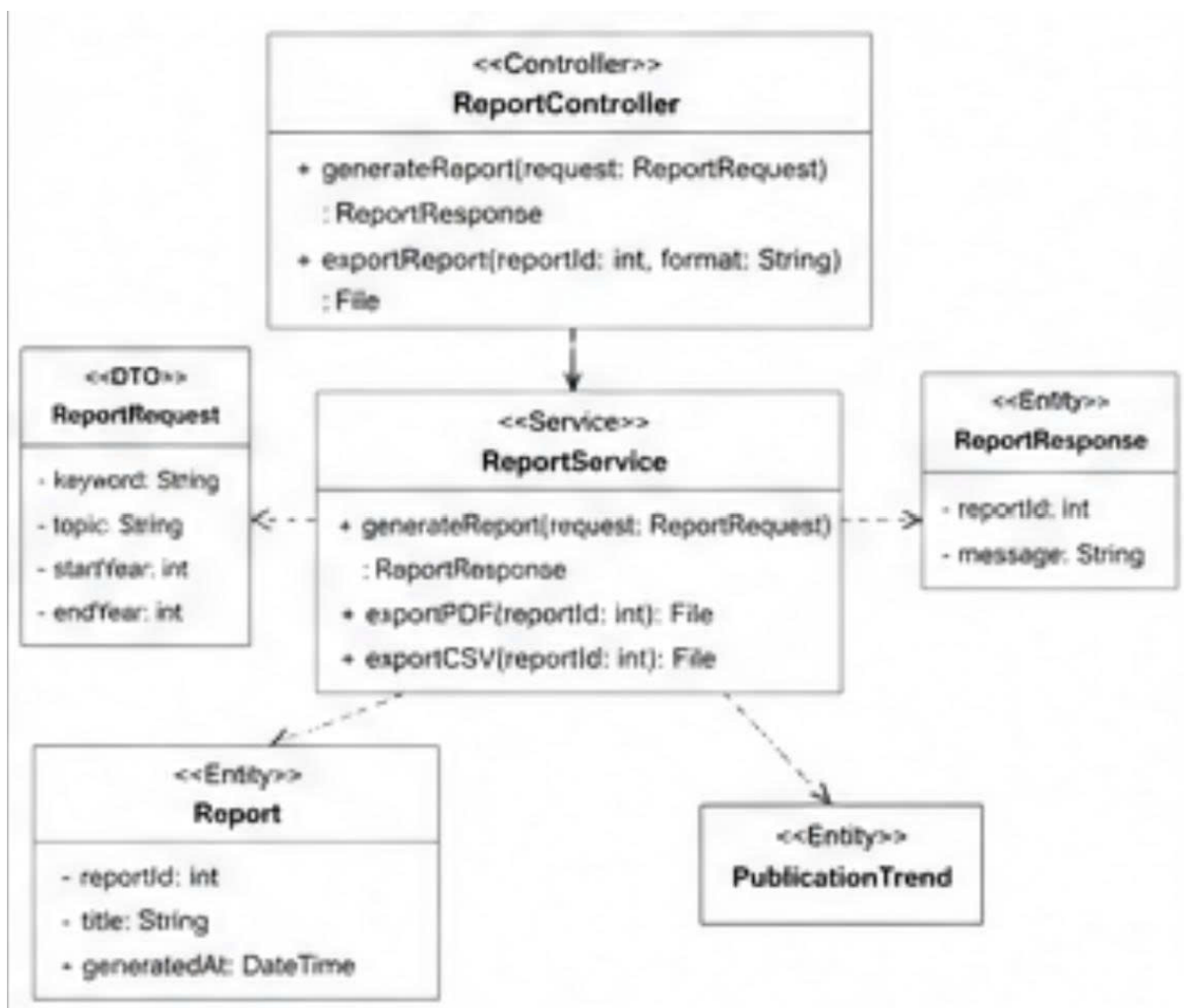
## 3.10 Generate Reports

### 3.10.1 Sequence Diagram



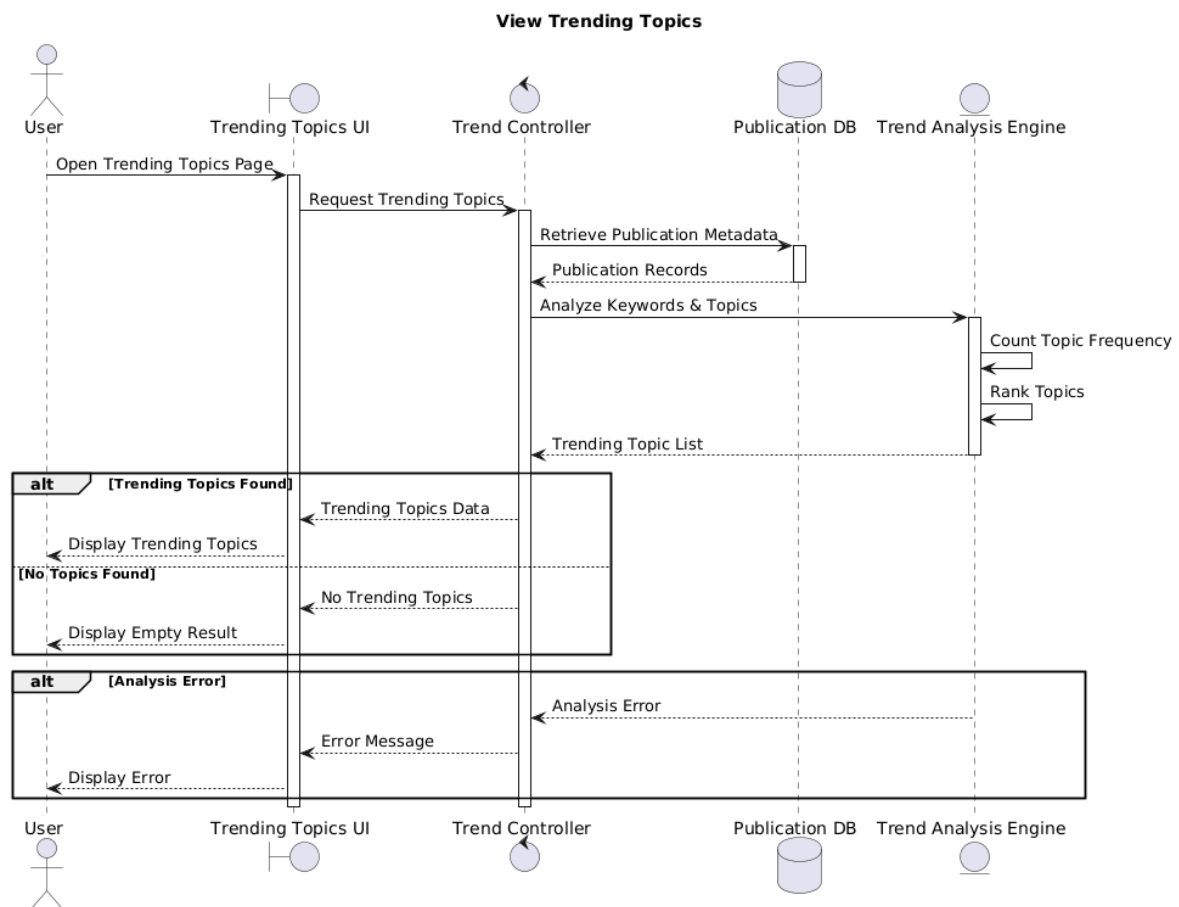
## 3.11 View Trending Topics

### 3.11.1 Class Diagram



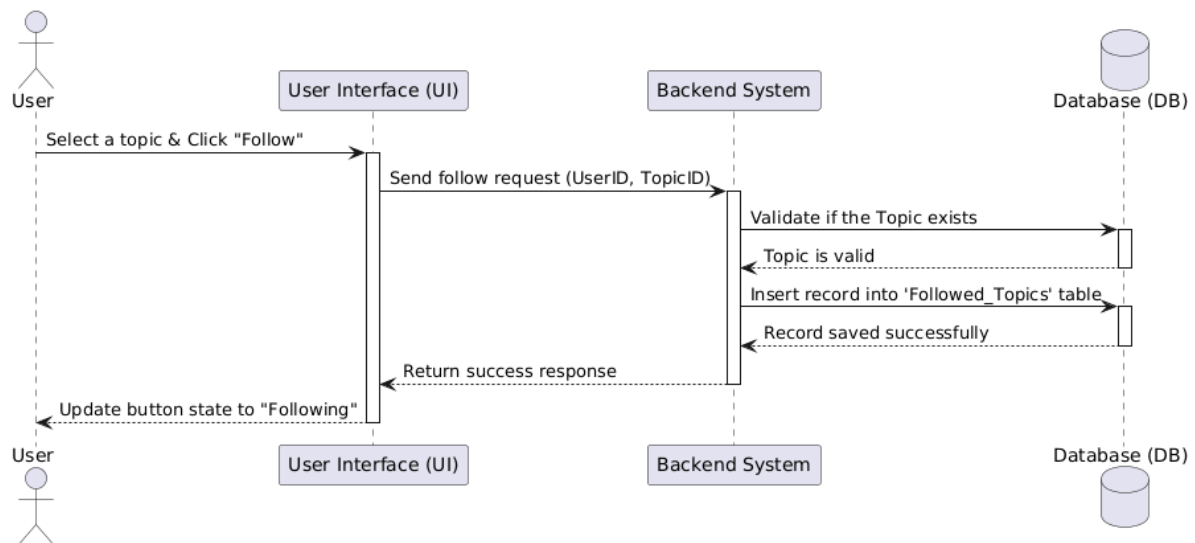
## 3.12 View Trending Topics

### 3.12.1 Sequence Diagram



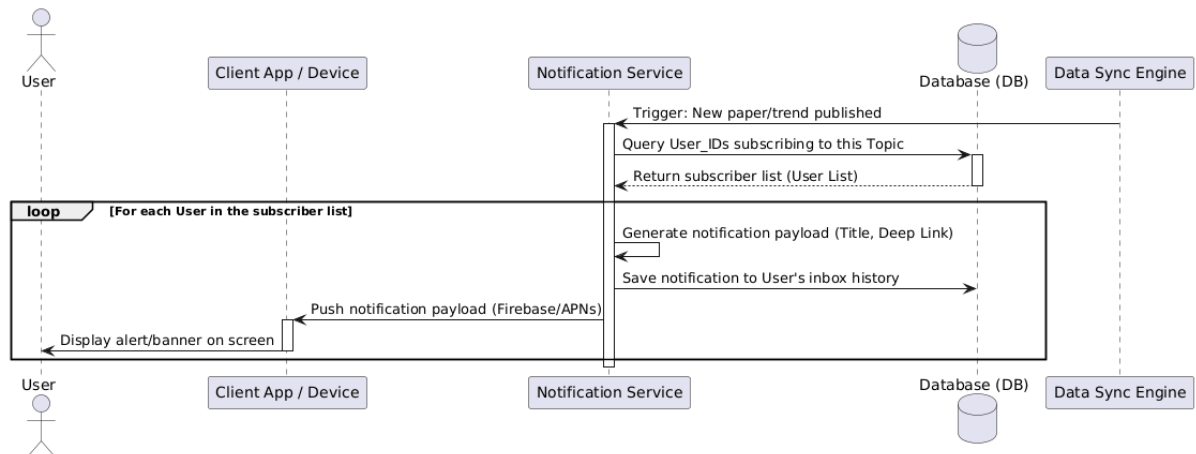
## 3.13 Follow Topics

### 3.13.1 Sequence Diagram



## 3.14 Receive Notifications

### 3.14.1 Sequence Diagram

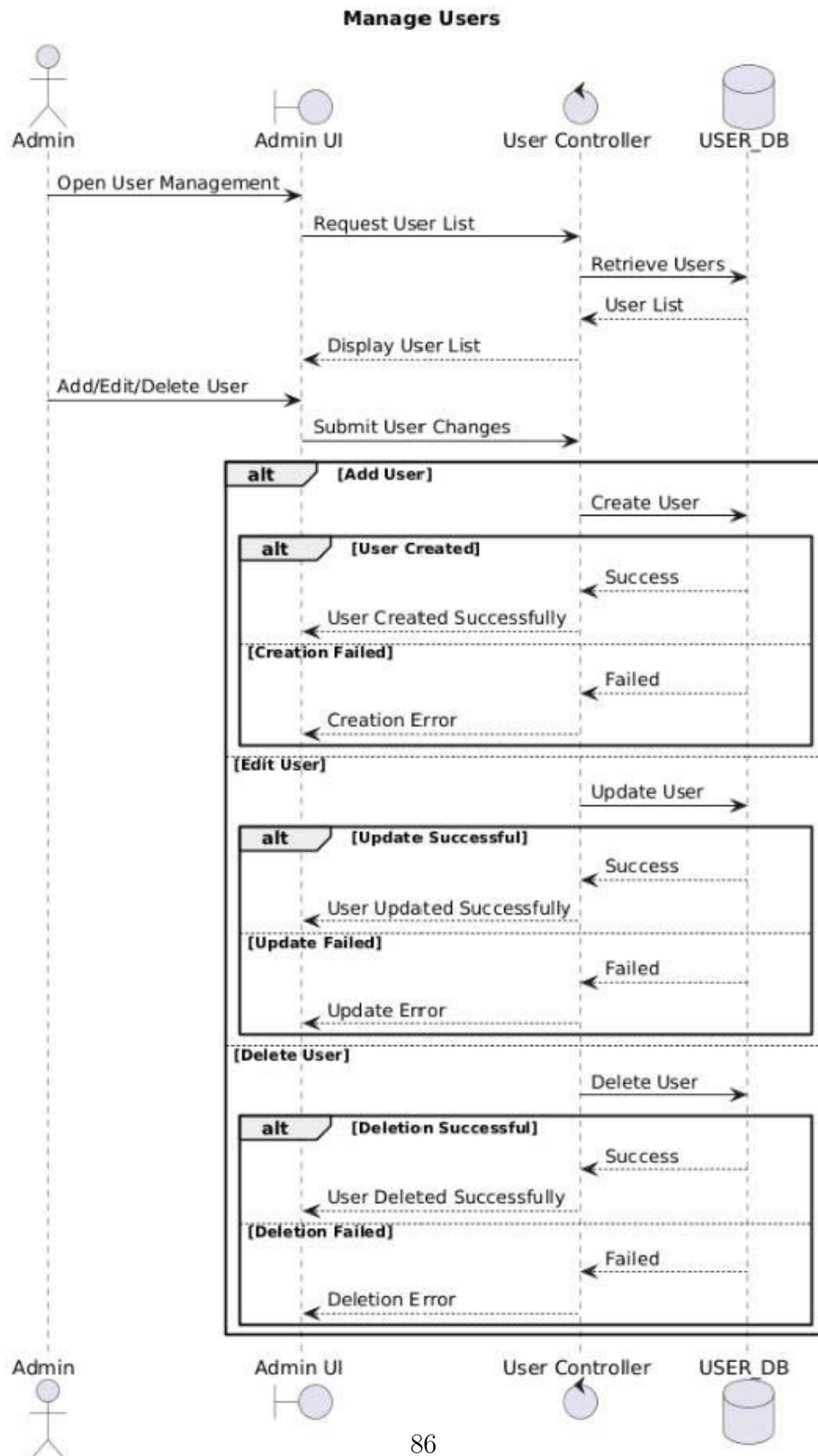




### 3.15 Manage System

### 3.16 Manage User

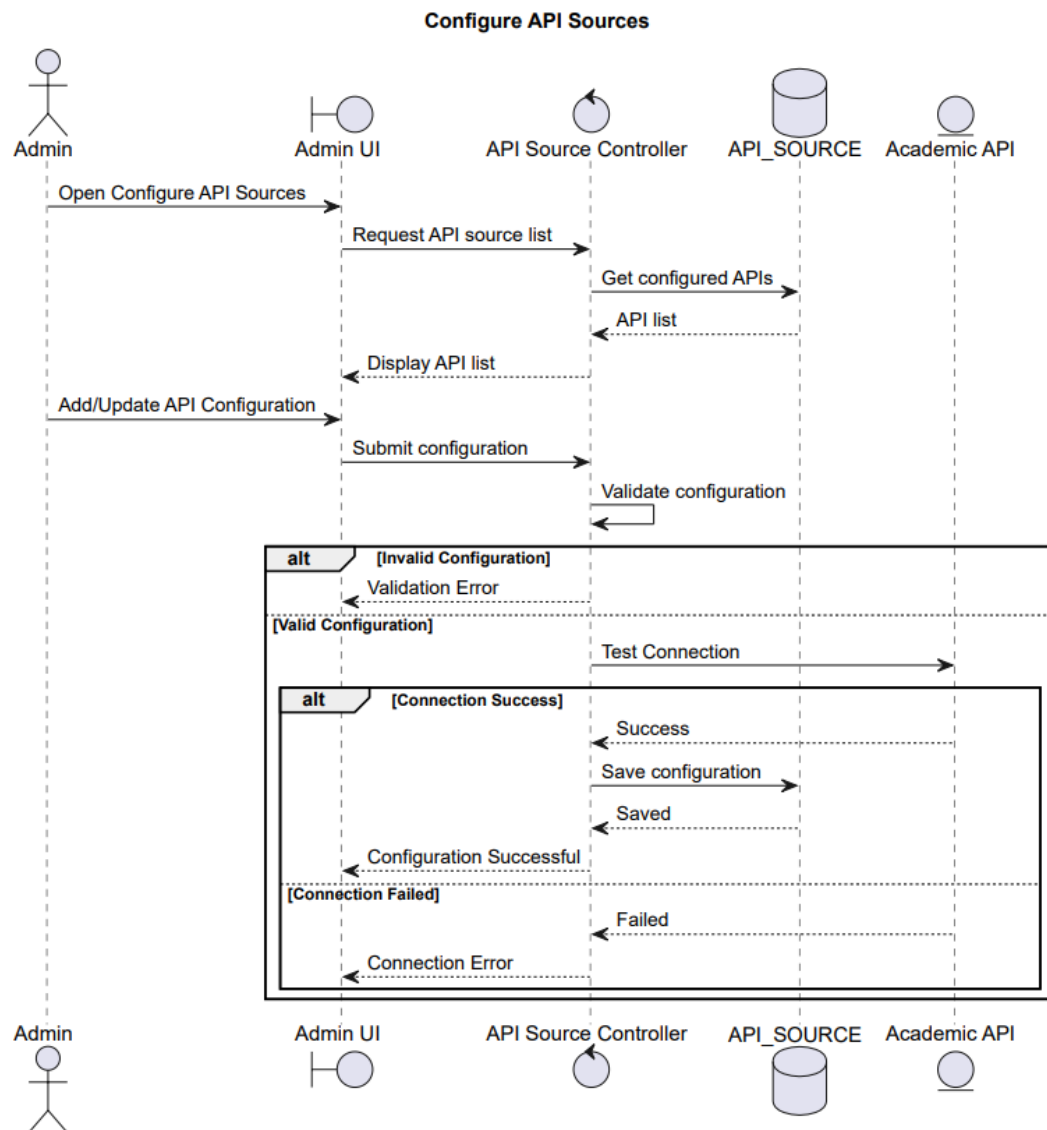
#### 3.16.1 Sequence Diagram



### 3.17 Configure API Sources

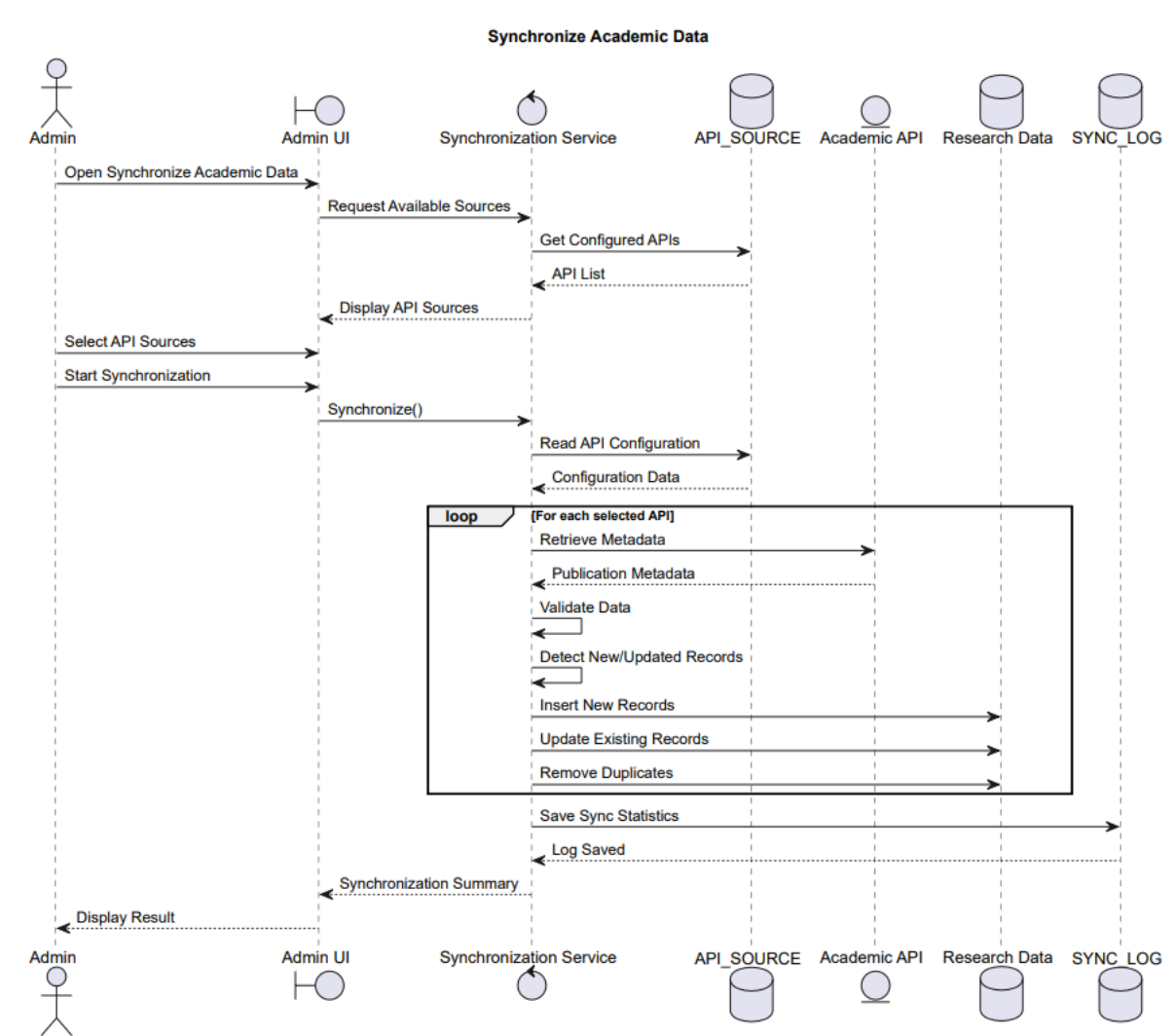
### 3.18 Configure API Sources

#### 3.18.1 Sequence Diagram



## 3.19 Sync Academic Data

### 3.19.1 Sequence Diagram





# V. SOFTWARE TESTING DOCUMENTATION

## 1 Testing Scope

The testing scope covers all major functionalities of the Scientific Journal Publication Trend Tracking System. The following modules and features are included within the scope of testing:

- User Registration
- User Login and Authentication
- Search Scientific Publications
- View Publication Details
- Save and Manage Bookmarks
- Track Publication Trends
- View Statistical Dashboard
- Generate Reports (PDF/CSV)
- View Trending Topics
- Follow Research Topics
- Receive Notifications

The following features are excluded from the testing scope:

- Third-party services and external systems that are outside the project scope.
- Real-time publication trend prediction using Artificial Intelligence (AI) or Machine Learning (ML) techniques.
- Performance testing under extremely large-scale environments involving more than 10,000 concurrent users.

## 2 Test Strategy

### 2.1 Testing Types

- **Functional Testing:** Verify that all system functions operate according to the specified requirements.
- **Integration Testing:** Verify communication and data exchange between the frontend, backend, database, and external APIs.

- **System Testing:** Validate complete business workflows and end-to-end system functionality.
- **Performance Testing:** Evaluate system response time and behavior under different workloads.

## 2.2 Testing Levels

- **Unit Testing:** Verify the correctness of individual modules using white-box testing techniques.
- **Integration Testing:** Verify interactions among integrated components and services.
- **System Testing:** Validate complete business processes using black-box testing techniques.
- **Performance Testing:** Observe system behavior and collect performance metrics under different workloads.

## 2.3 Supporting Tools

| Tool               | Purpose                                         |
|--------------------|-------------------------------------------------|
| Visual Studio Code | Application development and source code editing |
| MySQL              | Database validation and data storage testing    |
| phpMyAdmin         | Database administration and query verification  |
| Google Chrome      | User interface and functional testing           |
| Postman            | API testing and response validation             |
| GitHub             | Source code version control and collaboration   |

## 3 Test Plan

| Item          | Description                                     |
|---------------|-------------------------------------------------|
| Environment   | Windows 11, WampServer, MySQL                   |
| Browser       | Google Chrome                                   |
| Database      | MySQL                                           |
| External APIs | OpenAlex, CrossRef                              |
| Test Period   | During system development and before deployment |

### 3.1 Entry Criteria

- Source code has been completed.
- Database configuration has been completed.
- External APIs are accessible.

### 3.2 Exit Criteria

- All major functionalities have been tested.
- No critical defects remain unresolved.

## 4 Test Case

| Module                     | Number of Test Cases |
|----------------------------|----------------------|
| User Registration          | 5                    |
| User Login                 | 3                    |
| Publication Search         | 4                    |
| Bookmarks                  | 3                    |
| Follow Research Topics     | 3                    |
| Notifications              | 2                    |
| Publication Trend Analysis | 2                    |
| <b>Total</b>               | <b>22</b>            |

## 4.1 User Registration

| Field           | Description                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------|
| Module          | User Registration                                                                            |
| Test Case ID    | TC_REG_01                                                                                    |
| Test Case Name  | Valid Registration                                                                           |
| Objective       | Verify that a user can successfully create a new account.                                    |
| Preconditions   | Email and username do not exist in the database.                                             |
| Test Data       | Full Name: Group Six<br>Username: Group6<br>Email: group6@gmail.com<br>Password: Group654321 |
| Test Steps      | 1. Open Registration page.<br>2. Enter all required information.<br>3. Click Register.       |
| Expected Result | Account is created successfully and saved in the database.                                   |
| Actual Result   | Account created successfully.                                                                |
| Status          | Pass                                                                                         |

| Field           | Description                                                                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | User Registration                                                                                                                            |
| Test Case ID    | TC_REG_02                                                                                                                                    |
| Test Case Name  | Empty Required Fields                                                                                                                        |
| Objective       | Verify validation for mandatory fields.                                                                                                      |
| Preconditions   | None                                                                                                                                         |
| Test Data       | All fields are left blank.                                                                                                                   |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Open Registration page.</li> <li>2. Leave all fields empty.</li> <li>3. Click Register.</li> </ol> |
| Expected Result | System displays a validation message requiring all mandatory fields to be filled.                                                            |
| Actual Result   | Validation message displayed correctly.                                                                                                      |
| Status          | Pass                                                                                                                                         |

| Field           | Description                                                                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | User Registration                                                                                                                                    |
| Test Case ID    | TC_REG_03                                                                                                                                            |
| Test Case Name  | Invalid Email Format                                                                                                                                 |
| Objective       | Verify email format validation.                                                                                                                      |
| Preconditions   | None                                                                                                                                                 |
| Test Data       | Email: group6123                                                                                                                                     |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Open Registration page.</li> <li>2. Enter an invalid email address.</li> <li>3. Click Register.</li> </ol> |
| Expected Result | System displays "Invalid email address".                                                                                                             |
| Actual Result   | Validation message displayed correctly.                                                                                                              |
| Status          | Pass                                                                                                                                                 |

| Field           | Description                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | User Registration                                                                                                                           |
| Test Case ID    | TC_REG_04                                                                                                                                   |
| Test Case Name  | Weak Password                                                                                                                               |
| Objective       | Verify password strength validation.                                                                                                        |
| Preconditions   | None                                                                                                                                        |
| Test Data       | Password: 123456                                                                                                                            |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Open Registration page.</li> <li>2. Enter a weak password.</li> <li>3. Click Register.</li> </ol> |
| Expected Result | System displays a password strength validation message.                                                                                     |
| Actual Result   | Validation message displayed correctly.                                                                                                     |
| Status          | Pass                                                                                                                                        |

| Field           | Description                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | User Registration                                                                                                                             |
| Test Case ID    | TC_REG_05                                                                                                                                     |
| Test Case Name  | Existing Email                                                                                                                                |
| Objective       | Verify duplicate email validation.                                                                                                            |
| Preconditions   | The email already exists in the database.                                                                                                     |
| Test Data       | Email: group6@gmail.com                                                                                                                       |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Open Registration page.</li> <li>2. Enter an existing email.</li> <li>3. Click Register.</li> </ol> |
| Expected Result | System displays "Email already registered".                                                                                                   |
| Actual Result   | Validation message displayed correctly.                                                                                                       |
| Status          | Pass                                                                                                                                          |

## 4.2 User Login

| Field           | Description                                                                     |
|-----------------|---------------------------------------------------------------------------------|
| Module          | User Login                                                                      |
| Test Case ID    | TC_LOG_01                                                                       |
| Test Case Name  | Successful Login                                                                |
| Objective       | Verify that users can log in with valid credentials.                            |
| Preconditions   | User account exists and is active.                                              |
| Test Data       | Username: Group6<br>Password: Group654321                                       |
| Test Steps      | 1. Open Login page.<br>2. Enter valid username and password.<br>3. Click Login. |
| Expected Result | User is redirected to the dashboard successfully.                               |
| Actual Result   | User logged in successfully.                                                    |
| Status          | Pass                                                                            |

| Field           | Description                                                                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | User Login                                                                                                                                                                        |
| Test Case ID    | TC_LOG_02                                                                                                                                                                         |
| Test Case Name  | Invalid Login Credentials                                                                                                                                                         |
| Objective       | Verify system behavior when username or password is incorrect.                                                                                                                    |
| Preconditions   | User login page is accessible.                                                                                                                                                    |
| Test Data       | <div>Scenario 1:</div> <div>Username: Group6</div> <div>Password: WrongPassword123</div> <div>Scenario 2:</div> <div>Username: UnknownUser</div> <div>Password: Group654321</div> |
| Test Steps      | <div>1. Open Login page.</div> <div>2. Enter invalid credentials.</div> <div>3. Click Login.</div>                                                                                |
| Expected Result | System displays “Invalid username or password”.                                                                                                                                   |
| Actual Result   | Error message displayed correctly.                                                                                                                                                |
| Status          | Pass                                                                                                                                                                              |



| Field           | Description                                                                                                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------|
| Module          | User Login                                                                                                                         |
| Test Case ID    | TC_LOG_03                                                                                                                          |
| Test Case Name  | Empty Credentials                                                                                                                  |
| Objective       | Verify validation for empty login fields.                                                                                          |
| Preconditions   | None.                                                                                                                              |
| Test Data       | Username and Password are empty.                                                                                                   |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Open Login page.</li> <li>2. Leave all fields blank.</li> <li>3. Click Login.</li> </ol> |
| Expected Result | System displays validation messages for required fields.                                                                           |
| Actual Result   | Validation messages displayed correctly.                                                                                           |
| Status          | Pass                                                                                                                               |

### 4.3 Search Scientific Publication

| Field           | Description                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------|
| Module          | Search Publications                                                                                              |
| Test Case ID    | TC_SEARCH_01                                                                                                     |
| Test Case Name  | Search by Valid Keyword                                                                                          |
| Objective       | Verify that users can search publications using a valid keyword.                                                 |
| Preconditions   | Publication data exists in the database.                                                                         |
| Test Data       | Keyword: Machine Learning                                                                                        |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Enter keyword "Machine Learning".</li> <li>2. Click Search.</li> </ol> |
| Expected Result | A list of relevant publications is displayed.                                                                    |
| Actual Result   | Relevant publications displayed successfully.                                                                    |
| Status          | Pass                                                                                                             |

| Field           | Description                                                                                                    |
|-----------------|----------------------------------------------------------------------------------------------------------------|
| Module          | Search Publications                                                                                            |
| Test Case ID    | TC_SEARCH_02                                                                                                   |
| Test Case Name  | Search with No Results                                                                                         |
| Objective       | Verify system behavior when no publication matches the search keyword.                                         |
| Preconditions   | System is available.                                                                                           |
| Test Data       | Keyword: abcdeftopic123                                                                                        |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Enter keyword "abcdeftopic123".</li> <li>2. Click Search.</li> </ol> |
| Expected Result | System displays "No publications found".                                                                       |
| Actual Result   | No results displayed correctly.                                                                                |
| Status          | Pass                                                                                                           |

| Field           | Description                                                                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | Search Publications                                                                                                                                |
| Test Case ID    | TC_SEARCH_03                                                                                                                                       |
| Test Case Name  | Filter by Publication Year                                                                                                                         |
| Objective       | Verify filtering publications by publication year.                                                                                                 |
| Preconditions   | Publications from multiple years exist.                                                                                                            |
| Test Data       | Year: 2024                                                                                                                                         |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Open Search page.</li> <li>2. Select publication year = 2024.</li> <li>3. Click Apply Filter.</li> </ol> |
| Expected Result | Only publications published in 2024 are displayed.                                                                                                 |
| Actual Result   | Publications filtered correctly by year.                                                                                                           |
| Status          | Pass                                                                                                                                               |

| Field           | Description                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | Search Publications                                                                                                                           |
| Test Case ID    | TC_SEARCH_04                                                                                                                                  |
| Test Case Name  | Filter Open Access Publications                                                                                                               |
| Objective       | Verify filtering publications with Open Access status.                                                                                        |
| Preconditions   | The database contains both Open Access and non-Open Access publications.                                                                      |
| Test Data       | Open Access = True                                                                                                                            |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Open Search page.</li> <li>2. Enable Open Access filter.</li> <li>3. Click Apply Filter.</li> </ol> |
| Expected Result | Only Open Access publications are displayed.                                                                                                  |
| Actual Result   | Open Access publications displayed correctly.                                                                                                 |
| Status          | Pass                                                                                                                                          |

#### 4.4 Manage Bookmarks

| Field           | Description                                                                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | Bookmark (Favorites)                                                                                                                                                                             |
| Test Case ID    | TC_BM_01                                                                                                                                                                                         |
| Test Case Name  | Add Publication to Favorites                                                                                                                                                                     |
| Objective       | Verify that a publication can be added to the bookmark list successfully.                                                                                                                        |
| Preconditions   | User has logged into the system.                                                                                                                                                                 |
| Test Data       | Publication: "Deep Learning in Healthcare"                                                                                                                                                       |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Search for publication "Deep Learning in Healthcare".</li> <li>2. Open the publication details page.</li> <li>3. Click the Bookmark button.</li> </ol> |
| Expected Result | The publication is added to the bookmark table successfully.                                                                                                                                     |
| Actual Result   | Bookmark added successfully.                                                                                                                                                                     |
| Status          | Pass                                                                                                                                                                                             |

| Field           | Description                                                                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | Bookmark (Favorites)                                                                                                                                 |
| Test Case ID    | TC_BM_02                                                                                                                                             |
| Test Case Name  | Add Duplicate Bookmark                                                                                                                               |
| Objective       | Verify that duplicate bookmarks are not created.                                                                                                     |
| Preconditions   | The publication has already been bookmarked.                                                                                                         |
| Test Data       | Publication: “Deep Learning in Healthcare”                                                                                                           |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Open the details page of a bookmarked publication.</li> <li>2. Click the Bookmark button again.</li> </ol> |
| Expected Result | System displays “This publication is already in your favorites” and no duplicate record is created.                                                  |
| Actual Result   | Message displayed correctly and no duplicate record created.                                                                                         |
| Status          | Pass                                                                                                                                                 |

| Field           | Description                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | Bookmark (Favorites)                                                                                                                                                        |
| Test Case ID    | TC_BM_03                                                                                                                                                                    |
| Test Case Name  | Remove Bookmark                                                                                                                                                             |
| Objective       | Verify the bookmark removal functionality.                                                                                                                                  |
| Preconditions   | At least one publication exists in the bookmark list.                                                                                                                       |
| Test Data       | Bookmark ID: 2                                                                                                                                                              |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Open the Favorites page.</li> <li>2. Select the publication with Bookmark ID = 2.</li> <li>3. Click the Remove button.</li> </ol> |
| Expected Result | The publication is removed from the bookmark list.                                                                                                                          |
| Actual Result   | Bookmark removed successfully.                                                                                                                                              |
| Status          | Pass                                                                                                                                                                        |

## 4.5 View Trending Topics

| Field           | Description                                                                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | Topic Following                                                                                                                                                       |
| Test Case ID    | TC_TOPIC_01                                                                                                                                                           |
| Test Case Name  | Follow New Topic                                                                                                                                                      |
| Objective       | Verify that users can successfully follow a new research topic.                                                                                                       |
| Preconditions   | User has logged into the system.                                                                                                                                      |
| Test Data       | Topic: Artificial Intelligence                                                                                                                                        |
| Test Steps      | <ol style="list-style-type: none"><li>1. Open the Trend Topics page.</li><li>2. Select topic "Artificial Intelligence".</li><li>3. Click the Follow button.</li></ol> |
| Expected Result | The topic is stored in the UserFollowTopic table and a new notification is created.                                                                                   |
| Actual Result   | Topic followed successfully and notification created.                                                                                                                 |
| Status          | Pass                                                                                                                                                                  |

| Field           | Description                                                                                                                          |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Module          | Topic Following                                                                                                                      |
| Test Case ID    | TC_TOPIC_02                                                                                                                          |
| Test Case Name  | Follow Existing Topic                                                                                                                |
| Objective       | Verify that duplicate topic subscriptions are not created.                                                                           |
| Preconditions   | The topic has already been followed previously.                                                                                      |
| Test Data       | Topic: Artificial Intelligence                                                                                                       |
| Test Steps      | <ol style="list-style-type: none"><li>1. Select topic "Artificial Intelligence".</li><li>2. Click the Follow button again.</li></ol> |
| Expected Result | System displays "You are already following this topic" and no new record is created.                                                 |
| Actual Result   | Correct message displayed and no duplicate record created.                                                                           |
| Status          | Pass                                                                                                                                 |

| Field           | Description                                                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | Topic Following                                                                                                                                                                |
| Test Case ID    | TC_TOPIC_03                                                                                                                                                                    |
| Test Case Name  | Unfollow Topic                                                                                                                                                                 |
| Objective       | Verify that users can successfully unfollow a topic.                                                                                                                           |
| Preconditions   | The topic is currently being followed by the user.                                                                                                                             |
| Test Data       | Topic: Artificial Intelligence                                                                                                                                                 |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Open the Followed Topics list.</li> <li>2. Select topic "Artificial Intelligence".</li> <li>3. Click the Unfollow button.</li> </ol> |
| Expected Result | The topic is removed from the UserFollowTopic table.                                                                                                                           |
| Actual Result   | Topic unfollowed successfully.                                                                                                                                                 |
| Status          | Pass                                                                                                                                                                           |

## 4.6 Notification Management

| Field           | Description                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | Notification Management                                                                                                                                     |
| Test Case ID    | TC_NOTI_01                                                                                                                                                  |
| Test Case Name  | View Notification List                                                                                                                                      |
| Objective       | Verify that users can view all notifications associated with their account.                                                                                 |
| Preconditions   | User has logged into the system and has at least one notification.                                                                                          |
| Test Data       | User ID: 1                                                                                                                                                  |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Log in to the system.</li> <li>2. Click the Notification icon.</li> <li>3. Open the Notification page.</li> </ol> |
| Expected Result | The system displays all notifications belonging to the user, including title, content, and status.                                                          |
| Actual Result   | Notification list displayed successfully.                                                                                                                   |
| Status          | Pass                                                                                                                                                        |

| Field           | Description                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | Notification Management                                                                                                                                                         |
| Test Case ID    | TC_NOTI_02                                                                                                                                                                      |
| Test Case Name  | Mark Notification as Read                                                                                                                                                       |
| Objective       | Verify that users can mark a notification as read.                                                                                                                              |
| Preconditions   | User has at least one unread notification.                                                                                                                                      |
| Test Data       | Notification ID: 5                                                                                                                                                              |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Open the Notification page.</li> <li>2. Select an unread notification.</li> <li>3. Click the notification to view details.</li> </ol> |
| Expected Result | The notification status changes from “Unread” to “Read”.                                                                                                                        |
| Actual Result   | Notification status updated successfully.                                                                                                                                       |
| Status          | Pass                                                                                                                                                                            |

## 4.7 Trend Analysis

| Field           | Description                                                                                                                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module          | Trend Analysis                                                                                                                                                                                     |
| Test Case ID    | TC_TREND_01                                                                                                                                                                                        |
| Test Case Name  | Successful Trend Analysis                                                                                                                                                                          |
| Objective       | Verify that the system correctly generates publication trend analysis for a valid research topic based on synchronized publication metadata stored in the database.                                |
| Preconditions   | Publication metadata for the topic "Machine Learning" has been synchronized into the database from Semantic Scholar, OpenAlex, and Crossref.                                                       |
| Test Data       | Topic: Machine Learning<br>Year From: 2020<br>Year To: 2024                                                                                                                                        |
| Test Steps      | 1. Log in to the system.<br>2. Access "Track Publication Trends".<br>3. Enter topic "Machine Learning".<br>4. Select year range from 2020 to 2024.<br>5. Click Analyze.                            |
| Expected Result | The system aggregates publication counts by year from the database, calculates growth rate, and displays trend statistics and visual charts showing increasing or decreasing publication activity. |
| Actual Result   | Trend analysis results displayed correctly.                                                                                                                                                        |
| Status          | Pass                                                                                                                                                                                               |



| Field           | Description                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------|
| Module          | Trend Analysis                                                                                                          |
| Test Case ID    | TC_TREND_02                                                                                                             |
| Test Case Name  | Trend Analysis with No Data                                                                                             |
| Objective       | Verify system behavior when no trend data is available.                                                                 |
| Preconditions   | System is available and database contains no matching publication data for the topic                                    |
| Test Data       | Topic: abcdeftopic                                                                                                      |
| Test Steps      | <ol style="list-style-type: none"> <li>1. Enter non-existing topic "abcdeftopic".</li> <li>2. Click Analyze.</li> </ol> |
| Expected Result | An empty result list is displayed and no system error occurs.                                                           |
| Actual Result   | Empty result list displayed correctly.                                                                                  |
| Status          | Pass                                                                                                                    |